



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)



International Conference
on

Sustainable Development in Chemical and Environmental Engineering SDCEE-2024

February 22-24, 2024



BOOK OF ABSTRACTS

Edited by : Dr. Avinash Chandra

Organized By

**Department of Chemical Engineering
Thapar Institute of Engineering and Technology
Patiala, Punjab 147004, India**



About the Institute

Thapar Institute of Engineering & Technology (TIET) is among India's oldest and best universities, producing graduates who are in high demand both at home and abroad. Thapar Institute of Engineering & Technology in Patiala was established in 1956 and boasts a campus that spans an impressive 250 acres, making it one of the world's largest institutions of its kind. Our group comprises creative problem-solvers who are determined to improve the world through their ground-breaking ideas and methods. Thapar Institute of Engineering & Technology holds an "A+" accreditation from the National Assessment and Accreditation Council (NAAC), making it one of the country's best private universities and technical institutes in terms of innovation. We consider all of our students as individuals with their own interests and goals, so we work hard to tailor the lessons we teach to them. We believe in inspiring millions of young minds to succeed in India and worldwide. Therefore, we maintain a curriculum, faculty, and infrastructure that are unmatched in excellence. Our alums have made their names in various professions, including government, nonprofits, academia, and the private sector.

About Department of Chemical Engineering

The Department of Chemical Engineering at Thapar Institute of Engineering and Technology (Deemed-to-be University) was founded in 1992 with a primary mission to offer comprehensive undergraduate and postgraduate and Ph.D. programs in Chemical Engineering. The department is supported by DST-FIST and accredited by the Accreditation Board for Engineering and Technology (ABET), U.S.A., and the National Board of Accreditation (NBA). The department upholds pillars of innovation, excellence, and global research standards in the field of chemical engineering.

The Department offers three distinctive modes of B.E. programs, including B.E. Chemical Engineering (4 years duration) with an elective focus on Energy, Materials, Petroleum, and Chemical Process Engineering; B. Chem. Engg. (2+2) degree from the University of Leeds, U.K.; and an Integrated Master's degree (B.E. from TIET and Masters from the University of Leeds, UK).

A total of 13 faculty members comprises our department, each holding a PhD from prestigious institutions. Our esteemed faculty is deeply engaged in cutting-edge research, boasting an impressive track record of approximately 244 high-impact publications in SCI journals over the last 5 years, alongside 21 research projects totaling 281 lacs. With an average of 3.75 SCI publications per faculty member per year in the last 5 years, our department excels in thrust areas such as Polymeric materials, Modelling and Simulation, Energy and Environmental Engineering, Nanomaterials & Nanofluids, CO₂ Sequestration, Bio-energy and traditional areas like Heat Transfer, Mass Transfer, and Reaction Engineering.

The department facilitates entrepreneurial ventures and soft skills training through the Centre for Training and Development, while the Centre for Industrial Liaison and Placement connects students with top companies. Alumni excel in esteemed organizations like Reliance Industries Ltd, Honeywell UOP, and HPCL-Mittal Energy Limited. We adapt our curriculum to industry trends, ensuring graduates thrive in sectors like manufacturing, pharmaceuticals, and healthcare, promising a prosperous future.

About SDCEE 2024

The International Conference on “Sustainable Development in Chemical and Environmental Engineering” (SDCEE-2024) holds significant relevance in promoting sustainable practices, facilitating knowledge exchange, fostering collaborations, and outreach activities. It focuses on various areas of Chemical Engineering, including Materials, Energy, Environment, Sustainable Processing, Green Technologies, etc. It emphasizes sustainable development and encourages multidisciplinary approaches to tackle field challenges. The conference offers networking opportunities, research dissemination, and publication prospects in Journals of international repute. Overall, the conference will play a crucial role in driving innovation and sustainability in chemical and environmental engineering, making it an important event for researchers and industry professionals.

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Conference Themes

- Chemical & Process Engineering
- Sustainable Environmental Solutions
- Advancements in Energy Engineering
- Materials in Engineering
- Modeling and Simulation of Processes
- Bioprocessing and Food Engineering
- AI, Machine Learning, and Data Analytics in Process Design/Optimization

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Message From Chairperson

Prof. Rajeev Mehta

*Professor and Head,
Department of Chemical Engineering,
Co-ordinator, Centre of Excellence in Emerging Materials,
TIET, Patiala*



On behalf of the Department of Chemical Engineering, Thapar Institute of Engineering and Technology (TIET), Patiala, I am immensely honored to welcome all the delegates from across the country who have joined us for the International Conference on Sustainable Development in Chemical and Environmental Engineering (SDCEE-2024).

This conference (SDCEE-2024) serves as a vital platform to motivate researchers to collaborate and contribute their expertise towards the latest innovations and advancements in the field of Chemical and Environmental Engineering and its diverse applications. We deeply appreciate the invaluable contributions of researchers worldwide who have taken time out of their busy schedules to share their knowledge and insights with us.

Furthermore, I extend my sincere congratulations to the entire organizing team for their tireless efforts in making this conference a success. Their dedication and hard work have undoubtedly added vibrancy and significance to this international event.

Finally, I express my heartfelt gratitude to everyone who has played a role in making SDCEE-2024 a reality. Your invaluable efforts and support have contributed significantly to shaping this successful gathering.

Message from Convener

Dr. Avinash Chandra

*Associate Professor,
Department of Chemical Engineering,
TIET, Patiala*



"The future will either be green or not at all." - Bob Brown

"The greatest threat to our planet is the belief that someone else will save it." - Robert Swan

In our pursuit of sustainable development, engineering solutions must delicately balance environmental, social, and economic factors, safeguarding the prosperity of future generations.

Education transcends mere literacy; its true worth lies in the kaleidoscope of experiences it brings, enriching our lives in multifaceted ways. With this ethos in mind, we are delighted to extend this message for the "International Conference on Sustainable Development in Chemical and Environmental Engineering (SDCEE-2024)," organized by the Department of Chemical Engineering.

The primary objective of this conference is to provide a platform for young researchers, students, academicians, and industry professionals to showcase their research and professional development endeavors. It serves as a forum for not only engaging in informal mathematical discussions with experts but also delving into the forefront of research areas spanning contemporary sciences and engineering.

Our distinguished keynote and invited speakers, comprising eminent academicians, industrialists, and esteemed researchers, add immense value to the inaugural session of SDCEE-2024. We sincerely anticipate that your participation in SDCEE-2024 will be enriching and fruitful. Whether through presenting your research work or engaging with our esteemed speakers and faculty members, the experience garnered at this conference is bound to be transformative.

We extend our heartfelt gratitude to our Director, Prof. Padmakumar Nair, and Deputy Director, Prof. Ajay Batish, for their unwavering support and patronage of this event. Furthermore, we express our sincere appreciation to the dedicated faculty members of the department for their cooperation and encouragement. Special recognition is also due to all our undergraduate, postgraduate, and PhD students for their invaluable contributions, which have been instrumental in ensuring the success of this event.

OUR SPONSORS

We extend our heartfelt gratitude to our esteemed sponsors, including the Saraswati Group, SERB, BRNS, DRDO, INSA, and CSIR, whose generous contributions have played a pivotal role in ensuring the resounding success of the International Conference in “Sustainable Development of Chemical and Environmental Engineering” (SDCEE-2024). Their unwavering dedication to advancing sustainable development in chemical and environmental engineering is deeply appreciated.



OUR GOLD SPONSORS



SARASWATI GROUP: A LEADING MANUFACTURER OF FULL RANGE OF FORMULATIONS OF AGRO CHEMICALS

The Indian agrochemicals' market is significantly contributing to the growth of the Asia Pacific agrochemicals market. In the recent 5-6 years, 50% growth of agrochemicals manufacturing is due to increased export demands. Both production and export of agrochemicals have substantially increased during the last decade, and this trend is likely to continue. Government support to the farmers/agriculture sector through various packages under the "Aatmanirbhar Bharat" will likely boost agriculture further. Recently, there has been an increased focus on modernizing farming-related services by major agrochemical companies. The positive trend and the integration of farming practices in the country are expected to propel the industry growth rate of agrochemicals in India within the forecast period. Under the Green revolution movement, in 1987, Saraswati Group of Companies had its humble beginning by starting up a new Agro Chemical formulation unit in the name of 'Punjab Pesticides' at Kharar, Punjab.

The agrochemical industry is undergoing rapid growth, and it has its own set of challenges. The market may find hindrance due to low awareness of the benefits of agrochemicals among the farming community and their low acceptance of modern-day farming practices. With a safety-first approach and diverse portfolio, Saraswati Group has strengthened its foundations by fostering goodwill with global cross-sectoral partnerships across Agrochemicals, Packaging and Polycarbonate Sheet businesses. The promise of quality without sacrificing quantity drives the Saraswati Group to succeed every step of the way. It has led the company to amass 2.5 lakh square feet of warehouse and distribution facilities in India and different parts of the world. As it continues to grow, the company is touching millions of lives and turning obstacles into milestones through its strong values, and meticulous business planning.

Promising Quality Without Sacrificing Quantity

Saraswati Group has established several units in the last three decades in the Agro Chemicals sector. The company owns a total approved production capacity of 116.7 million liter/kilograms per annum across all its sites. Saraswati Agro Chemicals has registration with the US Environmental Protection Agency. "We have a world-class infrastructure for formulating Water Dispersible Granules (SDGs), Suspension Concentrates (SC), Emulsifiable Concentrates (EC), Soluble Liquids (SL), Wettable Powders (WP), Granules (GR), Water Soluble Granules (SG), Plant Growth Regulators (PGR), Capsulated Suspension (CS), Concentrated Aqueous Emulsions (EW) and Plant Nutrients. Saraswati

Group's latest establishment is 'Saraswati Plastotech,' enabling the company to shine in the industry with its 'UltraLite' Polycarbonate sheets. Automated plants have made Saraswati Group a pioneer in its industry", speaks Lalit Bansal, Managing Director, Saraswati Agro Chemicals.

The Saraswati Group empire stands tall and strong with leading accreditations such as ISO 9001, 14001, 45001, and NABL with Indian and International standards. Apart from these striking accomplishments, the association of this generational business with the likes of Bayer, BASF, Syngenta, Gowan, Corteva, FMC and Verdesian has truly made it unique. Saraswati Group has earned a great degree of trust and the capacity to produce an annual quantity of 15,000 metric tonnes for Multiwall, Compact, Textured, Corrugated, Standing Seam Panel Systems, PET-G, Light Diffusers and also Polycarbonate sheets for Green Houses.

SAIPL's mission is to contribute to India's agriculture sector, to make it a food- surplus nation and to be a leading partner in national agricultural growth and farmers' prosperity. With its values centered on offering farmers and growers - Biological, Nutritional, Seed Treatments and Inoculant Technologies that maximize performance on high-value row crops and specialty crops as well as turf and ornamental plants, Saraswati Group has gained recognition around the world.

Quote

"Growth is our habit. Quality is our DNA. Ethics are the pillars of our profits. Proof of this is visible in our long-standing partnerships. Not only did we save lives, but we also got almighty's blessings for fulfilling our duty as responsible business owners."

SDCEE-2024: Conference Schedule (In Person)

Day 1: February 22, 2024			
8:30-9:30		Registration (Auditorium)	
9:30-10:00		Inauguration (30 min)	
10:05-10:40		Chief Guest: Prof. A. B. Pandit, ICT Mumbai, INDIA Title: Progress and Nature	
10:40-11:10		High Tea break (30 min)	
Session-1: Parallel Technical Sessions (Auditorium)			
Session Chair		Prof. Chinmay Ghoroi, IIT Gandhinagar, INDIA	
Session Co-Chair		Prof. B. K. Chudasama, DSPMS, TIET, Patiala, INDIA	
11:10-11:40		Keynote-1: Prof. Animesh Jha, University of Leeds, UK Title: Semiconductor Doped Quantum dot Glass Structures for Solar Photocatalysis and Photothermal Applications	
11:45-12:15		Keynote-2: Prof. Fulvia Chiampo, Politecnico di Torino, ITALY Title: Wastes from crops and livestock as a methane resource: The role of engineering in the whole supply chain	
12:20-12:50		Keynote-3: Prof. S. Bedi, University of Waterloo, CANADA	
12:55-13-25		Keynote-4: Dr. Prhashanna Ammu, Mondelez International, USA Title: Circularity of Plastics in Packaging: A Brand Owners' perspective	
13:25-14:25		Lunch break (60 min)	
Session-2: Parallel Technical Sessions (LT-201)		Session-3: Parallel Technical Sessions (LT-202)	
Theme: Advancements in Energy Engineering (AEE)		Theme: Sustainable Environmental Solutions (SES)	
Session Chair		Session Chair	Dr. Ravi Kumar Arun, IIT Jammu, INDIA
Session Co-Chair		Session Co-Chair	Prof. H. Bhunia, DCHED, TIET, Patiala, INDIA
14:30-15:00		14:30-15:00	Keynote 6: Prof. Anil Verma, IIT Delhi, INDIA Title: Electrochemical Energy Conversion and Storage Technologies
15:05-16:05	Oral 1-4 (10 min each)		Oral 5-9 (10 min each)
	Paper ID: 32; "Performance analysis of hybrid solar desalination system with thermal storage materials at different water depth"; Authors: Mukesh Kumar, Dhananjay Singh, Suresh Kumar Patel, Ravi Kumar Sonwany, Balendu Shekher Giri, Sunita Singh		Paper ID: 19; Hydrogen Embrittlement: Mechanisms, Impacts, and Remedial Measures; Authors: Deepak Juneja, Satish Kumar Sharma
	Paper ID: 98; Intensified production of levulinic acid as the building block from the sugars; Authors: Ketaki S. Nalawade, Parag R. Gogate		Paper ID: 28; An Investigation for Malathion Pesticide Detection Using Pure Cu Electrode Leading to Electrochemical Sensor; Authors: Ashirbad Khuntia, Madhusree Kundu, Kamalakanta Mahapatra, Adhidesh Kumawat
	Paper ID: 101; Fabrication and Characterization of Shape-Stable Composite Phase Change Materials Supported by Biomass-Derived Activated Carbon for		Paper ID: 45; Advancing Sustainable Solutions: Harnessing Polyaniline/BiOCl/GO Ternary Nanocomposites for Solar-Powered Degradation of Organic Pollutant and

	Energy Storage.; Authors: Sheetal Sharwan, Bidyut Mazumdar, and Prerana Sikarwar		Photocatalytic Hydrogen Generation; Authors: Pritam Hait, Rajeev Mehta, Soumen Basu
	Paper ID: 206; An experimental study of the performance of passive type single basin double slope solar still; Authors: Mukesh Kumar, Suresh Kumar Patel, Dhananjay Singh, Vinay Mishra, Avinash Chandra, Balendu Shekher Giri, Sunita Singh		Paper ID: 70; Synthesis and Characterization of Biomass Derived Activated Carbon-Based Sulfur Cathode for Lithium Sulfur Batteries; Authors: Sakshi Tiwari, A. K. Poonia, Dharm Pal
			Paper ID: 112; Development of composite biohydrogels using gellan gum and glucomannan, and their potential application in sustainable agriculture and controlled release of bioactives in the gut.; Authors: Somnath Basak, Rekha S. Singhal
16:05-16:25 Tea break (20 min)			
16:25-17:15 Poster Session-1 (LT Block-First Floor)			
Session Chair	Dr. Mrunalini Gaydhane, BITs, Goa, INDIA		
Session Co-Chair	Dr. Diptiman Choudhury, DSCBC, TIET, Patiala, INDIA		
16:25-17:15	Poster (1-14)	16:25-17:15	Paper ID: 31; g-C3N4 loaded Bimetallic bismuth oxide nanocomposite for photocatalytic degradation of toxic pollutants.; Authors: Priti Rohilla, Dr. Bonamali Pal, Dr. Raj Kumar Das
	Paper ID: 7; Efficiency of bacteria source for bio-stabilization of soil; Authors: Shagun Ishara, Aditya Parihar, and M.S. Reddy		Paper ID: 43; Effect of nano-modified epoxy on the bond behavior of old-new concrete interface; Authors: Mandeep Kaur, Dr. Himanshu Chwala and Dr. Naveen Kwatra
	Paper ID: 8; Core-shell structure of reduced graphene oxide@Ag-TiO2 for photocatalytic H2O splitting and CH3OH dehydrogenation under UV light irradiation; Authors: Shikha Rani, Satnam Singh, Bonamali Pal		Paper ID: 88; Comparative study on supercooled silicon by using different force field model; Authors: Udisha Pathak, Arvind Kumar Gautam, Avinash Chandra
	Paper ID: 9; Comparison of machine learning models to predict the phase equilibrium of N2-CO2 system for cryogenic carbon capture; Authors: Emel V Kurian, Pavitra Sandilya		Paper ID: 115; Heterogeneous Mn3O4/CuO Interface for Boosted Oxygen Evolution Reaction: A Comprehensive Study; Authors: Sachin Pundir, Abhishek Chandel, Piyush Sharma, O.P. Pandey
	Paper ID: 11; Optimal selection of energy storage medium for cryogenic energy recovery using a multicriteria-based decision analysis; Authors: Chamarthi Bharat Surya, Pavitra Sandilya		Paper ID: 116; Bacterial Ingenuity: Synthesis of Rhamnolipid Biosurfactant Utilizing Glucose as a Growth Enhancer and Waste Engine Oil as Economical Substrates by a Non-Pathogenic Bacterium; Authors: MOHAN JUJARU AND DR. AMIT JAIN
	Paper ID: 15; Detection of H2S Gas Using Metal-Decorated and Metal-Doped Graphene Using Density Functional Theory; Authors: Diya Mehta and Robin Singla		Paper ID: 117; Beyond Graphene: Novel Perspective on 2D MBene structure; Authors: Monika, Piyush Sharma, and O.P. Pandey
	Paper ID: 17; Agro-Food Waste-Derived Bioglasses: Engineered Structures for Optimal Bone Regeneration; Authors: Navneet Kaur Mattu, Kulvir Singh		Paper ID: 119; Antibacterial activity investigation of ZnO-Ag nanocomposites prepared by thermal decomposition method against E. coli and S. aureus; Authors: Rahul Goyal, Partha Roy, P. Jeevanandam
	Paper ID: 25; Synthesis and application of bio-adsorbent for the removal of Tetracycline (TTC) antibiotics from its aqueous solution: A Equilibrium kinetic and thermodynamic study; Authors: Ashutosh		
19:00-22:00 Conference/Gala Dinner			

Day 2: February 23, 2024			
8:30-9:00		Registration (LT-201)	
Session-4: Parallel Technical Sessions (LT-201)		Session-5: Parallel Technical Sessions (LT-202)	
		Theme: Materials in Engineering (MAT)	
Session Chair	Dr. Sarang P. Gumfekar, IIT Ropar	Session Chair	Dr. Manish Kumar Dhiman, NIT Hamirpur, INDIA
Session Co-Chair	Dr. Amit Dhir, DEE, TIET, Patiala, INDIA	Session Co-Chair	Prof. Anoop Verma, DEE, TIET, Patiala, INDIA
9:00-9:30	Keynote 7: Dr. Prasun Roy, CFES, DRDO, Delhi, INDIA Title: Protective coating for mitigation of blast effects	9:00-9:30	Keynote 9: Prof. V.C. Srivastava, IIT Roorkee, INDIA
9:35-10:05	Keynote 8: Dr. Bhanu P. Singh, NPL, Delhi, INDIA Title: Carbon Nanotube Macroassemblies for Engineering Applications	9:35-10:05	Keynote 10: Prof. Chinmay Ghoroi, IIT Gandhinagar, INDIA Title: Impact of Atmospheric Particulate Matter on Energy Transition: Demystifying Solar PV Soiling Loss and its Mitigations
10:05-11:05	Oral 10-14 (10 min each)	10:05-11:05	Oral 15-19 (10 min each)
	Paper ID: 21; Synthesis of Coal-derived Graphene oxide and its Application in Glass-fiber based Polymer Composites for Improved Mechanical Performance; Authors: Anushka Garg, Soumen Basu, Rajeev Mehta, Roop L. Mahajan		Paper ID: 70; Synthesis and Characterization of Biomass Derived Activated Carbon-Based Sulfur Cathode for Lithium Sulfur Batteries; Authors: Sakshi Tiwari, A. K. Poonia, Dharm Pal
	Paper ID: 69; Fabrication and characterization of graphene-reinforced MMC by FSP for Lightning Strike Protection of aircraft; Authors: Anoop Singh, Dr Satish Kumar Sharma, Dr. Ajay Batish		Paper ID: 89; Development of epoxy/flyash composites containing Halloysite nanotubes: Thermal stability analysis and degradation kinetic; Authors: Abrar, Ahamad, Pradeep Kumar, Brajesh Kumar
	Paper ID: 73; State-of-the-art review on Rheology of Styrene-Butadiene-Styrene modified bitumen; Authors: Shubham Bansal, Vivek Gupta, Tanuj Chopra, Rajeev Mehta		Paper ID: 99; Alkaline extraction of phenolics from rice straw and their characterization; Authors: Harmeet Kaur, Dinesh Goyal
	Paper ID: 91; Styrene-Butadiene-Styrene polymer modified bitumen: A review on compatibility and physico-chemical properties; Authors: Shubham Bansal, Vivek Gupta, Tanuj Chopra, Rajeev Mehta		Paper ID: 100; Optimizing p-Nitrophenol (PNP) Degradation with Iron-Doped Bismuth and Tungsten Photocatalyst: Exploring Water Matrix Effects for Enhanced Performance; Authors: Manmeet Kaur, Dr. Amit Dhir, Dr. Shilpi Verma
	Paper ID: 93; Development and thermomechanical analysis of hybrid epoxy nanocomposite reinforced with reduced graphene oxide and HNT; Authors: Archana Arya, Pradeep Kumar, Avinash Chandra		Paper ID: 113; Synthesis of degradable copolymer through chemical sequestration of carbon dioxide using salcomine complexes; Authors: Akanksha Ranade, Prof. Rajeev Mehta, Prof. Amjad Ali
11:05-11:25		Tea break (20 min)	
Session-6: Parallel Technical Sessions (LT-201)		Session-7: Parallel Technical Sessions (LT-202)	
		Theme: Modeling and Simulation of Processes (MSP)	
Session Chair	Dr. Neelkanth Nirmalkar, IIT Ropar, INDIA	Session Chair	Dr. Shushil Kumar, MLNIT, Allahabad, INDIA
Session Co-Chair	Prof. S. K. Ahuja, DCHED, TIET, Patiala, INDIA	Session Co-Chair	Prof. D. N. Ratha, DCED, TIET, Patiala, INDIA

11:25-11:55	Keynote 11: Prof. Amit Dhiman, IIT Roorkee, INDIA; <i>Title: Computational Fluid Dynamics with Case Studies</i>	11:25-11:55	Keynote 12: Dr. Snadip Patil, E-Spin NanoTech, Kanpur, INDIA; <i>Title: Nanofiber Technology & its Commercial Applications: Future Start-Ups</i>
12:00-13:40	Oral 20-28 (10 min each)	12:00-12:30	Keynote 18: Prof. P. R. Gogate, ICT Mumbai, INDIA <i>Title: Intensified production of biofuels from sustainable raw materials using ultrasonic reactors</i>
	Paper ID: 62; Aqueous Two-Phase Flow Dynamics in Millichannels; Authors: Joseph Abhisheka Rao B, Babajan Bakthar Khan, Sunil Kumar Thamida, Anil B. Vir	12:35-13:40	Oral 29-33 (10 min each)
	Paper ID: 76; Mixed convective viscous dissipative flow of Casson hybrid nanofluid with variable thermal conductivity at the stagnation zone of a rotating sphere.; Authors: Tanvi Singla, Bhuvaneshvar Kumar, Sapna Sharma		Paper ID: 142; Fabrication and characterization of cost-effective ceramic membrane utilizing; Authors: Sandhya Khatri, Abhishek Keshav Sharan Saxena, A.B. Soni
	Paper ID: 78; Numerical Investigations on the hydrodynamic force for a Janus Cylindrical particle; Authors: Manish Dhiman		Paper ID: 147; Fungi-based Microbial Fuel Cells (MFCs) an alternative source of energy and waste management; Authors: Vaanie Godbole, Pankaj Gautam, Avinash Sharma, Manoj Kumar Pal
	Paper ID: 80; Numerical investigation on an in-line array of trimmed annular finned tubes for weight reduction in a crossflow heat exchanger; Authors: Mohit Raje, Amit Dhiman		Paper ID: 160; Utilizing Kapok seed for Mn (II) removal from synthetic wastewater solution through adsorption; Authors: Mansi Sahu, V.K. Singh
	Paper ID: 84; Identification of uniform thermodynamic equilibrium in tetrahedral liquids at liquid-liquid phase transition temperature; Authors: Arvind K. Gautam		Paper ID: 176; Treatment Of Textile And Pharmaceutical Wastewater Using Low-Cost Adsorbents For Single And Multi-Component Systems"; Authors: Umesh Kumar, Kesheo Prasad
	Paper ID: 183; Forced Convection heat transfer from confined circular/ semi-circular heaters and coolers with various orientations; Authors: Rajvinder Kaur, Sapna Sharma, Avinash Chandra		Paper ID: 196; Advance Corrosion Resistant Coatings In Recent Era: A Review of Advances And Perspectives; Authors: Sonika Lamba1, Rajendra Vishwakarmaa*
	Paper ID: 221; Effects of viscous dissipation, temperature dependent thermal conductivity and local thermal non-equilibrium on the heat transfer in a porous channel to Casson fluid; Authors: Rajvinder Kaur, Sapna Sharma, Avinash Chandra		
	Paper ID: 209; Study of phase change materials in a spherical enclosure under periodic boundary conditions: Applications in thermal energy storage and heat management systems; Authors: Tanvi Singla, Sapna Sharma, Bhuvaneshvar Kumar, Avinash Chandra		
	Paper ID: 223; Use of spent ballast as a column material for enhancing load carrying capacity of soft clay bed: Numerical Analysis; Authors: Srijan , Pooja Kharra , Kevin Somra , Gautam Narula, Aditi Sharma		
13:40-14:30 Lunch break			

Session-8: Parallel Technical Sessions (LT-201)		Session-9: Parallel Technical Sessions (LT-202)	
Theme: Applied Radiation Technology (ART) & Materials in Engineering (MAT)		Theme: AI, Machine Learning, and Data Analytics in Process Design/Optimization (AML)	
Session Chair	Prof. Amit Dhiman, IIT Roorkee, INDIA	Session Chair	Dr. Chandi Sasmal, IIT Ropar, INDIA
Session Co-Chair	Prof. B. C. Mohanty, DPMS, TIET, Patiala, INDIA	Session Co-Chair	Prof. S. S. Mallick, DME, TIET, Patiala, INDIA
14:30-15:00	Keynote 13: Dr. Rajesh K. Upadhyay, IIT-BHU, Varanasi, INDIA Title: Radioactive Particle Tracking Technique and Its Application on Different Multiphase Flow Reactors	14:30-15:00	Keynote 14: Prof. Kishalay Mitra, IIT Hyderabad, INDIA; Title: Scope of Machine Learning in Chemical Process Systems Engineering
15:05-16:15	Oral 34-39 (10 min each)	15:05-16:15	Oral 40-44 (10 min each)
	Paper ID: 123; Surface modification of titanium alloy using graphene oxide for biomedical applications; Authors: Pawandeep kaur, Jyotsana Mehta, Dinesh Goyal		Paper ID: 42; GUI development for Data Treatment in Radiotracer based RTD Analysis of Industrial Pulp digesters using Data Science; Authors: Vibhuti Sharma, Avinash Chandra, Sharad Saxena, Anu Bajaj
	Paper ID: 139; Fabrication and characteristic analysis of Graphene oxide-based Humidity sensor with an aspect for moisture sensing in transformer oil; Authors: Vikash Ranjan, Shailesh Kumar, Prasenjit Basak		Paper ID: 44; Prediction of Residence Time Distribution (RTD) models using Machine Learning Techniques for Continuous Flow Pulpig Digester; Authors: Vibhuti Sharma, Sharad Saxena, Avinash Chandra, and Anu Bajaj
	Paper ID: 181; Synthesis and characterization of novel crystalline polyaniline/Kaolin composite; Authors: Vineeta gautam, Umesh kumar, Kesheo Prasad		Paper ID: 203; "ENELP: An Ensemble Node Embedding based Link Prediction in Protein-Protein Interaction Networks by Attributed Graph Embedding"; Authors: Mukesh Kumar, Dharmendra Kumar, Sushma Kumari, Biswajit Sahoo
	Paper ID: 177; Synthesis of nano membranous composite material based on Polyaniline/MWCNTs/Cellulose acetate for selective electrochemical detection of Ascorbic Acid and Catechol"; Authors: Vineeta Gautam, Umesh Kumar		Paper ID: 79; Blockchain Technology: Catalyzing Sustainability in Chemical Industry Processes and Practices; Authors: Sonal Saxena, Seema Bawa, Rajiv Kumar
	Paper ID: 111; Comparative study on various length scale and chirality of single walled carbon nanotubes by using molecular simulation approach; Authors: Arvind K. Gautam, Rahul Saha, and Manish Kumar Dhiman		Paper ID: 30; LiFi over WiFi as a viable and health cognizant indoor Communication Technology; Authors: Anushka, Dr. Sharad Saxena, Dr. Rajesh Kumar
	Paper ID: 137; Soft, flexible and skin-compatible electrodes for monitoring electrophysiological parameters; Authors: Sunita Mehta, Purva Singal, Sunita Mishra		
16:15-16:35	Tea break (20 min)		
16:35-17:15	Poster Session-2 (LT Block-First Floor)		
Session Chair	Dr. A. K. Tiwari, Dr. BRANIT, Jalandhar, INDIA		
Session Co-Chair	Dr. Vivek Gupta, DCED, TIET, Patiala, INDIA		
16:35-17:15	Poster (14-27)		

	Paper ID: 120; Artificial Intelligence: scope and demand in Bioprocess; Authors: Naincy Tiwari, Dr. Anand Kumar Pandey	16:35-17:15	Paper ID: 138; Sustainable Integration of Rice Stubble Biochar in Cementitious Composites: A Novel Approach to Mitigate Stubble Burning Impact; Authors: Sarmad Rashid, Dr. Arpit Goyal, Dr. A.B Danie Roy, Dr. Manpreet Singh
	Paper ID: 121; Microbial Engineering: A Boon to Enhance Food Sustainability; Authors: Prachi Paharwar		Paper ID: 145; Bio-based Surfactant from Biofuels: Production and optimization using Response Surface Methodology; Authors: Neha Rawat, Aman Kumar Bhonsle, Parvez Alam, Serges Bruno Lemoupi Ngomade, Tushar Agarwal, Raj Kumar Singh, Neeraj Atray
	Paper ID: 122; Bioelectrochemical systems can reduce antibiotic-resistant bacteria in soil ecosystems: Emerging approaches for soil remediation; Authors: Sakshi Singh		Paper ID: 180; A Green approach towards synthesis and waste management of Biological MOF for drug delivery applications; Authors: Khushboo Yadava, Pawan Kumar
	Paper ID: 124; Artificial Intelligence: role and significance in pathway designing; Authors: Dr. Anand Kumar Pandey, Alsaiyya Choudhary		Paper ID: 187; Waste to Wealth: Carbon coated ZrSiO ₄ Nanocomposites from Recycled Plastic and natural Zircon For oxygen Evolution Catalysis; Authors: Abhishek Chandel and O P Pandey
	Paper ID: 125; Bioprocess Industries: The Future with AI/ML and Data Mining; Authors: Rudrakshi Mishra, Anand Kumar Pandey.		Paper ID: 213; A comprehensive study on rooftop rainwater harvesting and post-harvest treatment using Acoustic and Hydrodynamic Cavitation; Authors: Aniruddha Pandit, Ketan Desai
	Paper ID: 132; Approaches for The Synthesis of Silicon Nanoparticles; Authors: Ravinder Pal Singh and Mohit Agarwal		Paper ID: 217; Synergetic Effect of Self-Healing Tung-Oil Microcapsules with Nano-Clay in Epoxy Coatings for Corrosion Inhibition; Authors: Nikhil Sharma, Dr. Shruti Sharma, Dr. Sandeep Kumar Sharma, Dr. Rajeev Mehta
	Paper ID: 134; Sustainable trends in the development of firefighting foam; Authors: Pragya Malik, Durgesh Nandini, Bijay P. Tripathi		

Day 3: February 24, 2024

8:30-9:00 Registration (LT-201)

Session-10: Parallel Technical Sessions (LT-201)

Theme: Chemical & Process Engineering (CPE)

Session Chair Dr. A. K. Gautam, NIT Hamirpur, INDIA

Session Co-Chair Dr. Rakesh Gupta, DCHED, TIET, Patiala, INDIA

9:00-9:30 Keynote 15: Dr. Sachin Kimar, SSS-NIBE Kapurthala, INDIA
Title: *R&D Advancement in Biogas Technology for Sustainable Development*

9:35-10:05 Keynote 16: Prof. S.K. Kansal, SSBSC PTU Chandigarh, INDIA
Title: *Industrial and Municipal Wastewater- Generation and Treatment*

10:10-10:30 Invited Talk 1: Dr. S. K. Behera, VIT Vellore, INDIA

Session-11: Parallel Technical Sessions (LT-202)

Theme: Bioprocess and Food Engineering (BFE)

Session Chair Dr. R. K. Arya, Dr. BRANIT, Jalandhar, INDIA

Session Co-Chair Dr. Sapna Sharma, DOM, TIET, Patiala, INDIA

9:00-9:30 Invited Talk 2: Manish K Dhiman, NIT Hamirpur, INDIA
Title: *Modelling of Gas Explosion: #Hydrogen*

9:35-10:05 Invited Talk 3: Dr. Ashutosh Yadav, IIT Jammu, INDIA
Title: *Machine Learning-based CFD Modeling of Multiphase Reactors*

10:05-12:00 Oral 53-61 (10 min each)

	<i>Title: Industrial Symbiosis – An excellent platform for decoupling industrial growth from environmental degradation</i>		
	Oral 45-52 (10 min each)		Paper ID: 67; Effect of Drying on Extraction of Anthocyanins from Syzygium cumini (Jamun); Authors: Amritpreet Kaur, Darshanjot Kaur, Ovais Shafiq Qadri
	Paper ID: 24; Study of Interaction of Serum Albumin with Pharmacological Drugs Using Isothermal Titration Calorimetry Technique; Authors: Vani Arora, Apoorva Pathania, Harmandeep Kour, Anu Radha Pathania		Paper ID: 95; Atmospheric plasma mediated quality changes in finger millet starch and its mechanism; Authors: Parinder Kaur, Prof Uday S. Annapure
	Paper ID: 83; Ultrasound-assisted cooling crystallization of ciprofloxacin; Authors: Yagna S. Hirpara, Parag R. Gogate		Paper ID: 110; Extraction of mango (Mangifera indica) seed kernel starch: Process optimization for extraction using response surface methodology (RSM) and its comparative characterization with commercial corn (Zea mays) starch; Authors: Bharat Singh Irom, Uday S. Annapure, Jyoti S. Gokhale
10:35-12:00	Paper ID: 94; Unlocking the potential of cashew apples: Bioactives preservation and kinetic modelling; Authors: Vikramaditya R. Shirsat; Prakash D. Vaidya; Vishwanath H. Dalvi; Rekha S. Singhal; Anil K. Kelkar; Jyeshtharaj B. Joshi		Paper ID: 103; "Determination of the bovine serum albumin subjected to microwave interaction by liquid chromatography-tandem mass spectrometry"; Authors: Dr. Sushma Kumari, Mr. Dharmendra Kumar, Dr. Sunil Kumar Singla
	Paper ID: 170; Greener Production of Styrene via Low-Temperature Dehydration of Phenethyl Alcohol over Co-Supported P/ZrO ₂ ; Authors: Neha Dhiman		Paper ID: 157; Biosurfactant Production by free and immobilized cells of P.aeruginosa using waste engine oil; Authors: Mohan Jujaru, Kajol Pradhan, Shailee Gaur, Sushil Kumar and Amit Jain
	Paper ID: 28; An Investigation for Malathion Pesticide Detection Using Pure Cu Electrode Leading to Electrochemical Sensor; Authors: Ashirbad Khuntia, Madhusree Kundu, Kamalakanta Mahapatra, Adhidesh Kumawat		Paper ID: 12; A Technical Analysis of Biofuel Production from Dairy Waste Sludge for Sustainable Environment; Authors: Rohit Pathak, Sandeep Singh, Inderpreet Singh Ahuja
	Paper ID: 218; Computational evaluation of optimum tumor size-dependent MNP dose for realistic breast tumor models with different injection strategies in magnetic hyperthermia; Authors: Sandeep Nain, Dr. Neeraj Kumar, Dr. Pramod Kumar Avti		Paper ID: 220; Comparative analysis on the effect of aeration and agitation on quality and quantity of algal biomass for its further valorization to value-added products; Authors: Bimalpreet Singh, Amit Dhir and Prangya Ranjan Rout
	Paper ID: 175; Techno-economic analysis for the innovative process of		Paper ID: 149; Impact of Different Preprocessing Treatments on The

	pre-treatment of paddy straw for biofuel production; Authors: Lalitha Shree, Dr. Anurag K. Tiwari, Dr. Sachin Kumar		Germination of Bottle Gourd Seeds for the Application of Hydroponic; Authors: Pranav Prakash Dr.Mohit Agarwal Dr.Amit Munjal
	Paper ID: 150; Characterization and Morphological study of foulant in water-cooled heat exchanger of a fibre manufacturing industry; Authors: Gursimranjot Singh Mavi, Dr. Vineet Srivastava, Dr. Anoop Verma		Paper ID: 193; Effects of recycled SAW slag and the process parameters on the chemistry of stainless steel claddings; Authors: Jatinder Garg , Sonu Bala Garg , Kulwant Singh , Sunil Kumar Singla
12:00-12:20	Tea break (20 min)		
12:25-13:00	Valedictory Session (LT-201)		
13:00-14:00	Lunch break		

SDCEE-2024: Conference Schedule (Virtual)

Day 1: February 22, 2024	
8:30-9:30	Registration (Auditorium)
9:30-10:00	Inauguration (30 min)
10:05-10:40	Chief Guest: Prof. A. B. Pandit, ICT Mumbai, INDIA Title: <i>Progress and Nature</i>
10:40-11:10	High Tea break (30 min)
Virtual Parallel Technical Session-1 (Hall-C)	
<i>Theme: Advancements in Energy Engineering (AEE)</i>	
Session Chair	Dr. V.C.S. Palla, IIP Dehradun, INDIA
Session Co-Chair	Dr. Sudhir K. Singh, DCHED, TIET, Patiala, INDIA
11:10-13:30	Oral (Virtual) 1-11 (10 min each)
	Paper ID: 50; Phase Change Material Based Passive Cooling of a Single Pouch Li-Ion Cell Under Different Ambient Conditions; Authors: Rajesh Kumar, Anoop K. Gupta
	Paper ID: 54; Effects of Segmentation in Composite Phase Change Material/Solidification On Melting Performance Of Triplex-Tube Thermal Energy Storage Systems; Authors: Md Tabrez Alam, Anoop K. Gupta
	Paper ID: 81; Life Cycle Assessment of Biodiesel Produced from Cereal Byproducts; Authors: Mohd Faizullah, Gaurav Matiwari, Aman Kumar, Dr. Anil Kumar
	Paper ID: 204; Thermodynamic Analysis of CO ₂ Recompression Cycle for Critical Analysis Of Recompression Fraction And Pressure Ratio To Optimize Pressurized Water Reactor Plant Efficiency.; Authors: Garima Bharti, Manjunath K And Anil Kumar
	Paper ID: 210; Performance Evaluation of Single Slope Solar Still Augmented with Hanging Cotton Wick; Authors: Maneesh Kumar Shivhare, Samsher And Anil Kumar
	Paper ID: 197; An Experimental Study of The Deactivation of Supported Ni Catalyst Due To Sulfur Poisoning During Tri-Reforming of Biogas; Authors: Garima Bharti, Manjunath K And Anil Kumar
	<i>Theme: Bioprocess and Food Engineering (BFE)</i>
	Paper ID: 52; Application of Freeze Drying to Extend the Shelf Life Of Golden Delicious Apples; Authors: Asma Iqbal, Faiz Iqbal, Adnan Ali, Jibraan Asim
	Paper ID: 92; Optimizing Drying Parameters for Enhanced Nutritional Retention in Dried Apples: A Scientific Investigation Into Temperature, Humidity, and Pre-Treatment Methods; Authors: Tanushree Bhattacharjee, Sanchay More, Nikita Chaudhari, Maithili Chaware, Hitesh Mukunde, Diya Misal
	Paper ID: 130; Nutritional Enrichment of Salad Dressing with Flax Microgreens: Formulation and Sensory Evaluation; Authors: Dr. Soumya Rathore, Dr. Anand Kumar Pandey
	Paper ID: 168; Thermodynamic Evaluation of Calcium in Milk using Graphical Analysis; Authors: Tanushree Bhattacharjee, Subhadeep Chakrabarti, Shreyash Koyale, Sumit kumar Prasad, Pruthviraj Kadam, Mayuresh Naikwade
	Paper ID: 174; Biotransformation of Hexavalent Chromium from Wastewater Using Bacillus firmus; Authors: Pragatisheel, Abhishek Kumar, Leela Manohar Aeshala, Tapas Palai
13:25-14:25	Lunch break (60 min)
Virtual Parallel Technical Session-2 (Hall-C)	

Session Chair	Dr. Dhananjay Singh, IET Lucknow, INDIA
<i>Theme: Materials in Engineering (MAT)</i>	
Session Co-Chair	Dr. Bonamali Pal, DSCBC, TIET, Patiala, INDIA
14:30-16:00	Oral (Virtual) 12-19 (10 min each)
	Paper ID: 14; Effect of silicate units on optical and structural properties for various automotive windshield glasses; Authors: Santosh Kumar, K. Singh, Devender Kumar
	Paper ID: 40; "Effect of Superplasticizer Dosage on Fresh and Mechanical Properties of Blended Self-Compacting Geopolymer Concrete"; Authors: NARESH
	Paper ID: 53; Experimental Investigation of Sustainable Lightweight Precast Structural Elements; Authors: Durga Chaitanya Kumar Jagarapu, Arunakanthi Eluru, Chiranjeevi Rahul Rollakanti And Shalin Prince
	Paper ID: 68; Study of Chemically Treated PET Fiber with Micro Silica in M25 Grade Concrete Mix; Authors: Rishabh Joshi, Anurag Singh, Anjali Singh, Rifat Resatoglu, Shaib Abdulazeez Shehu, Mohd Zain
	Paper ID: 85; Variation of chirality cause significant thermodynamic changes in multi walled carbon nanotubes: A computational study; Authors: Arvind K. Gautam, Nandlal Pingua, Udisha Pathak, Kumud Pandey, Avinash Chandra
	Paper ID: 165; "Corrosion inhibition of mild steel using Azadirachta indica leaf extract in an alkaline medium containing Cl ⁻ ions"; Authors: Payal K.Baitule, R. Manivannan
	Paper ID: 214; Parametric Study, Optimization and Ranking Analysis of Grewia Optiva Fibre/Dolomite Based Hybrid Composites Using Hybrid MCDM Technique; Authors: Shashi Kant Verma, Ashutosh Gupta, Brijesh Gangil
16:05-16:25	Tea break (20 min)
Virtual Parallel Technical Session-3 (Hall-C)	
<i>Theme: Chemical & Process Engineering (CPE)</i>	
Session Chair	Dr. Pradeep Kumar, IET Lucknow, INDIA
Session Co-Chair	Dr. Jyotsana Mehta, DBT, TIET, Patiala, INDIA
16:25-17:25	Oral (Virtual) 20-24 (10 min each)
	Paper ID: 22; Separation of azeotropic mixture of ethyl acetate and isopropyl alcohol by extractive distillation: entrainer selection, simulation, sensitivity analysis and experimental validation"; Authors: Ms. Pooja Pandurang Sanap, Dr. Yogesh. S. Mahajan
	Paper ID: 23; Transesterification as a means to produce esters: A critical review; Authors: Dr. Reshma R. Devale, Dr. Yogesh S. Mahajan
	Paper ID: 141; Catalytic Thermo Liquefaction (CTL) of Multilayer Plastic Waste into Liquid Hydrocarbon Oil; Authors: Amol T. Waghmode, Shehbaz M. Ansari, Yogesh J. Badgujar, Vishwanath H. Dalvi, Aniruddha B. Pandit, Hitesh S. Pawar .
	Paper ID: 144; Process for biodiesel production from used cooking oil using ultrasonic intensification; Authors: Aman Kumar Bhonsle, Neha Rawat, Tushar Aggarwal, Serges Bruno Lemoupi Ngomade, Ajay Kumar, Neeraj Atray
	Paper ID: 166; Adsorption of cobalt ion from aqueous metal ion solution by thermally activated punica granatum carbon -adsorption isotherm and kinetic study; Authors: Sushma,Amit keshav,R.Manivannan
19:00-22:00	Conference/Gala Dinner
Day 2: February 23, 2024	
8:30-9:00	Registration (LT-201)
Virtual Parallel Technical Session-4 (Hall-C)	
Session Chair	Dr. V.C.S. Palla, IIP Dehradun, INDIA
<i>Theme: Sustainable Environmental Solutions (SES)</i>	
Session Co-Chair	Dr. Bunushree Behera, DBT, TIET, Patiala, INDIA
9:00-11:05	Oral (Virtual) 25-34 (10 min each)

	Paper ID: 10; Thermodynamic Modeling of Binary Binders with Agricultural Residues for Hydration Analysis; Authors: Ajeesh kumar N S,Radhika K.L,Sunil
	Paper ID: 25; Synthesis and application of bio-adsorbent for the removal of Tetracycline (TTC) antibiotics from its aqueous solution: A Equilibrium kinetic and thermodynamic study; Authors: Ashutosh
	Paper ID: 36; Osmotic Computing: A Catalyst for Cloud-Edge-IoT Integration; Authors: Akashdeep Kaur, Sharad Saxena, Rajesh Kumar
	Paper ID: 39; Experimental studies on Sustainable concrete using oil palm industrial waste; Authors: MADDULA ANJALI, Durga Chaitanya Kumar Jagarapu, Sunil Nandipati
	Paper ID: 41; Experimental investigation of Sustainable Light Weight Concrete Made with Oil Palm Industrial Waste; Authors: Gurram Likitha, Durga Chaitanya Kumar Jagarapu, Sunil Nandipati
	Paper ID: 51; "Potential utilization of bambusa balcoa in cement mortar as a reinforcing filler to improve flexural and compressive strength; Authors: Hemlata*, Harnisha Rana
	Paper ID: 57; Sustainable Clothing: Analyzing the Need and Viability of This Greener Alternative; Authors: Nidhi Sharma, Mudita Nagpal, Tanuja Nautiyal
	Paper ID: 82; Low cost bio-adsorbent for Cr(VI) contaminated effluent discharge from leather Industry; Authors: Archana Kushwaha, Zeenat Arif
	Paper ID: 126; Fabrication and synthesis of biodegradable films from cellulose waste to be used as an alternative for Plastic; Authors: R. Lokesh, Amit Keshav
	Paper ID: 127; Degradation of Congo Red Dye by using Synthesized Photocatalysts of CdO, SnO ₂ and CdO-SnO ₂ Nanocomposites in the presence of Visible Light; Authors: L. Nawin, Chandrakant Thakur
11:05-11:25	Tea break (20 min)
Virtual Parallel Technical Session-5 (Hall-C)	
<i>Theme: Sustainable Environmental Solutions (SES)</i>	
Session Chair	Dr. Dhananjay Singh, IET Lucknow, INDIA
Session Co-Chair	Dr. Vivek Pawar, DCHED, TIET, Patiala, INDIA
11:25-13:25	Oral 35-44 (Virtual) (10 min each)
	Paper ID: 131; Sustainable Lead Removal from Produced Water Using Cassava Stem Biochar: An Eco-Friendly Approach"; Authors: Vasu Gajendiran, Prabu Deivasigamani, Selvaraju Sivamani
	Paper ID: 135; Revealing Urban Atmosphere Dynamics: Understanding New Particle Formation in the Kanpur Region through Analysis of Fresh Emissions and Long-Range Transport; Authors: Vaibhav Chaturvedi, Ravish Singh Rajput, Anubhav Kumar Dwivedi
	Paper ID: 155; Quantification of Urban Growth Dynamics using Spatial Metrics and Relative Entropy in Roorkee Planning Area; Authors: Anjali Singh, Rishabh Joshi, Akash Srivastava, Shaib Abdulazeez Shehu
	Paper ID: 158; Sonochemical CuS Synthesis: Unveiling Its Importance in the Remediation of Malachite green; Authors: Akashdeep Dey, S. Noyel Victoria, Mahendra S.Gaikwad
	Paper ID: 161; 'Unravelling Arsenic Contamination in the Ganga-Ghaghara Basin, India: Reconnaissance using Geographical Information System"; Authors: Namrata Pandey,Abhishek Saxena, GeetgovindSinam, Ravi Kumar Tiwari
	Paper ID: 167; Water Footprint Assessment of Paddy based Cropping System in High Altitude Temperate Zone of Kashmir, India; Authors: Shafaq Hassan and Aditya Rana
	Paper ID: 171; Green synthesis of Copper oxide nanoparticles using Solanum tuberosum extract to mediate photocatalytic degradation of Methylene blue; Authors: Dr. Lata Sheo Bachan Upadhyay, Pratistha Bhagat
	Paper ID: 182; Facile Synthesis of Nickel Oxide Nanoparticles and Its Application Photocatalytic Remediation of Crystal Violet Dye; Authors: Anita Tirkey, Lata Sheo Bachan Upadhyay
	Paper ID: 186; Hydrothermally treated Tea Waste: A Novel Approach for Methylene Blue Photodegradation and Antimicrobial Treatment of Wastewater; Authors: Manish Pant*, Jaya Sethi, Prayas Singh, Samta Manori, Ravi Shukla, Jabrinder Singh

	Paper ID: 223; Use of spent ballast as a column material for enhancing load carrying capacity of soft clay bed: Numerical Analysis; Authors: Srijan, Pooja Kharra , Kevin Somra , Gautam Narula, Aditi Sharma
13:30-14:30	Lunch break
Virtual Parallel Technical Session-6 (Hall-C)	
<i>Theme: Sustainable Environmental Solutions (SES)</i>	
Session Chair	Dr. Pradeep Kumar, IET Lucknow, INDIA
Session Co-Chair	Dr. D. Choudhury, DSCBC, TIET, Patiala, INDIA
14:30-16:15	Oral 45-54 (10 min each)
	Paper ID: 195; Experimental Investigation of Bi-Component Particle Segregation Studies in a Spiral Concentrator; Authors: Prudhvith Reddy, Narasimha Mangadoddy
	Paper ID: 173; Environmentally friendly remediation of hazardous malachite green dye by utilizing cashew nut shell powder as bio-adsorbent: a sustainable approach; Authors: Poonam Mandale, Kavita Kulkarni, Kishor Jadhav, Anand Kulkarni,
	Paper ID: 192; Evaluating the effectiveness of eggshells in remediation of lead metal ions from aqueous solution; Isotherms and Kinetics Study; Authors: Vinay Raj, Mrityunjay Singh Chauhan, Sunder Lal Pal, Shubham Tomar
	Paper ID: 216; Optimization of microwave pyrolysis parameters to enhance yield and BET surface area of MSH char using RSM methodology; Authors; Akanksha Verma, Manoj Tripathi
	Paper ID: 201; Effect of cerium oxide as additive in biodiesel on the performance and emission characteristics of biodiesel fueled engine; Authors: Anil Kumar Yadav, Saurabh Sharma, Anil Kumar, Shailendra Sinha
	Paper ID: 202; Highly economical and efficient polyethylene separator for redox flow battery; Authors: Utsav Dalal, Anil Verma
	Paper ID: 211; Comparative study of wireless sensor networks for sustainable environmental surveillance; Authors: Upasna Joshi, Dr Rajiv Kumar
	Paper ID: 222; Groundwater quality assessment and management strategies in Mathura: Insights and Implications; Authors: Vinod Kumar Kushwah and Kunwar Raghvendra Singh
	Paper ID: 224; Application of demolished construction waste as a column infill to improve soft soil's strength: Numerical Analysis; Authors: Srijan, Gautam Narula, Abhisek Singh, Mansi Thakur, Sagar Singh, Aditi Sharma, Khawaish Gupta
	Paper ID: 225; Utilisation of fly ash as a top layer of foundation under embankment loading: Numerical Study; Authors: Gautam Narula, Srijan, Abhisek Gupta, Shashwat Singh, Rishikesh Kanitker, Aditi Sharma
16:15-16:35	Tea break (20 min)
Virtual Parallel Technical Session-7 (Hall-C)	
<i>Theme: Modeling and Simulation of Processes (MSP)</i>	
Session Chair	Dr. Ravi Kumar Arun, IIT Jammu, INDIA
Session Co-Chair	Dr. Shilpi Verma, DEE, TIET, Patiala, INDIA
16:35-17:35	Oral (Virtual) 55-60 (10 min each)
	Paper ID: 26; Polygeneration process for recovering waste cold energy in LNG regasification terminal; Authors: Hrithika G, Om Rajesh Gunjal, Satakshi Agarwal, Arnab Dutta
	Paper ID: 87; Simulation on propanol reforming for hydrogen production: A thermodynamic analysis; Authors: Brajesh Kumar, Pradeep Kumar, Arun Rai, Abhimanyu Kumar, Anuj Kumar
	Paper ID: 90; Comparative analysis of fluid flow behavior in helical coil, cylindrical pipe, and porous media through CFD simulations; Authors: Krishna Srihari Bonasi, Sehaj Anand
	Paper ID: 184; An Experimental and Computational Study of Indian Sub-Bituminous Coal for Pyrolysis of Vitrinite; Authors: Dharmendra Pandey
	Paper ID: 188; Modified Fractional Order PID Controller Design for Mixing Tank Process; Authors: Parmanand Maurya, Ram Saran Singh

	Paper ID: 207; CFD Modelling of Liquid-Liquid Batch Stirred Tank; Authors: Ms. Prafulli Mahakal, Prof. Ashwin W Patwardhan
Day 3: February 24, 2024	
8:30-9:00	Registration (LT-201)
Virtual Parallel Technical Session-7 (Hall-C)	
<i>Theme: AI, Machine Learning, and Data Analytics in Process Design/Optimization (AML)</i>	
Session Chair	Manish K. Dhiman, NIT Hamirpur, INDIA
Session Co-Chair	Dr. Sharad Saxena, DCSSED, TIET, Patiala, INDIA
9:00-11:50	Oral 61-68 (10 min each)
	Paper ID: 9; Comparison of machine learning models to predict the phase equilibrium of N ₂ -CO ₂ system for cryogenic carbon capture; Authors: Emel V Kurian, Pavitra Sandilya
	Paper ID: 27; Machine learning framework for predicting ignition properties of pure and fuel mixtures; Authors: Aditya Ravi Swami, Yashi Khandelwal, Shashank Sistla, Arnab Dutta, Iyman Abrar
	Paper ID: 37; Enhancing Osmotic Computing using Machine Learning for Dynamic Resource Optimization in Cloud Environments; Authors: Akashdeep Kaur, Sharad Saxena, Rajesh Kumar
	Paper ID: 47; Application of Opposite Point-based Differential Evolution (OPDE) algorithm for modeling VLE data; Authors: Swati Yadav, Rakesh Angira
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	Paper ID: 102; Prediction of reactant concentration in batch reactor through soft sensor using electrochemical parameters; Authors: Ashutosh Kumar Pathak; Madhusree Kundu
	Paper ID: 136; A new integrated color balance technique for underwater image Enhancement; Authors: Dr. Meenu
	Paper ID: 133; Emerging trends in the design of Non-bioaccumulable fluorinated surfactants and its application; Authors: Versha Joshi, Dr. Durgesh Nandini and Prof. Roli Purwar
	Paper ID: 128; Optimal design of coolant jacket for cryogen transfer pipelines; Authors: S.Sivasree; B.Nitin
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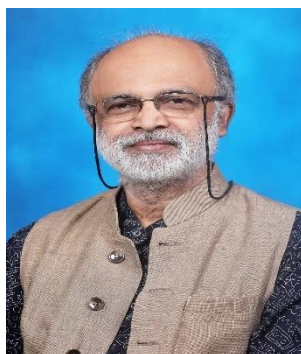
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Keynote Speakers SDCEE- 2024

Chief Guest



Prof. A. B. Pandit

*Vice Chancellor, UGC Professor
Department of Chemical Engineering
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PROGRESS AND NATURE

Sustainable development is among the most pressing and urgent challenges facing humanity today. The need for research and education to meet this challenge has been identified in virtually every recent study on engineering research needs. Fortunately, many corporations are realizing that sustainability is essential for their long-term prosperity and survival and makes good business sense. However, achieving sustainability is a formidable challenge since it cuts across all disciplinary boundaries and requires a broad, systems view that integrates social, environmental and economic aspects of technology and human activity. It requires fundamental rethinking or “creative destruction” (Hart and Milstein, 1999) of long-accepted ways of doing business and human activity, along with technological breakthroughs, methods for holistic analysis, and transformation from a reliance on growth of physical capital to a growth of just intellectual capital. Among chemical engineering specialities, Process System Engineering (PSE) is best positioned to address these challenges due to its broad systems view and multidisciplinary nature. Achieving sustainability requires a new generation of engineers who are trained to adopt a holistic view of processes as embedded in larger systems. Engineering can no longer be performed in isolation, and must consider interactions among industrial processes, human and ecological systems. Therefore, the quest for sustainability requires research and educational innovation and advances not just in engineering but also in many other disciplines, including management, economics, and sociology. Ultimately, this may be the most effective way of initiating the necessary long-term social, political, economic and technological changes for assuring human prosperity and ecological integrity throughout the world.

Prof. Animesh Jha



Professor Animesh Jha is a full-time professor of materials science at the University of Leeds. One of his main areas of research is in Photonic Materials & Devices, Energy Harvesting, and Photo-thermal energy Conversion. He has 30+ years of experience in RE-doped glass materials, processing, and applications in laser and amplifier engineering and laser-tissue matter interaction. He has published more than 230 articles in peer-reviewed journals in relevant subject areas, trained over 35 PDRFs and 45 PhD and Master's students for dissertation.

SEMICONDUCTOR DOPED QUANTUM DOT GLASS STRUCTURES FOR SOLAR PHOTOCATALYSIS AND PHOTOTHERMAL APPLICATIONS

M. Al-Murish¹, V. Autade², J. Mottram¹, N Petrou¹, R P. Panmand², S W. Gosavi³, B B. Kale², A. Scott¹, E K Barimah¹ and A. Jha^{1*}

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The amount of accessible potable water in the world is just around 1% and a significant part of it is trapped in snowfields and glaciers. The unpredictable rain, drought, and floods are becoming more frequent because of the climate change. Consequently, these unpredictable climate factors are affecting water resourcing adversely in several low- and middle-income countries situated between the Tropics of Cancer and Capricorn. The tropical regions on earth also receive abundant rain and sunlight, however, the lack of water resource management and infrastructure continues to exasperate the 2/3rd of the global population. Consumption of unclean water causes major health issues, more specifically, the women and children are the worst affected by becoming the victims of dirty water consumption.

Our goal in this talk has been to harness the sun's energy for boiling water in an engineered Q-dot containing glass heat exchanger which captures the sun's energy via blackbody radiation, and transfer effectively into heat for boiling water. Our first project vision is to produce drinking water for domestic, community, and hospital use after filtration which is free from the presence of pathogens and harmful industrial effluents. Our goal is to provide a low-cost affordable solution.

We will present the results of control of crystallisation of Q-dot semiconductors (CdS, MnS) with rare-earth ions as co-dopant for controlling the radiation absorptivity across the terrestrial solar spectrum which spans from 350 nm to 2500nm. In our investigations, we have controlled the Q-dot size for maximizing the absorbance in the visible and near-IR for enhancing the photothermal efficiency of glass heat exchanger. The bulk Q-dot glass samples were prepared by melt quenching and heat treatment. The results from heat transfer properties of heat-treated Q-dot glass samples, using solar radiation simulator, were analysed for designing a water heat exchanger with low Reynolds number for flowing water. The results of heat transfer model with Newtonian cooling rates are analysed and compared with the properties of standard Pyrex glass for designing an efficient heat exchanger system for boiling water.

Prof. Fulvia Chiampo



Fulvia Chiampo has been an associate professor of Chemical Equipment at Politecnico di Torino (Torino, Italy) since 1998. She is a chemical engineer, and at present, her scientific activity is mainly focused on environmental features, with a particular focus on solid waste treatment and soil remediation. In the past, she did research activities in the field of chemical reactors and food engineering. Her expertise on environmental topics supports local authorities and municipalities involved in waste treatment projects during the authorization procedure. Fulvia Chiampo was and is involved in international cooperation both for scientific research and for higher education programs, with India, Uzbekistan, Kazakhstan, Azerbaijan, Vietnam, and North Macedonia, in the frame of European Union programs. Currently, she is an adjunct professor at Vellore Institute of Technology – School of Civil Engineering.

WASTES FROM CROPS AND LIVESTOCK AS A METHANE RESOURCE: THE ROLE OF ENGINEERING IN THE WHOLE SUPPLY CHAIN

Fulvia Chiampo

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The use of renewable source energy and environmental respect are two of the challenges of the Third Millennium. At the same time, sustainable development is one duty and one of the goals for humankind, as adopted by the United Nations Member States in 2015 and provided by the 2030 Agenda for Sustainable Development. Energy consumption has grown more quickly in the last century than ever before. The drawback has been the use of fossil and non-renewable fuels, with emissions of huge amounts of Greenhouse Gases, responsible for climate change and linked natural disasters. The search for alternative sources is not new, however, at present it is urgent. Moreover, these sources must be renewable, to support the achievement of net-zero emissions. To this purpose, India has the target of net-zero emissions by 2070. In this framework, crop residue and livestock manure represent the raw material for anaerobic digestion, which becomes a dual tool: 1. To transform waste in biogas, i.e. fuel, thanks to its methane concentration always higher than 60% by volume. 2. To treat biodegradable waste, that otherwise would be discharged or burnt into the environment with evident impacts for air, water, and soil. The anaerobic process stabilizes biodegradable material (first target) and generates biofuel (second target), solving several environmental issues together. In this, India has a huge potential, and in particular, some States can play a relevant role as biogas producers due to their agricultural vocation. It is well known that biogas collection and treatment is compulsory to avoid accidental explosions caused by its methane content and limit its emission which has global warming effects. Given its flammability, collected biogas can be burnt for energy recovery, in thermal or electrical form, or both. In other words, one impact can be transformed into an energy resource. Moreover, looking at its composition, biogas can become a source of methane, after separation and purification with physical or chemical treatments (upgrading), to be used for transport sectors or fed to the natural gas grid. It is well evident that the supply chain from waste to biogas involves different engineering sectors, whose effective cooperation is essential to achieve the best results in terms of both energy efficiency and environmental sustainability.

Keywords: livestock manure, crop waste/residue, biogas, energy recovery, methane, Sustainable Development Goals.

Dr. Prhashanna Ammu

"Prhashanna Ammu is a principal scientist at Mondelez International. In his role, he leads the polymer material technology group to build and deliver innovation pipelines to create future polymer-based packaging materials. Prhashanna has a PhD in Chemical Engineering from NUS, Singapore. He has worked on a variety of topics related to materials during his career in academia, multinational organizations and startups. His current interests are in sustainability, AI for materials and bioplastics".



CIRCULARITY OF PLASTICS IN PACKAGING: A BRAND OWNERS' PERSPECTIVE

Prhashanna Ammu

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Plastic packaging has gained an increasing amount of attention in all aspects of society. Over the past several decades, plastics became the material of choice due to their excellent properties, performance, and economics. However, the end of life of plastics is not well managed, posing environmental threats. The pervasive environmental impact of plastic packaging has spurred a paradigm shift towards circularity, marking a crucial juncture in sustainable materials management. While the shift toward circular models holds promise for environmental benefits, technical challenges persist in material innovation and recycling technologies, necessitating continued efforts to overcome hurdles in the journey toward a more sustainable and circular economy. As guardians of both products and customer commitments, brand owners are leading the way in transforming the landscape of plastics from choosing raw materials to managing end-of-life processes. In this talk, I will highlight the existing obstacles and approaches being used by brand owners as they adopt circularity principles. This involves collaborative efforts throughout the value chain, with a focus on sustainable materials and recyclability.

Prof. Jhuma Sadhukhan



Professor Jhuma Sadhukhan is the Director of Research & Innovation in the School of Sustainability, Civil and Environmental Engineering at the University of Surrey, UK. Jhuma obtained a PhD from UMIST, UK in 2002. Jhuma specialises in biorefineries, life cycle assessment (ISO14040-44), carbon foot printing (ISO14064-65) and life cycle sustainability assessment (ISO26000). She wrote the only authored biorefinery book "Biorefineries and Chemical Processes: Design, Integration and Sustainability", Wiley, 2014.

BUILD BIOREFINERIES AND LET THE NATURAL WORLD DRIVE THE ECONOMY

Jhuma Sadhukhan

School of Sustainability, Civil and Environmental Engineering at the University of Surrey, UK.

We can save our planet by building biorefineries powered by the natural world. We need a country-wide bioeconomy plan to optimise renewable and bio-based resources (biomass) to produce products from food and pharmaceuticals through consumer chemicals to biofuel and bioenergy. In the drive of net-zero, it is easy to fixate on growing crops to power our homes or cars. However, we can do so much more with alternative resources. Around the world, businesses are turning biomass into shoe soles or skin creams. With the right investment and the right plan, a country can grow its economy sustainably based on renewables and biomass. There are plenty of reasons to prefer biological material from agricultural, forestry or municipal waste over fossil fuels. Plants are renewable and absorb carbon dioxide while they grow. The key is to build biorefineries – large plants that turn organic matter into chemicals. The more products made in the same facility, the better for the environment and the economy. Even better – they could generate energy and heat at the same time. A biorefinery producing 220 kilotonnes per annum costs about the same as a petrochemical facility: \$750 million.

Prof. Chinmay Ghoroi



Chinmay Ghoroi, PhD (IIT Bombay) is Associate Professor. His educational qualification is B.Sc. (Hons.): Calcutta University, 1995 (Ramakrishna Mission, Narendrapur), B.Tech.: Calcutta University, 1998, M.E.: Jadavpur University, 2000, Ph.D.: Indian Institute of Technology Bombay, 2007. His research interests are Particle Engineering for Dry Powder Processing, Powder Coating and Flow Characterization of Cohesive Fine Particles, Multi-phase Reactions Involving Particulate Solids.

IMPACT OF ATMOSPHERIC PARTICULATE MATTER ON ENERGY TRANSITION: DEMYSTIFYING SOLAR PV SOILING LOSS AND ITS MITIGATIONS

Chinmay Ghoroi

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India has been going through an enormous energy transition in the last couple of years, which got a significant push in recent years due to current climate change policies. With increased stress on the Net-zero emissions and the thrust on renewable energy generation, especially solar energy from photovoltaic (Solar PV) is becoming more critical than before. In fact, solar energy comprises ~54.5% share of our renewable capacity (about 73.3 GW) at the end of 2023. The country has set a goal of about 280 GW of solar energy addition by the end of 2030. However, several of our cities are currently facing massive air pollution and are listed as the most polluted cities in the world. The atmospheric particulate matter (PM) in the form of solid and liquid aerosol particles, which includes dust and other components such as ions, elemental carbon (EC), and organic carbon (OC), poses a critical challenge to energy generation through solar PVs. In fact, it is a significant threat to solar energy generation and can cause 17-25% or more reduction due to dust deposition on the PV panel. The recent research on the topic has brought significant insight into this critical problem of energy transition initiatives. The talk will cover challenges posed by dust deposition on the solar panel (known as ‘soiling’) and quantification of the threat through a few field-scale experiments. It will also discuss a few mitigation techniques and ways forward.

Keywords: Particulate Matter, Energy Transition, Renewable Energy, Solar PV, Soiling Loss.

Dr. Bhanu Pratap Singh



Dr. Bhanu Pratap Singh is Senior Principal Scientist at Advanced Carbon Products and Metrology Section of CSIR- National Physical Laboratory, New Delhi, India, and Professor at the Academy of Scientific and Industrial Research (AcSIR), India. He has over 19 years of research experience in the development of Advanced Carbon Products. His research interests include large-scale synthesis of carbon nanostructures and their advanced nanocomposites for structural applications, anode materials for Li-ion batteries, efficient EMI shielding materials, ballistic materials, flexible thermoelectrics. Dr. Singh has published 155+ research articles in prestigious international journals with 9600+ citations and 51 h-index, two books, as well as 14 book chapters and 4 patents.

CARBON NANOTUBE MACROASSEMBLIES FOR ENGINEERING APPLICATIONS

Bhanu Pratap Singh

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Scientists and engineers have been fascinated by carbon nanotubes (CNTs) for a long period of time because of their remarkable mechanical, electrical, and thermal properties. The emphasis has recently switched to utilizing unique properties of CNTs by assembling them into macrostructures, which is creating new opportunities for engineering applications. This talk aims to explore the transformative potential of CNT macroassemblies over traditional CNT powder forms. The synthesis methods that enable the creation of these macroassemblies will be discussed [1,2,3]. Further our exploration of these macroassemblies spans vital domains including structural composites [4,5], high impact resistance applications [6,7] and flexible thermoelectrics, will be discussed. CNT macroassemblies have also been explored in ballistic protection by sandwiching them between existing aramid fabrics and showed significant improvement over the existing fabrics. In the field of flexible thermoelectrics, we will discuss how CNT macroassemblies enable efficient heat-to-electricity conversion in wearable devices, leading in an era of sustainable, energy-efficient technologies. Throughout the talk, we will highlight real-world examples and ongoing research activities to demonstrate how CNT macroassemblies break down traditional barriers and redefine engineering in a variety of crucial applications. The presentation will be concluded by discussing future potential.

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Dr. P. R. Gogate



Professor Parag Gogate is currently working as Professor of Chemical Engineering at ICT, Mumbai. He is a versatile Chemical Engineer with research in the area of use of alternate energy sources, process intensification, wastewater treatment including biological oxidation, and converting lignocellulosic waste into value added products. Prof. Gogate has contributed extensively to publishing in journals of high repute with around 410 International journal publications along with over 23000 citations (h-index of 82) as per Scopus as well as has written 25 chapters in edited books. Dr. Gogate has active consultancy projects with many National/International industries and also collaborations with many research groups worldwide. He is Fellow of Indian National Academy of Engineering, The National Academy of Sciences in India and Institution of Chemical Engineers, UK.

INTENSIFIED PRODUCTION OF BIOFUELS FROM SUSTAINABLE RAW MATERIALS USING ULTRASONIC REACTORS

Parag R. Gogate

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Instability of oil prices and the imminent depletion in the traditional fossil fuels have directed research into alternate fuels and use of sustainable resources has been a key feature especially with an objective of reducing costs of production of the alternate fuels and avoiding the conflict of food against energy commonly seen for the first generation biofuels. The current talk will discuss different aspects into ultrasound assisted enzymatic biofuel production i.e. bioethanol production from sustainable biomass such as newspaper, pistachio shells, peanut shells etc. and biodiesel production mainly from waste cooking oil based on the transesterification and interesterification reactions. Initially the basics of ultrasound induced cavitation and mechanisms of intensification will be discussed followed by inputs on the possible reactor designs that can be applied for the biofuel production. The bioethanol production from lignocellulosic biomass involves pretreatment followed by hydrolysis and subsequent fermentation. Use of ultrasound can intensify each of these processing steps giving the main benefits as reduction in the time and enzyme requirement. It is important to optimize the various operating parameters to maximize the process intensification benefits that will be elucidated using various case studies. In the case of biodiesel production, ultrasound significantly influences the heterogeneous enzyme catalyzed reactions resulting in the intensification of biodiesel production. The talk will clearly bring out that ultrasound holds good promise for the intensification of biofuels from sustainable resources and combined efforts from the design engineers and biotechnologist are required for translating the successful laboratory scale projects into large scale operations.

Keywords: Ultrasound; Process Intensification; biofuels; Sustainable biomass.

Dr. Manish Dhiman

Dr. Manish Kumar Dhiman obtained his PhD degree in Chemical Engineering from Indian Institute of Technology Guwahati. He worked as a National Post-doctoral fellow in IIT Madras. He worked as the Research Scientist in Refining R&D of Reliance Industries at Jamnagar Manufacturing Division. In reliance he was part of the Modelling and Simulation team which was involved in the scale up processes for various technologies developed by Reliance through Computational Methodologies. His research interests are in Computational Fluid Dynamics, Gas Explosion Modelling, Turbulent Combustion, Fluid dynamics and Granular suspensions.



MODELLING OF GAS EXPLOSION: #HYDROGEN

Manish Dhiman

Computational Fluid Dynamics (CFD) has gained a lot of popularity in the consequence assessments of gas explosions in the past couple of decades. The Govt. of India mandate for using Green Hydrogen as a replacement to conventional energy sources is known to all. Hydrogen comes with a lot of challenges such as its transportation and its high calorific value. Keeping such intrinsic properties in the hindsight, explosion modeling pertaining to the Hydrogen leaks in an establishment is explored. This work reviews Porosity/ Distributed Resistance (PDR) approach-based CFD solver (PDRFoam) for gas explosions. PDR-based modified Favre averaged equations for mass, momentum, enthalpy, and regress variables are solved using PDRFoam solver, which is developed as a new application in OpenFOAM. The evaluation of the solver is done against medium-scale standard benchmark experiments. Compared to standard body-fitted grid simulations, the PDR approach is computationally economical; The predicted results are in good agreement with experiments.

Dr. Sandip Patil



Dr. Sandip Patil is a noteworthy entrepreneur, philanthropist, technologist & motivational speaker. He is a PhD from IIT Kanpur in Chemical Engineering with more than 13 years of research experience. His expertise and passion for nanotechnology motivated him to establish E-Spin Nanotech Pvt. Ltd. in 2010. He has authored an impressive list of scientific publications, some of which have been covered by international CNN U.S. edition, Mint, Wall Street Journal, and prominent Indian newspapers.

NANOFIBER TECHNOLOGY & ITS COMMERCIAL APPLICATIONS: FUTURE START UPS

Dr. Sandip Patil

Director, E-Spin Nanotech Pvt. Ltd., Kanpur, India

The paradigm of material science research is rapidly shifting towards the realm of nanofiber technology, signalling the arrival of innovative solutions to complex problems. This presentation delves into the pivotal domain of translational research, outlining the path from research findings to business ventures in the vast field of nanofiber technology. The discussion navigates through diverse applications and commercial product developments, spanning critical engineering sectors. Key focal points encompass energy, bio technology and filtration devices. By exploring these specialized fields, this discussion seeks to shed light on the multitude of options inherent in nanofiber technology. Furthermore, the session endeavours to provide actionable insights, guiding future startups on harnessing the unmatched potential of nanotechnology. Through an overview of its applications, this presentation aims to inspire entrepreneurial endeavours that capitalize on the limitless possibilities offered by nanofiber technology.

Dr. Sachin Kumar



Dr. Sachin Kumar, an alumnus of IIT Roorkee and HBTU, Kanpur, has an experience in the field of biomass and bioenergy. His primary research interest is in integrated biorefinery including biogas, biohydrogen, bioethanol, lignin valorization for value-added products. He has published more than 75 papers in peer reviewed journals, book chapters and papers in conference proceedings and 9 edited books. He has one granted US patent and 3 filed Indian patents. His achievements include recipient of 2016 ASM-IUSSTF Indo-US Research Professorship and selected as Bioenergy-Awards for Cutting Edge Research (B-ACER) Fellow 2016 by DBT and IUSSTF. He has also secured the place among the top 2% researchers in the world based on Scopus reported by Stanford University, USA.

R&D ADVANCEMENT IN BIOGAS TECHNOLOGY FOR SUSTAINABLE DEVELOPMENT

Sachin Kumar

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Energy is the most essential commodity, which plays an important role in the development of any country. Environment is another concern due to increased greenhouse gas (GHG) emissions, which leads to global warming. Energy consumption in India is rising rapidly with the increase in population that has grown beyond 1.4 billion. Moreover, India has committed to generate 500 GW non-fossil energy by 2030 and to cut its emission to net zero by 2070. Hence, India has to explore all the suitable and available energy sources including bioenergy to meet the energy targets. Bioenergy is a renewable energy created from materials that were once alive, known as “biomass.” Bioenergy production from agricultural leftovers, animal manure, and municipal waste has the potential to meet the rising need for sustainable energy. Biogas is one of the alternatives to the fossil fuels which is sustainable in respect to environmental, economic, and social dominant. Biogas is a mixture of gases i.e., methane, hydrogen, carbon dioxide, etc. produced by the anaerobic digestion of organic matter/ lignocellulosic biomass. In India, about 754.50 million tonnes of agricultural waste is generated annually, out of which 228.52 million tonnes is in surplus, which can be used as feedstock for biogas production.

To contribute to renewable energy generation, SSS-NIBE, Kapurthala has developed a thermophilic consortium THERMI-NIBE V.1 for anaerobic digestion of different feedstocks including agro-residues, grasses, kitchen waste, municipal solid waste, etc. for enhanced biogas and biomethane production. A thermophilic consortium constitutes a group of microorganisms that work together to break down organic matter at high temperatures. These microorganisms thrive in high-temperature environments such as geothermal areas, hot springs and deep-sea hydrothermal vents. The development of a thermophilic consortium involves the isolation and selection of different thermophilic microorganisms that can work together in a synergistic matter to enhance the efficiency of the substrate. The use of consortium has several advantages such as a fast rate of substrate utilisation, low hydraulic retention time (HRT), lesser risk of pathogens and high biogas yield. Moreover, this consortium works well on a variety of feedstocks including lignocellulosic biomass and generates biogas with high methane content.

Keywords: Agricultural residues, Anaerobic digestion, Thermophilic consortium, Biogas.

Dr. Anil Verma

Dr. Verma did his Ph. D. (Chemical) from IIT Delhi and B. Tech., M. Tech. (Chemical) from H.B.T.I., Kanpur. His research interests are Redox Flow Batteries; CO₂ Electrochemical Reduction to Hydrocarbons; Microbial Fuel Cell; PEM Fuel Cell; Graphene Synthesis and C/C Composite for Energy Devices; DSSC.



ELECTROCHEMICAL ENERGY CONVERSION AND STORAGE TECHNOLOGIES

Anil Verma

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Excessive utilization of fossil fuel derived energy has caused adverse situations of climate change. To address such alarming situations, nations across the globe are putting collective efforts to promote the usage of renewable energy. In this effort, India has promised to cut its emissions to net zero by 2070. Moreover, it is committed that by 2030, 50% of India's electricity requirement will be fulfilled by renewable energy. Though, adopting renewable energy for a greener and cleaner environment is practically the most viable solution to tackle climate change but the intermittent nature of the renewable energy makes the task challenging. Therefore, the success and the sustainability in the current ecosystem is primarily dependent on the availability of technologically and commercially viable renewable energy conversion and storage technologies. Developing such technologies is one of the most researched problems by the scientific community till-date. We at sustainable environergy research lab (SERL), IIT Delhi, are aggressively working on methods to efficiently store and utilize renewable energy. Majorly, the research activities at SERL are focused towards development of efficient and economical storage of renewable energy, utilization of renewable energy for electrochemical reduction of CO₂, fuel cell, H₂ generation research etc. The group's contribution towards development of efficient and economical vanadium redox flow battery has been well known in the scientific community. The battery can efficiently store and utilize renewable energy for a wide range of applications such as rural electrification, e-vehicle charging station, and commercial power back-up etc.

Prof. Sushil Kumar Kansal



Sushil Kansal is a Professor of Chemical Engg. at Panjab University, Chandigarh. And Ex- Dean International students at PU, Chandigarh. He has 25 years of teaching as well as research experience in Chemical Engg./Environmental Science and Engg. He has many fellowships and awards in his credit including DAAD Fellowship, Germany; Visiting Scientist to South China University of Technology, Guangzhou, China; Nominated Visiting Scholar to visit Mauritius by UGC, Govt. of India; Visiting Faculty to University of Waterloo, Canada under Indo-Shastri Canadian Fellowship; Mentor, Newton-Bhabha India-UK Advanced Training Course in Waste Water Treatment at University of Hull, UK.

TOPIC:

INDUSTRIAL AND MUNICIPAL WASTEWATER- GENERATION AND TREATMENT

Prof. Amit Kumar Dhiman



Dr. Amit Kumar Dhiman is a Professor in the Department of Chemical Engineering, IIT Roorkee, India. He is a joint faculty member in the Centre of Excellence in Disaster Mitigation & Management (CoEDMM), IIT Roorkee. He has also worked for the Steel Authority of India Limited (SAIL). He published more than 95 research articles in refereed journals. His research interests include CFD, Heat transfer, non-Newtonian fluid mechanics, Bluff bodies, Nanofluids, Porous media, Fire research & engineering, and Safety. Dr. Dhiman is a member of the National Academy of Sciences, India (NASI), an Editorial Board member of Heat Transfer (Wiley), and the Journal of Computational and Applied Mechanics (JCAM), Hungary.

COMPUTATIONAL FLUID DYNAMICS WITH CASE STUDIES

Amit Kumar Dhiman

Department of Chemical Engineering, IIT Roorkee, India

This presentation focuses on the philosophy of computational fluid dynamics (CFD), a review of equations governing fluid flow and heat transfer, boundary conditions, types of fluid flow, methods of investigation, methods of discretization, methods of solutions, numerical errors, accuracy of solution, etc. Basic steps for analyzing a CFD problem, such as problem identification and preprocessing, solver execution, and post-processing, will be discussed. Furthermore, various case studies will be addressed using the Finite Difference Method (FDM) and the Finite Volume Method (FVM), such as heat conduction in an isolated rectangular horizontal fin, heat transfer to non-Newtonian fluids in a channel with a single built-in object, flow studies around more than one built-in objects (multiple objects), flow of non-Newtonian fluids in a T-channel, three-dimensional CFD analysis around novel fin tubes.



Prof. Rajesh Kumar Upadhyay

Prof. Rajesh Kumar Upadhyay has pursued his Bachelor of Engineering degree in Chemical Engineering with specialization in Polymer Technology in the year 2004 from Punjab Technical University. Currently he is serving as Associate Dean Student Affair and Chairman of council of wardens in IIT (BHU) Varanasi. He has received prestigious 'Bharatratna Sir Mokshagundam Vishveshwarayya Engineering Award' in Year 2019.

Radioactive Particle Tracking Technique and Its Application on Different Multiphase Flow Reactors

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Multiphase flow reactors are critically important in industrial applications across many sectors like petroleum refining, petrochemicals, bulk and fine chemicals, mineral processing, pharmaceuticals, energy and power, paper manufacturing, food processing, etc. Despite the wide-spread usage of multiphase reactors, the methodology used for scale-up of these reactors is mainly based on some heuristic "rules-of-thumb" and "correlations". The main reason for this state of affairs is that the local flow fields in such type of reactors is extremely complex and occurs at different length and time scales, and the link between these various scales has not been clearly understood. The successful approach towards the understanding of such complex flows requires reliable experimental information which is collected with a clear view towards the multi-scale character of the flows. It has been realized in recent decades that a proper investigation of multiphase reactors demands the use of sophisticated measuring techniques with capability to provide the required information over the entire flow field, as well as time, at multiple scales. Aside from the mean velocities and volume fractions, information about the flow fluctuations is also desirable. Several experimental techniques have been reported in literature to quantify the flow field in multiphase flow reactors, with each technique having its own advantages and disadvantages [3]. One of the techniques that have shown a lot of promise in recent times is the 'Radioactive Particle Tracking' (RPT) technique. The current work focuses on installation of a next generation RPT technique and its implementation on various multiphase flow reactors like gas-solid, gas- liquid and liquid-liquid systems.

In the RPT technique, a single radioactive particle (gamma ray emitter) is used as the marker of the 'phase' whose velocity field is to be mapped. The 'phase' refers to one of the phases in a multiphase reactor of interest. In case a liquid phase is being tracked, the tracer particle is made neutrally buoyant with respect to the liquid. In case a solid phase (granular particles) is being tracked, the size, shape and density of tracer particle are made identical to the solids present in the flow. The tracer particle motion is interrogated by an array of scintillation detectors that are strategically placed around the vessel of interest. Subsequently, the Lagrangian trace of this particle is used to decipher the instantaneous position time series of any "typical" fluid element (an element of liquid phase or a solid particle), and from that the instantaneous velocity time series. From this information, a rich database of flow quantities such as mean velocity fields, kinetic energies of the turbulence, shear stresses, dispersion coefficients and other parameters that represent the prevailing flow regimes and flow characteristics are evaluated. In the current work RPT is installed and implemented on different multiphase flow systems to study their flow field. Different parameters like mean axial and radial velocities, RMS velocities, granular temperature, kinetic energy of fluctuations are calculated for all the cases.

Dr. Shishir Kumar Behera



Dr. Shishir Kumar Behera is working as a professor in School of Chemical Engineering at Vellore Institute of Technology (VIT), Vellore, India. He holds Master's Degree in Chemical Engineering from Indian Institute of Technology (IIT), Madras and Ph.D. in Environmental Engineering from University of Ulsan, South Korea. He has more than 20 years of teaching and research experience in India and abroad. He has more than 50 publications in Science citation indexed journals and reputed conferences, one book and 10 book chapters to his credit.

INDUSTRIAL SYMBIOSIS – AN EXCELLENT PLATFORM FOR DECOUPLING INDUSTRIAL GROWTH FROM ENVIRONMENTAL DEGRADATION

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Industrial symbiosis, a subset of industrial ecology, engages traditionally separate industries in a collective approach to competitive advantage involving physical exchange of materials, energy, water, and/or by-products. It links to broader agendas in the fields of circular economy, eco-innovation, material and energy security, and climate change in relation to sustainable industrial development. Industrial symbiosis encompasses initiatives in which two or more industrial entities develop mutually beneficial relationship. Examples include the productive use of waste materials (also called secondary resources) by one industry generated and supplied by another industry, shared utilization of specific equipment or facilities. Environmental benefits result from reduced consumption of resources and reduced amounts of waste and emissions. Thus, industrial symbiosis promotes economic growth, while at the same time it generates environmental benefits. Implementation of industrial symbiosis, therefore, holds a great potential to unlock powerful mechanisms that foster sustainable industrial development.

Dr. Ashutosh Yadav



Dr. Yadav has completed his B.Tech. from Harcourt Butler Technological Institute, Kanpur, U.P., India, M.Tech. from Indian Institute of Technology, Delhi, India, 2012 and Ph.D. from Indian Institute of Technology, Delhi, India, 2018. His research interests are Multiphase Reactors, Machine Learning, CFD, Process Optimization, Petroleum Refining, Environmental Engineering. His teaching interests are Artificial Intelligence for Chemical Engineers, Chemical Reaction Engineering, Transport Phenomena, Process Design and Economics.

MACHINE LEARNING-BASED CFD MODELING OF MULTIPHASE REACTORS

Ashutosh Yadav

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Multiphase flow reactors are crucial in diverse industries such as pharmaceuticals, chemicals, mineral processing, petrochemicals, and food processing. Bubble column, fluidized bed, and three-phase slurry reactors have proven to be widely used among the various reactor types. However, the design and scale-up of these reactors are complex due to the complex hydrodynamics and coupled heat and mass transfer. Various Computational Fluid Dynamics (CFD) based numerical techniques are employed to delve into the hydrodynamics of these reactors, with popular approaches including Euler-Euler and Euler-Lagrangian methods. Despite their efficacy, challenges persist in reducing the high computational cost and required skill set while efficiently utilizing CFD simulation data. This study addresses these challenges by exploring three case studies involving two-phase and three-phase reactors. The focus will be on applying Machine Learning to data generated by CFD models to enhance understanding and optimize performance in these reactors. Key hydrodynamics critical to the design and scaleup of these reactors are predicted at a significantly reduced computational time.

Prof. Kishalay Mitra



Dr. Kishalay Mitra received B.E. from NIT Durgapur in 1995, M. Tech from IIT Kanpur in 1997, and Ph.D. from IIT Bombay in 2009 (all in Chemical Engineering). He worked at several engineering lead positions at General Electric Global Research, Bangalore and Tata Research Development & Design Centre, Pune for close to 15 years. He is currently a Professor in the Department of Chemical Engineering and associated faculty in Departments of Artificial Intelligence and Climate Change at IIT Hyderabad, India. His work interests lie in the interface of data analysis and process optimization such as machine learning, evolutionary optimization, optimization under uncertainty, planning and scheduling of supply chain and analysis of systems involving biology, climate change and renewable energy. His research has been supported by several leading agencies in India like DST, DBT, MoE, DRDO, Royal Academy of Engg, UK and Tata Steel. He has more than 200 international journal and conference publications to his credit. He led the implementation of the first advanced control project in the mineral processing sector funded by the Ministry of Mines and DST and earned the best project implementation award.

SCOPE OF MACHINE LEARNING IN CHEMICAL PROCESS SYSTEMS ENGINEERING

Kishalay Mitra

Department of Chemical Engineering and associated faculty in Departments of Artificial Intelligence and Climate Change at IIT Hyderabad, India.

Over the last several decades, significant progress in theoretical sciences, experimental facilities, business competition under regulatory restrictions, automation, and ubiquitous connectivity across the world has forced the conventional fields of chemical engineering to go through a humongous phase of transition. Handling and management of data of huge variety was the most noteworthy outcome of this transition, which a chemical process engineer has never experienced before. With data science, the fourth paradigm of scientific discoveries, the first three being empirical, theoretical and computational, acting as an enabler for rapid transfer of knowledge from academics to industry and then industry to society in the fields of computer science, there is a strong need for exploring and exploiting its potential towards solving various types of process engineering problems. In this talk, we are going to see a variety of applications of machine learning in the field of chemical process systems engineering performed at the Global Optimization and Knowledge Unearthing Lab (GOKUL) in the recent past.

Dr. Prasun Roy



Dr. Prasun Roy is presently working as Scientist 'F' and is heading the Protective Clothing and Materials group at the Centre for Fire, Explosive and Environment Safety, DRDO, Delhi. He did his graduation from Hindu College, Delhi university followed by M.Sc. from IIT Delhi, Chemistry Department, followed by M.Tech and Ph.D., both from Centre for Polymer Science and Engineering, IIT Delhi. He is also the recipient of the BOYSCAST fellowship from DST, under which he did his post-doctoral research in Royal Institute of Technology, Stockholm. He has guided 7 Ph.D. students till date. His research interests include development of degradable polymers, polymers for blast mitigation and fire protective clothing materials. He has 93 international research papers and 6 patents to his credit.

PROTECTIVE COATING FOR MITIGATION OF BLAST EFFECTS

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Elastomeric coatings are being advocated as excellent retrofit materials for strategic applications, particularly for blast mitigation and ballistic protection. Polyurea, an elastomer formed by the reaction of isocyanate and amine, possesses hard domains dispersed randomly within the soft domains, forming a heterogeneous landscape with a nano-segregated microstructure, with each domain exhibiting its own characteristic glass transition temperature. Commercialised in the late eighties, this relatively new entrant in the field of elastomers has received enormous attention in view of its excellent blast mitigation properties and ballistic protection. Although the literature is abundant with studies demonstrating the potential of polyurea for retrofitting applications, the underlying mechanism behind its exceptional properties has not yet been fully comprehended. The ballistic protection ability is attributed to the dynamic transition from "rubber to glass", which occurs when the material is subjected to extremely high strain rates, while the blast mitigation potential is attributed to a phenomenon more commonly referred to as "shock wave capture and neutralization". Since the blast mitigation and ballistic protection ability is decided by the hard and soft domains of polyurea, respectively, the polymer needs to be tuned for a particular application through judicious choice of the raw materials.

Contributed Talks

SDCEE- 2024

Chemical & Processing Engineering (CPE)

SEPARATION OF AZEOTROPIC MIXTURE OF ETHYL ACETATE AND ISOPROPYL ALCOHOL BY EXTRACTIVE DISTILLATION: ENTRAINER SELECTION, SIMULATION, SENSITIVITY ANALYSIS AND EXPERIMENTAL VALIDATION

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It is difficult to separate azeotropic mixtures by methods like simple distillation. Use of azeotropic and extractive distillation (AD and ED), membranes, ionic liquids (ILs) and deep eutectic solvents (DESs) are some alternatives used by industry and academia. The ampicillin sodium production process involves ethyl acetate (EtAc) and isopropyl alcohol (IPA) mixture forming an azeotrope (74 mass % EtAc, 348.45 K). This work has devised a methodology to separate this mixture into pure streams by using an entrainer in the ED column. Binary vapor liquid equilibrium (VLE) data was obtained from reliable sources and various thermodynamic models were used for prediction. Heuristics and rules of thumb were used to shortlist the model. Error in predicting the VLE and also the ability of the model to correctly predict the azeotrope were used for selection of the best model. This validated model was then used to develop a ED process for separation. VLE data for many probable entrainers was used to select dimethyl sulfoxide (DMSO) and dimethyl formamide (DMF) as the suitable entrainers. Modeling and simulation was performed to select the better of the two. Sensitivity analysis was used to obtain a sensible range of parameters. Experiments were also performed to verify the predictions. It was observed that almost pure components and entrainer can be obtained. Overall flowsheet for the separation sequence is suggested.

Keywords: azeotrope separation; complete separation flowsheet; ethyl acetate and isopropyl alcohol; extractive distillation with entrainer; simulation and experiments.

TRANSESTERIFICATION AS A MEANS TO PRODUCE ESTERS: A CRITICAL REVIEW

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Esters are important chemicals used in fine and bulk chemical industry with numerous applications: solvents, paints, varnishes, dyes and cosmetics. Ester formation is dominated mainly by Fischer esterification and transesterification. Fischer esterifications are straightforward reactions which cater to the need of most ester production but in certain cases, transesterification offers unmatched advantages. It is useful when the acid is less soluble in the alcohol or in the solvent used thus forming two layers. Water formation creates purification problems due to azeotrope formation, so transesterification can be useful. Commercially, cheaply available methyl and ethyl esters can be conveniently used as raw materials for value added ester production by transesterification. Transesterification is also useful when the parent acids are highly reactive and pose difficulty in separation. Typically, transesterifications are slow and a catalyst is needed: acids and bases, ion exchange resins, zeolites and clays. Homogeneous catalysts were used in the past which are now replaced by their heterogeneous counterparts. Heterogeneous catalysts offer advantages like reusability, reduction in corrosion and ease of separation. Esters with a number of important industrial applications can be conveniently produced by transesterification: acrylics, biodiesel and β -keto esters. This review presents an account of all these aspects among others. Tables and figures are used wherever appropriate for brevity and clarity. A large literature set was scanned and its judicious extract is presented.

Keywords: alternative to Fischer esterification; circumventing azeotrope; heterogeneous catalysts; production of industrially important chemicals; reduction in downstream separation

STUDY OF INTERACTION OF SERUM ALBUMIN WITH PHARMACOLOGICAL DRUGS USING ISOTHERMAL TITRATION CALORIMETRY TECHNIQUE

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Through literature survey, Isothermal Titration Calorimetry is an elegant physical technique. It is largely used to quantify the variety of biomolecular interactions by directly measuring the heat produced or consumed during a biomolecular process. The interaction between drugs and serum proteins plays a pivotal role in determine the pharmacokinetics, bioavailability, and therapeutic efficacy of pharmaceutical compounds. Isothermal Titration Calorimetry has been widely employed in the development of prospective medications. The uses of isothermal titration calorimetry in nanotechnology, kinetics, drug design and characterization, soft matter and polymer science, bacterial chemotaxis, food science, and antibody-antigen binding are discussed in this study. The Isothermal Titration Calorimetry study adds to our biophysical knowledge of protein-drug interactions by giving binding affinity K_a , enthalpy H , entropy S , Gibbs free energy G , and binding stoichiometry n . This review attempted to compile a collection of studies on the interactions of various pharmacological medicines with serum.

Keywords: Protein-drug interaction, Drug characterization, Thermodynamics, Human serum albumin, Calorimetry.

TREATMENT OF PHARMACEUTICAL WASTEWATER CONTAINING DOXYCYCLINE ELECTROCHEMICAL ADVANCED OXIDATION METHODS

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Pharmaceutical wastewater contains a high amount of harmful toxic compounds. Doxycycline is a semi-synthesis form of tetracycline antibiotic. Doxycycline (DOXY) antibiotics kill gram-positive and negative bacteria. 50 mgL⁻¹ concentration of doxycycline solution has been prepared for electrochemical oxidation. Ti/RuO₂/IrO₂/TiO₂ electrode was used as an anode, and stainless steel was used as a cathode for the electrochemical oxidation process of doxycycline on batch mode operations pH 3, time 110 minute, and current 1 A were used in the electrochemical oxidation process. The electrochemical advanced oxidation process effectively degrades doxycycline compounds. The electrochemical advanced oxidation process is based on the charge transfer in anode and cathode to the generation of hydroxyl radical in wastewater. Doxycycline is degraded by direct charge transfer on the electrode and indirectly degraded by the intermediate reaction during the electrochemical oxidation process. At this optimum condition, 92% of doxycycline was degraded from the synthetic pharmaceutical wastewater. Ti/RuO₂/IrO₂/TiO₂ electrode successfully degrades doxycycline in the electrochemical oxidation process.

Keywords: electrode, anode, degrade, harmful compound, hydroxyl radical.

BLOCKCHAIN TECHNOLOGY: CATALYZING SUSTAINABILITY IN CHEMICAL ENGINEERING PROCESSES AND PRACTICES

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Blockchain technology extends far beyond its initial application in the realm of digital currencies, offering transformative possibilities for a variety of industries. This research delves into the emerging field of 'Blockchain for Sustainable Chemical Engineering,' which lies at the intersection of Industry 4.0, environmental sustainability, and the digital revolution. Our paper investigates how blockchain can be applied in chemical engineering and related areas, contributing to the growing but essential conversation about its potential to enhance the sustainability of supply chains, waste management, and material tracking, among other aspects. The adoption of blockchain in environmentally sustainable supply chain processes has been relatively unexplored in academic and practical contexts. Addressing this gap, this paper examines the implementation of blockchain in managing and tracking hazardous waste from inception to disposal within the chemical industry. This process encompasses the critical aspects of safe handling and efficient recycling. Utilizing blockchain technology in this context serves as a bedrock for accessible, traceable, and verifiable data, thereby significantly bolstering the credibility and validity of sustainability claims. Our research offers a novel perspective on sustainability communication, underscored by the robust traceability inherent in blockchain. Furthermore, it presents invaluable insights and implications for both academia and industry practitioners navigating the evolving landscape of Industry 4.0.

Keywords: Blockchain technology, Chemical Engineering, Hazardous waste, Industry 4.0, Sustainability

ULTRASOUND-ASSISTED COOLING CRYSTALLIZATION OF CIPROFLOXACIN

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Ciprofloxacin is an essential oral fluoroquinolone drug used in various medications for bacterial infections and diseases. For any pharmaceutical drug, crystallization is an important step to decide on the final crystal size and any deviations result in problems in downstream processing. Size reduction of active pharmaceutical ingredients (APIs) via milling is generally applied but has drawbacks of a probable change in API polymorph, higher energy requirements and contamination possibilities. Using ultrasound during the crystallization can help to achieve the desired crystal sizes. Ultrasound passage results in generation of the cavity and its collapse that helps in micro-mixing and nucleation rate, which help in controlling the nucleation and crystal growth. Ultrasound-assisted crystallization batch of ciprofloxacin hydrochloride (CIPFX) was studied and changes in operating conditions as ultrasound power, duty cycle and irradiation time were shown to affect the crystallization. At 50% duty cycle and three irradiation cycles, $D[4,3]$ increased with an increase of power from 60W to 120W while yield increased from 60W to 80W and then reduced. Changes in duty cycle from 20% to 100% showed a decrease in $D[4,3]$ while an increase in yield was seen. With increase in irradiation time from one to five cycles, $D[4,3]$ decreased till four cycles, then increased, and yield increased till three cycles, then decreased. The optimized conditions established in the work were 80W sonication power, 80% duty cycle and three irradiation cycles which resulted in $2.90\mu\text{m}$ as the average particle size and 87.98% as yield. Conventional batch crystallization of CIPFX resulted a higher average particle size of $D[4,3]=4.86\mu\text{m}$. The characterization studies elucidated that there is no change in chemical structure after the ultrasound treatment and the crystal morphology is also unaffected by the use of ultrasound. The overall outcome of the study can be considered as improved cooling crystallization of CIPFX using ultrasound.

Keywords: API crystallization; Ciprofloxacin hydrochloride; Crystal morphology; Particle size reduction; Sonication.

MOLECULAR DYNAMICS STUDY ON TiO₂ TO INVESTIGATE THE THERMAL AND MECHANICAL BEHAVIOR FOR SELECTED ENERGY CONVERSION APPLICATIONS

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Titania (TiO₂) is an oxide material which shows many industrial applications including food, ceramics, cosmetics, photovoltaics and photocatalysis. It is manufactured in huge quantities worldwide due to the significant industrial applications at large scale. TiO₂ generally exists in three different crystalline phases such as anatase, rutile and brookite. Although it shows scientific importance of all three phases as mentioned above. However, rutile is one of the most chemically stable crystalline phase among all others. Due to the chemically stable structure, the TiO₂ in rutile phase presents the significant energy conversion application in the field of solar cell and photovoltaic industries. By considering all above facts and importance of rutile TiO₂, we would like to examine the thermal and mechanical properties of this oxide material below the melting temperature. To perform this analysis, we have followed recently developed Modified Embedded Atomic Method (MEAM) potential for Titania (TiO₂). Furthermore, to compute the thermal and mechanical properties followed by MEAM potential model we have used LAMMPS open-source software and performed molecular dynamic simulations. From the above mentioned methodology and approach a comparison between NVT, NPT and NVE ensembles has been examined for Titania (TiO₂) by following MEAM potential model in this work. The predicted results over this study are consistent with the literature which may be referred to understand the thermal and mechanical behaviors of rutile TiO₂ below the melting temperature in more significant way.

Keywords: LAMMPS; MEAM; Melting Temperature; Rutile Phase; Titania

OPTIMIZING DRYING PARAMETERS FOR ENHANCED NUTRITIONAL RETENTION IN DRIED APPLES: A SCIENTIFIC INVESTIGATION INTO TEMPERATURE, HUMIDITY, AND PRE-TREATMENT METHODS.

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This comprehensive research paper delves into innovative strategies for apple drying, with a primary focus on enhancing product quality and sustainability in response to the escalating global demand for dried apples. The imperative to devise methods that not only elevate organoleptic attributes but also reduce energy consumption and environmental impact underscores the urgency of this study. The investigation spans a spectrum of drying methodologies, encompassing traditional approaches like sun drying and hot air drying, as well as cutting-edge technologies such as vacuum drying and infrared drying. The study meticulously examines critical parameters such as drying time, temperature, and pre-treatment techniques to systematically pinpoint optimal conditions conducive to preserving the innate color, flavor, and nutritional richness of apples. Moreover, a comprehensive assessment is conducted to gauge the influence of diverse drying methods on the overall energy efficiency and environmental footprint of the operation. The research's findings provide fascinating details on the realm of fruit processing, offering a robust foundation for the advancement of commercially viable and sustainable apple drying techniques. With consumer preferences aligning increasingly with healthier and environmentally conscious choices, the findings of this study represent a pivotal stride in meeting the evolving demands of the dried fruit market. As a catalyst for innovation, this research serves as a guide for industry stakeholders and researchers alike, fostering the development of practices that harmonize both product excellence and environmental responsibility.

Keywords: Energy efficiency; Fruit processing; Gala cultivar; Global demand; Sustainability.

INTENSIFIED PRODUCTION OF LEVULINIC ACID AS THE BUILDING BLOCK FROM THE SUGARS

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Levulinic acid is a short-chain fatty acid, considered as a useful and sustainable feedstock for various products such as maleic anhydride, succinic acid, 1,4-butanediol, 2-methyl tetrahydrofuran, ethyl levulinate and γ -valerolactone. Levulinic acid can be synthesized via dehydration of hexose sugars using different homogeneous and heterogeneous catalysts. The present study deals with understanding the effect of different parameters on converting glucose to levulinic acid during hydrothermal process. The consideration of glucose for Levulinic acid production was based on the abundant availability of glucose in the lignocellulosic biomass. Different salts (NaCl, AlCl₃, FeCl₃) and acids (HCl, H₃PO₄, H₂SO₄) were applied in work along with variation in the various operating parameters as time (0-180 min), temperature (140- 180°C), nitrogen pressure (0-25 bar), glucose concentration (3-10%), agitation (100-300 RPM) and acid concentration (2-6%). The reaction was performed in a batch autoclave reactor. The study showed that variations in salts and pressure do not impact sugar conversion. On the other hand, presence of acids, mainly as homogeneous catalysts, significantly affects the process with the best results for H₂SO₄. The initial nitrogen purging also played a vital role in the sugar conversion process. The best conditions for maximum levulinic acid production were established as 160°C, 5% sugar loading, and 5% H₂SO₄ concentration, giving the highest levulinic acid yield (39.7g/g) at 90 min, with almost complete sugar conversion. The present study revealed that temperature and time are the main operating parameters in this conversion process, showing the increasing sugar conversion trend with an increase in temperature and lowering the initial sugar concentration. Overall, the study elucidates an environment-friendly method for levulinic acid production from glucose (considered as source from sustainable biomass) with various applications in the fields of pharmaceuticals, agriculture, transport (fuel extenders), cosmetics, and textiles.

Keywords: Acid; catalyst; glucose conversion; salts, yield.

A HYBRID HEAT INTEGRATION OF REACTIVE DIVIDED WALL DISTILLATION COLUMN FOR SYNTHESIS OF METHYL ACETATE USING ASPEN PLUS

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Methyl acetate is a volatile solvent that has numerous applications in the field of chemical and petrochemical industries, such as polyurethane coatings and adhesives, solvents for agricultural solvents, general organic synthesis, paints strippers, battery electrolytes, etc. Distillation technology was one of the best separation processes in the current scenario. Divided wall distillation and reactive distillation are the two techniques that are primarily used in the chemical industries for the reaction and separation of binary component mixtures. In this paper, we proposed the heat integration of the intensified distillation column in which reaction and separation occurred, known as the reactive dividing wall distillation column (RDWC). In this study, ASPEN PLUS is used to design the RDWC and Heat integration process for the production of methyl acetate. The conventional study shows that methyl acetate production at the top product was 93.78 %, side product methanol was 76.98 %, and bottom product water was 97.26 %, with the reboiler duty of 5.0353 kW. After heat integration, it was observed that there was no change in the product purity at reduced reboiler heat duty, i.e., 1.6411 kW. This means it shows a 67.40 % reduction in energy consumption in reboiler duty.

Keywords: Heat Integration, Reactive dividing wall Column (RDWC), Methyl Acetate, Vapor Recompression (VRC)

CATALYTIC THERMO LIQUEFACTION (CTL) OF MULTILAYER PLASTIC WASTE INTO LIQUID HYDROCARBON OIL

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Plastic is a potential substrate to produce liquid fuel with the additional advantage of treatment as well as energy recovery if treated scientifically. The present work relates to a process for catalytic thermo liquefaction (CTL) for converting multilayer plastic waste (MLP) into hydrocarbon oil in presence of heterogenous transition metal catalyst at non-stringent operating conditions. The CTL process is having potential to convert multilayer plastic trash into liquid hydrocarbon oil (HC-Oil). The novel transition metal supported catalyst provides a complete conversion of plastic waste to produce > 90 % yield of hydrocarbon oil with negligible char and gas formation. The use of copper doped TiO₂ (Cu@TiO₂) catalysts provides several process economic benefits such as affordable cost, durability, non-toxic nature, ease of recyclability and selectivity for the liquefaction of polyolefinic multilayer plastic waste. The present study provides highly efficient, cost effective, high yield and easy to operate CTL process for converting MLP material into liquefied HC-Oil. Thus, it provides a novel approach to develop a sustainable and cost-effective process for converting plastic waste into energy, fuel and chemical products.

Keywords: Catalyst, Energy recovery, fuel oil, Liquefaction, Plastic waste.

PROCESS FOR BIODIESEL PRODUCTION FROM USED COOKING OIL USING ULTRASONIC INTENSIFICATION

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The depleting sources of crude oil drew attention to alternative fuels like biodiesel. Since biodiesel fuel generates less smoke, particulate and SO_x in exhaust gas than petroleum fuels, its wide use as a substitute for diesel is highly expected. Several methods have been suggested for biodiesel production. Ultrasound-assisted cavitation is a novel technique to intensify the reactions that are mass transfer limited particularly with the immiscible nature of the reactants. This cavitation results in the generation of hot spots as well as highly reactive free radicals leading to the intensification of many chemical reactions. The present work illustrates the use of cavitation to intensify biodiesel synthesis (transesterification) reaction. An alkali-catalyzed biodiesel production method with power ultrasonic has been developed that allows a short reaction time and high yield because of emulsification and cavitation of the liquid-liquid immiscible system. The reaction was carried out with the application of low-frequency (24 kHz) high-intensity ultrasound under atmospheric pressure and ambient temperature in a closed batch reactor. The different reaction operating parameters such as power input of ultrasound, molar ratio of triglyceride to alcohol and catalyst concentration have been optimized under acoustic cavitating conditions for the synthesis of biodiesel. It is found that using ultrasound will shorten biodiesel synthesis time compared with conventional methods leading to increased yield.

Keywords: Biodiesel; Transesterification; Ultrasound; Used Cooking oil; Yield

EXPERIMENTAL ANALYSIS ON PRODUCTION OF BIO METHANE FROM FOOD WASTE USING MICROBIAL CO-DIGESTION.

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Food waste, a readily available organic resource, holds immense potential for producing sustainable biofuel through anaerobic digestion. However, its inherent limitations, such as low nutrient content and imbalanced microbial communities, often impede efficient bio-methane production. This study explores the Combined potential of co-digesting kitchen waste with cow dung, a readily available microbial inoculum rich in nutrients and cellulose-degrading microbes. It propose a controlled experiment with digesters fed with varying ratios of kitchen waste and cow dung. Biogas production kinetics, methane yield, and microbial composition will be monitored throughout the process. Functional gene analysis will be employed to unravel the synergistic interactions between native kitchen waste microbes and the diverse microb of cow dung. This research contributes to the advancement of sustainable waste management by exploring the potential of synergistically utilizing readily available resources like kitchen waste and cow dung for biofuel production. The findings can further guide in optimizing co-digestion processes for maximum methane yield and stable reactor operation, contributing to a circular bioeconomy approach.

Keywords: Biomethane, Economical viable, anaerobic digestion, kitchen waste.

EXTRACTION OF HYDROGEN FROM WASTE COOKING OIL USING TRANSESTERIFICATION AND HAMMETT OXYGEN CARRIER CATALYST PROCESS

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The ever-increasing demand for clean energy alternatives has placed hydrogen fuel at the forefront of sustainable development. However, traditional hydrogen production methods, often relying on fossil fuels, pose significant environmental challenges. This research explores a novel and eco-friendly approach for hydrogen extraction from waste cooking oil (WCO), a readily available biomass with immense potential for valorization. The proposed method leverages a two-step process: transesterification followed by catalytic decomposition using Hammett oxygen carrier catalysts. In the first step, WCO is converted into fatty acid methyl esters (FAMES) through transesterification, a well-established and efficient process. The FAMES, essentially bio-oils, serve as the feedstock for the subsequent hydrogen generation step. The key innovation lies in the utilization of Hammett oxygen carrier catalysts for the catalytic decomposition of FAMES. These unique catalysts reversibly cycle between oxidized and reduced states, facilitating the oxygenation and subsequent cleavage of C-H bonds in the FAME molecules. This controlled fragmentation leads to the release of hydrogen gas along with carbon dioxide and water as byproducts. The advantages of this method are multifold. Firstly, it repurposes WCO, a common waste product, into a valuable resource for hydrogen production. This not only reduces environmental pollution but also alleviates the dependence on fossil fuels. Secondly, the Hammett oxygen carrier catalysts offer several benefits, including high activity, selectivity towards hydrogen production, and the ability to regenerate, thus ensuring long-term catalyst stability and economic viability.

Keywords: Transesterification, Oxygen carrier , Oxygenation waste cooking oil.

INVESTIGATING THE IMPACT OF RESERVOIR ROCK-FLUID PROPERTIES ON ADSORPTION OF SURFACTANTS ENCOUNTERED DURING SURFACTANT FLOODING OIL RECOVERY APPLICATIONS

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The enhancement in residual crude oil production from matured reservoirs to overcome the energy demand are encountered by positioning surfactant inside the reservoirs as a foreign agent. This process of surfactant flooding is a form of tertiary enhanced oil recovery schemes. The injected surfactant reduces the oil-water interfacial tension, emulsify the crude oil, and simultaneously improves micro as well as macroscopic oil displacement efficiency. However, the success of surfactant flooding depends widely on its interaction with the existing reservoir fluid and reservoir rock properties. In this study, we aimed to investigate the impact of reservoir fluid properties like salinity (0 – 5000 mg/L) of the reservoir formation water along with the reservoir temperature (30 – 70°C). Moreover, the influence of rock properties like surface charge, mineralogy of different types and surface area were further examined for rock collected from various oil field. The adsorption kinetics and isotherm model fittings were further inspected in detail to identify and understand the appropriate adsorption characteristics. The thermodynamics behaviour additionally predicts the feasibility of the adsorption process, enthalpy, entropy, and adsorption capacity as a function of temperature. The results shows that adsorption capacity is dependent on the presence of different types of minerals present in the reservoir rock. The adsorption followed Langmuir isotherm model and kinetics of pseudo second order both with R^2 value of approximately 0.99. The thermodynamics data revealed the adsorption process as feasible with exothermic in nature. The ultimate aim of this study is to derive or establish a general understanding of the adsorption system so that appropriate surfactant formulation can be achieved depending on the reservoir characteristics.

Keywords: Surfactant, Adsorption, Rock properties, Salinity, Temperature.

A GENERIC OBSERVATIONS EXPLORING THE STATUS OF DIFFERENT NANOFLUID FORMULATIONS IMPOSED FOR OIL RECOVERY APPLICATIONS

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Nanofluid produces a significant impact on the mechanisms which are held responsible for enhancing residual oil recovery from reservoirs. The mechanisms like interfacial tension behaviour between crude oil and aqueous phase, the capability to form nano-emulsion, the creaming behaviour, rheological properties and wettability characteristics of the solid reservoir rock are the main factors that governs the oil recovery factor. In this work, we have describes the status of various forms of nanofluid solutions prepared based on different reservoir characteristics in order to identify a generic pattern such that a screening guidelines can be organized. The interaction of nanofluid with crude oil belonging to different categories were observed and its correlations with the nano-emulsion formulation along with its creaming behaviour were identified. The interfacial tension reduces by more than 10 times with stable emulsion formulation retaining the average droplets size of approximately 5 microns. Moreover, the disjoining pressure were examined to illustrate the wettability alteration of the reservoirs from oil-wet to favourable water-wet. Additional, the storage and loss modulus for the nanofluid system were scrutinized to understand the rheological behaviour with an aim to induce favourable mobility ratio resulting in improved oil displacement efficiency. The oil recovery in terms of residual oil was found to be more than 16% and above when nanofluid were incorporated.

Keywords: Nanofluid, Interfacial tension, Nano-emulsion, Wettability alteration, Oil recovery.

ADSORPTION OF COBALT ION FROM AQUEOUS METAL ION SOLUTION BY THERMALLY ACTIVATED *PUNICA GRANATUM* CARBON – ADSORPTION ISOTHERM AND KINETIC STUDY

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Cobalt is used in the semiconductor industry as a connecting material in the fabrication of integrated circuits (ICs). The spent slurry in the chemical mechanical planarization of cobalt ion might contain the cobalt ion. The present study investigates the potential application of *Punica granatum* (pomegranate) peel activated carbon as adsorbent for the cobalt metal removal. *P. granatum* peel powder was thermally activated at 300°C for 30 min. Subsequently, the sample was washed thoroughly until the pH reaches 7 and utilized for the adsorption experiment. This study investigates the effect of cobalt ion removal efficiency (RE) and adsorption capacity (AC) on pH, contact time, starting cobalt ion concentration and adsorbent dosage. The aqueous solution's pH strongly influences the removal efficiency. The kinetics of Co(II) adsorption revealed that a pseudo-second order is a better fit with a rate constant of around 0.00358 g.mg/min as compared to the pseudo-first order kinetic model. SEM, EDX, FTIR, TGA analysis were performed to characterize the adsorbent used in this work. EDX composition revealed that the adsorbent after the adsorption experiment contained cobalt. The point of zero charge (pzc) study was carried out and the results revealed that a pzc lies near to pH ~5.5. Various adsorption isotherm models were fitted. However, the Freundlich isotherm model was better fitted for the adsorption of cobalt ion onto the PPAC. The maximum removal efficiency of cobalt ion is ~90% and the maximum adsorption capacity of cobalt ion ~275 mg/g was obtained at neutral pH and room temperature (~303 K).

Keywords: Adsorption capacity; Freundlich; pseudo-second order; kinetics; pomegranate peel.

GREENER PRODUCTION OF STYRENE VIA LOW-TEMPERATURE DEHYDRATION OF PHENETHYL ALCOHOL OVER CO-SUPPORTED P/ZRO₂

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The ever-rising demand for styrene forced us to find green, environmentally friendly alternative ways to manufacture it. Despite the rising need for styrene, the alternative method of synthesis from bio-based alcohols is little explored and poorly understood. We have investigated the option for the alternate produce styrene via vapour-phase dehydration of phenethyl alcohol using a cobalt phosphonate zirconia oxide. The Study demonstrates that a metallic element (cobalt) is essential for dehydrating phenethyl alcohol, restricting the decomposition of styrene to polystyrene acid. This Co phosphonate-zirconium-based catalyst exhibits selective styrene from phenylethanol showing a much greener route to producing styrene and lower activation energy ($E_a = 54.36$ KJ/mol) than other solid acid catalysts used so far. An 88.6% conversion of phenethyl alcohol was achieved with a selectivity of 75.1% over 4.5CoP/ZrO₂ catalyst. The new sustainable way to produce styrene addresses ever-growing environmental concerns to achieve UN SGD- 13.

Keywords: Styrene, Dehydration, Supported Catalyst, Phenethyl Alcohol, Sustainable Chemicals.

PROCESS INTENSIFICATION OF AROMA ABSORPTION FROM BIOREACTOR OFF-GASES

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The study explores process intensification using a Rotating Packed Bed (RPB) column and an Extractive Distillation Column (EDC). This study aims to estimate the cost of aroma absorption from bioreactor off-gas in a RPB and examine the separation of the propylene-oxide-methanol-water mixture in an extractive distillation column. This study also investigates the separation of ethyl acetate-ethanol mixture using different side stream extractive distillation configurations. The impact of rotational speed, liquid flow, and gas flow rate on efficiency in the separation of volatile organic compounds (VOCs) is also evaluated. In the separation of propylene-oxide-methanol-water mixture in an Extractive Dividing Wall Column (EDWC), the capital cost is reduced compared to Conventional Extractive Distillation (CED) by using only one reboiler for both the main column and side column, resulting in a total reduction of TAC of \$28000 per annum. In Side Stream Extractive Distillation (SSED), the capital and energy costs are lower than CED because the reboiler duty and stages of the distillation column decrease, reducing the height of the distillation column. This results in a reduction of TAC of \$40000 per annum. For the separation of ethyl acetate-ethanol mixture, the reduction of TAC in different configurations i.e. Feed -Economizer (F-E), Side -Economizer (S-E), and Feed-Side-Economizer (F-S-E) is 13.6%, 16.6%, and 18.1%, respectively.

Keywords: Extractive distillation column (EDC), Extractive dividing wall column (EDWC), Process Intensification, Rotated packed bed, Side stream extractive distillation (SSED).

ENVIRONMENTALLY FRIENDLY REMEDIATION OF HAZARDOUS MALACHITE GREEN DYE BY UTILIZING CASHEW NUT SHELL POWDER AS BIO-ADSORBENT: A SUSTAINABLE APPROACH

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In this study, the sorption potential of cashew nut shell powder was investigated as low cost adsorbent for removal of cationic dyes, malachite green (MG), from aqueous solution. In batch experiments, the effects of contact time, absorbent dose, solution pH, initial dye concentration, and temperature of MG ions on the cashew nut shell powder adsorption capacity was studied. The characterization of the cashew nut shell powder has been accomplished by electronic dispersive elemental analysis, scanning electron microscope, and Fourier transform infrared radiation. Equilibrium data were accurately fitted onto Langmuir, Freundlich, and Temkin isotherms. The results showed that the adsorption isotherm data were fitted well to the Langmuir isotherm and the maximum percentage removal calculated were 72% for MG at 25°C. The kinetic data for dye were better described by the pseudo-second-order kinetic model. It is expected that cashew nut shell powder can be considered as a low-cost adsorbent for the removal of MG dye from wastewater.

Keywords: Adsorption; Cashew nut shell powder; Malachite green; Kinetics.

EFFECT OF GAS VELOCITY AND SMALL PARTICLE FRACTION ON THE SOLID FLOW FIELD OF GAS-SOLID BINARY FLUIDIZED BED

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Gas-solid binary fluidized bed is widely used in many industries where the solids of different size and density are used. It is well established in the literature that the behavior of the binary fluidized bed is significantly different from the mono dispersed fluidized bed. The behavior of binary fluidized bed strongly depends on gas velocity and fraction of different size/density particles. Several authors have investigated the behavior of the binary fluidized bed. However, till date the dependency on the solid fraction when a mono dispersed bed start behaving like binary bed is not well understood. The current work investigate the behavior of binary gas-solid fluidized bed by using radiotracer based techniques. The experiments are performed in a cylindrical perspex column of 0.09 m ID and 1.5 m height. Compressed air is used as a gas phase and glass beads of two different sizes, 150 μm and 500 μm , of density 2500 kg/m^3 are used as solid phase. Experiments are performed for three different gas velocities and different small particle fraction (0, 3, 5 and 8%). For all the cases densitometry and radioactive particle tracking (RPT) experiments are performed to measure the solid flow filed. The solid fraction, mean solid velocity and rms velocities are calculated for all the cases to decipher the solid flow filed.

Keywords: Fluidized bed reactor, Solid flow filed, Gamma ray densitometry, Radioactive particle tracking.

AN EXPERIMENTAL STUDY OF THE DEACTIVATION OF SUPPORTED Ni CATALYST DUE TO SULFUR POISONING DURING TRI-REFORMING OF BIOGAS

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The upgradation of biogas to drop-in fuel and industrial feedstock is considered an economically viable process and an effective way to reduce greenhouse gas emissions. Tri-Reforming of Methane (TRM), also known as oxidative steam reforming of methane, has emerged as a promising way to produce renewable energy carriers such as syngas (CO + H₂). Despite various advantages of the Tri-Reforming of Methane reaction, the process is still at its infant stages because of challenges like catalyst deactivation. The deactivation of the commercially used supported Ni catalyst due to sulfur (H₂S) poisoning hinders the commercialisation of the complete process. The Tri-reforming of methane was carried out using model biogas with the introduction of sulfur poisoning to understand the effect of oxygen on the deactivation phenomena. The Tri-Reforming reactions were performed in a packed bed reactor using Ni/ γ -Al₂O₃ as the catalyst. Experimental observations were made on the deactivation of the catalyst due to sulfur poisoning both in the presence and absence of oxygen in the reacting mixture. The catalyst was observed to be stable in the absence of sulfur but deactivated rapidly in the presence of sulfur. Experimental investigation reveals that the catalyst was not oxidized when H₂S was introduced in the absence of oxygen in the reacting mixture. Catalyst regeneration was performed under steam reforming conditions, and the catalyst was subsequently reduced using hydrogen. The regeneration process in the presence of steam was slow because of the system's low hydrogen content, which was insufficient to reduce the oxidized Ni. The catalyst was able to trace back its original conversion after performing the reduction using hydrogen for 4 hours. The fresh and spent catalyst was characterized to understand the interaction between the sulfur and the active phase of the catalyst.

Keywords: Reforming; Syngas; Catalyst Poison; Steam Reforming; Catalyst Regeneration.

Sustainable Environmental Solutions (SES)

EFFICIENCY OF BACTERIA SOURCE FOR BIO-STABILIZATION OF SOIL

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Bio-stabilization by Microbially induced carbonate precipitation (MICP) is a relatively new ground improvement approach wherein the shear strength of soil is tremendously increased. The increment is due to calcium carbonate precipitation resulting from the ureolytic action of bacteria. The objective of the present study is to investigate the bio-stabilization efficiency of ureolytic bacteria isolated from three different locations. The locations are selected so as to have different soil types. The isolated ureolytic bacteria have been injected in both native and non-native soils. The cementation solution (CS) comprises of urea and calcium carbonate in a 1:1 ratio and at concentrations of 0.50M, 0.75M, and 1.00M. The CS is applied daily for 3 days and then the treated soil samples have been tested for Unconfined Compressive Strength (UCS) to determine the improvement in strength post-treatment. The soil specimens treated with different molarity CS have been assessed to determine the calcium carbonate formed in them. The morphology of both untreated and treated soil samples has been analyzed using scanning electron microscopy (SEM). Results indicate that strength improvement is more when injected with bacteria isolated from native soil.

Keywords: Calcium carbonate; Ground improvement; Indigenous bacteria; MICP; Ureolytic bacteria.

A TECHNICAL ANALYSIS OF BIOFUEL PRODUCTION FROM DAIRY WASTE SLUDGE FOR SUSTAINABLE ENVIRONMENT

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Ensuring a clean and secure environment has consistently presented a difficulty for researchers worldwide. The use of fossil fuels has made it more difficult to achieve this goal. In order to accomplish this objective, attention has been drawn to biofuels. Sustainable alternative fuel sources for diesel engines have been the focus of efforts in recent years. The current study investigates the synthesis of biodiesel from stimulated composite sludge, which is a blend of primary as well as secondary sludge. From dried sludge including free fatty acids and triglycerides, the unprocessed lipid was obtained. After that, it was exposed to suitable pre-treatment techniques such as removing of wax and degumming. A standard transesterification method was developed by examining the effects of process variables including preheating temperature, reaction time, methanol-to-oil ratio and catalyst concentration in order to estimate the lowest feasible viscosity and maximum ester recovery. The optimum kinematic viscosity (3.4 cSt) with ester recovery (78.93%) was obtained by preheating dairy waste sludge at a 5:1 molar ratio to 40 degrees celsius and keeping that temperature for 40 minutes in the existence of 1 % KOH. The biofuel was subsequently enabled to settle for 24 hours. These findings were based on the measurements of biofuel recovery and kinematic viscosity. The methyl ester and dairy waste sludge were also measured for their various fuel properties. The methyl ester of dairy waste sludge produced under ideal conditions is a great replacement for fossil fuels, according to the results.

Keywords: Sustainable Environment; Dairy waste sludge oil; ASTM Standard, Biofuel, Blend.

HYDROGEN EMBRITTLEMENT: MECHANISMS, PREVENTIVE COATINGS AND REMEDIAL MEASURES

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Hydrogen embrittlement (HE) is a well-known phenomenon for the manufacturing as well as processing industries to deal with. Hydrogen gets induced in the metal in a high gaseous pressure environment and electrochemical reactions. It causes loss of mechanical properties in material and grows subcritical cracks, hence, resulting in failures and catastrophic fracture. Therefore, to understand the HE phenomena, a summary of the literature on the proposed HE mechanism is presented. Additionally, to discuss the severity of losses and the current status of research, a detailed review of studies reporting structure failure due to HE in steel pipelines, welded joints, structures, wind turbine gear box and hydrogen storage tanks, and power plant components are analytically discussed. Moreover, to envisage the issues and challenges faced by researchers and practitioners dealing with HE, the current status of remedial measures and practices to control embrittlement caused by hydrogen are categorized into three i.e. coatings to prevent the entry of hydrogen in material, microstructural alteration, and traps to render the hydrogen immobile, also discussed in detail. Future scopes are presented which will definitely help the researchers and practitioners to pave the way for more effective preventive as well as remedial measures.

Keywords: Surface; failure; hydrogen damage; HE mechanisms; remedial measures.

LIFI FOR SUSTAINABLE COMMUNICATION ENVIRONMENT

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LiFi (Light Fidelity) technology, a revolutionary communication paradigm, utilizes visible light to transmit data wirelessly, offering a promising avenue for sustainable environmental solutions. This paper explores the intersection of LiFi technology and environmental sustainability, highlighting its potential to address key challenges in energy efficiency and connectivity. LiFi's reliance on light waves, as opposed to traditional radio frequency signals, contributes to reduced electromagnetic pollution and interference, fostering a cleaner and healthier environment. The energy efficiency of LiFi systems, achieved through the use of LED lights for data transmission, aligns with global efforts to minimize carbon footprints. The low power consumption and extended lifespan of LED bulbs contribute to energy conservation, aligning with sustainable design principles. This dual-purpose functionality enhances the economic viability of Li-Fi deployment. Moreover, LiFi's capability to operate in areas with electromagnetic sensitivity, such as hospitals and aircraft, positions it as a green alternative for communication technologies. In the context of sustainable smart cities, LiFi's integration can enhance energy management systems, optimize resource utilization, and enable efficient communication between IoT devices. Its deployment in urban environments can facilitate high-speed wireless communication without contributing to the radio frequency spectrum congestion. This can enhance connectivity in densely populated areas while minimizing the environmental impact associated with traditional wireless communication technologies. This paper sheds light on the transformative potential of LiFi in mitigating environmental impact while advancing connectivity in diverse applications, and how Li-Fi as a key player in the quest for greener and more sustainable technological solutions.

LIFI OVER WIFI AS A VIABLE AND HEALTH COGNIZANT INDOOR COMMUNICATION TECHNOLOGY

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This research paper explores the feasibility and health-conscious aspects of adopting Light Fidelity (LiFi) over traditional Radio Frequency (RF) technologies, namely Wireless Fidelity (WiFi) and Fifth Generation (5G), for indoor communication. A thorough literature review examines existing guidelines and discern the diverse effects associated along with an analysis that considers the potential risks associated with prolonged exposure to different frequency ranges. In addition, the paper presents the outcomes of a survey designed to gauge public awareness and perceptions regarding WiFi and 5G radiation. The survey having 416 respondents, including academicians and students, investigates factors such as familiarity, concerns, and preferences among users, providing valuable insights into the evolving landscape of wireless communication technologies. The integration of survey findings with the literature review enhances the comprehensiveness of the study, shedding light on the potential societal acceptance and adoption of LiFi technology. This research contributes to the ongoing discourse on health-conscious communication technologies and advocates for the exploration of LiFi as a viable indoor communication solution.

G-C₃N₄ LOADED BIMETALLIC BISMUTH OXIDE NANOCOMPOSITE FOR PHOTOCATALYTIC DEGRADATION OF TOXIC POLLUTANTS.

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Toxic materials are continuously released into the surrounding ecosystem, causing pollution, siltation, acidity, and the suppression of aquatic life. As a result, water contains various hazardous materials that are bad for human and environmental health. Therefore, removing the pollutants from the wastewater is essential. Nowadays, photocatalytic degradation is one of the most efficient methods to remove toxins from wastewater. Amongst the various photocatalysts binary heterojunction-like composite materials are attracting a lot of interest due to their narrow band gap, high surface area, high visible light response and facile electron transfer properties. Herein we report various g-C₃N₄ loaded BiVO₄ or Bi₂WO₆ binary composites using simple wet impregnation method. These composites act as efficient photocatalysts towards the different toxic pollutants such as rhodamine B or Reactive orange 19. Many characterisation techniques can prove the synthesised material like XRD, FESEM, XPS, HRTEM, DRS and BET analysis. All these catalysts have superior catalytic activity, high stability and excellent recyclability. As a result, these binary composites can be considered a promising catalyst for the photocatalytic removal of pollutants from wastewater.

Keywords: Binary composites; heterojunction structures; wet impregnation; wastewater treatment; emerging contaminants.

PERFORMANCE ANALYSIS OF HYBRID SOLAR DESALINATION SYSTEM WITH THERMAL STORAGE MATERIALS AT DIFFERENT WATER DEPTH

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Population of India has been reached 144 crore in 2023, out of which rural population is 65% of total population. Being the largest populated country in the world, it is very challengeable to provide potable water for the rural area of the country. Contaminated water can cause a lot of gastro intestine illness across the country and more than 60 billion cases have been reported annually due to consumption of contaminated water. Due to high treatment cost through chemical process, we are unable to provide pure drinking water to everyone. Solar energy can be harvested to provide safe and potable water to rural population of India. Solar desalination is the least expensive technology for producing potable water as it depends on solar energy which is available free of cost and certainly a good gift to human beings. Current study focus on experimental investigation of modified passive type solar desalination system by using low cost river pebbles as thermal storage materials (TSM). The Solar desalination unit has basin area of 3600 cm², thickness of glass reflector is 5.0 mm tilted at 20° with the horizontal surface. By using pebbles, an improvement in the yield has been observed in the range of 11% to 37%, depending on the charging wastewater. Analysis shows that the water quality parameters are under permissible limit as per drinking water standard mentioned in Bureau of Indian Standards. This paper presents a low cost technology used for the production of drinking water.

Keywords: Solar desalination; Solar Energy; Potable water; TSM; Performance.

EXPERIMENTAL STUDIES ON SUSTAINABLE CONCRETE USING OIL PALM INDUSTRIAL WASTE.

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Diving into the world of concrete, Normal concrete is widely used in construction, does pose some environmental challenges. Firstly, cement production is a major contributor to carbon dioxide emissions. This adds a significant carbon footprint to your conventional concrete. Incorporating Oil Palm Shells (OPS) and Palm Oil Fuel Ash (POFA) into concrete offers a sustainable alternative. These materials are waste products of the palm oil industry, utilizing waste resources rather than contributing to additional environmental strain. Based on experimental behavior of Agro- industrial waste into sustainable light weight concrete by oil palm shells (OPS) and palm oil fuel ash (POFA) are partial replacement of coarse aggregate and cement in concrete with 7, 28, and 56 days of curing. The mechanical, durability properties and Flexural behavior of beams are carried out. The test results are compared with conventional concrete. The results obtained and observations made clearly demonstrate that the replace of cement and aggregates by Oil Palm waste is advantageous, partially for mass concrete.

Keywords: Flexural Behavior, Mechanical and Durability properties, Oil Palm Shells, Palm Oil Fuel ash, Sustainable light weight concrete.

EXPERIMENTAL INVESTIGATION OF SUSTAINABLE LIGHT WEIGHT CONCRETE MADE WITH OIL PALM INDUSTRIAL WASTE

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In present days a very huge amount of raw materials are used by the construction sector which releases very large amount of harmful gases such as carbon dioxide (CO₂) into the environment. To avoid such problems research should be much focused on sustainable materials, among which Oil Palm industrial waste that can be used in concrete. The main purpose of this research paper is to utilize the palm oil fuel ash (POFA) as a replacement of cement and replacing fine aggregate with Oil Palm Shells (OPS) with different proportions. The test results for workability of concrete, Mechanical and durability properties are carried to produce sustainable light weight concrete for 7, 28 & 56 days of curing period. The results obtained and observations made clearly demonstrate that the replace of cement and fine aggregates by Oil Palm waste is advantageous, partially for mass concrete.

Keywords: Mechanical and durability properties, Oil Palm Shells, Palm Oil Fuel Ash, Sustainable Light Weight Concrete.

ADVANCING SUSTAINABLE SOLUTIONS: HARNESSING POLYANILINE/BIOCL/GO TERNARY NANOCOMPOSITES FOR SOLAR-POWERED DEGRADATION OF ORGANIC POLLUTANT AND PHOTOCATALYTIC HYDROGEN GENERATION

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In this study, we have synthesized a ternary nanocomposite (PANI/BiOCl/GO) through an oxidative polymerization approach. Different weight percentages of composites (GO/BiOCl: 0.5%, 1%, 2.5%, and PANI: 99.5%, 99%, 97.5%) were prepared as 0.5PBG, 1PBG, and 2.5PBG, respectively. UV-Vis-DRS, PL, XRD, XPS, FESEM, HRTEM, BET, and EDS analysis were carried out to analyze the structural and morphological characteristics of synthesized nanocomposite. The ternary nanocomposite was finally tested for its ability to degrade an organic dye, rhodamine-b (Rh_b), and in the process also generate solar-light-driven green hydrogen by water half-splitting. The composite achieved about 90% detoxification (assigned from GC-TCD by analyzing the evolved CO₂ gas after degradation) and 96% color removal of Rh_b dye within 120 minutes. The degradation of Rh_b by 1PBG displayed a first-order reaction, featuring a rate constant 7.25 times higher than that observed for pure PANI, 3.8 times higher than for pure GO, and 3.9 times higher than for pure BiOCl. Thus, the ternary composites achieve a good amount of synergy. Significantly, this reaction rate constant is 4.7 times greater than the rate observed with commercially used TiO₂-P25 photocatalyst. Various reaction parameters including solution pH, different illumination areas, catalyst dosage, and the study of scavengers were investigated to understand their effects on the degradation reaction. The photocatalyst's reusability effectiveness was evaluated over 6 cycles, and its stability was subsequently confirmed through XRD and ICP-OES analysis. The LC-MS study revealed the identification of various intermediates and end products following the degradation reaction. The nanocomposite also produced 1000 ppm of hydrogen gas with an apparent quantum efficiency (AQE) of 17.97% when CH₃OH was used as a sacrificial agent, 500 ppm (AQE of 9.69%) in the acidic conditions, and 600 ppm (AQE of 11.63%) in the basic conditions.

Keywords: Organic dye degradation; Polyaniline/Bismuth oxychloride/Graphene oxide; Photocatalytic hydrogen evolution; photocatalysis; Ternary nanocomposite.

ELECTRO-OXIDATION TREATMENT OF ACTUAL HUMAN URINE USING HIGHLY STABLE MMO ANODE IN A CONTINUOUS RECIRCULATING REACTOR

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The present work demonstrates the removal of micropollutants in actual human urine (AHU). One of the leading causes of water contamination is human urine. Urea, ammonia, creatinine, and other organic and inorganic substances constitute its constitution. Consequently, this untreated Urine is harmful to the environment. Therefore, it is essential to treat such pollutants present in the water. A low-cost novel mixed metal oxide (MMO) made of Titanium/Iridium oxide/Ruthenium oxide/Platinum was used to treat the pollutants electrochemically. Significant Findings: The experiments were conducted under optimal conditions using a photo voltaic-driven continuous recirculating reactor showing 69.09% urea, 67.95% creatinine, and 95.180% uric acid. Furthermore, the evidence demonstrates that the oxidation of organic components in AHU was caused by reactive chlorine species (RCS) and reactive oxygen species (ROS) created during the treatment process. After the analysis, MMO electrodes are efficient for the actual urine effluent treatment and could be expanded for industrial applications.

Keywords: Electrocatalytic treatment; Photovoltaic cell; Recirculating reactor; Response surface methodology; Urine metabolites.

POTENTIAL UTILIZATION OF BAMBUSA BALCOA IN CEMENT MORTAR AS A REINFORCING FILLER TO IMPROVE FLEXURAL AND COMPRESSIVE STRENGTH

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In present work, the bambusa balcoa species of bamboo as reinforcement bamboo fiber for the partial replacement of cement in mortar structure was analyzed. The bamboo fiber was treated with 10% hot alkaline solution. The bamboo fiber was varied at 0.25%, 0.5%, 1%, 1.5%, 2%, 2.5%, 3%, 5% and 10 wt% respectively, in cementitious mortar composites. The bamboo fiber reinforced cementitious mortar composites were evaluated by the means of compressive, flexural and water absorption test. The compressive strength of the composite increased up to 1wt% bamboo fiber and the highest value was achieved at 0.5wt% bamboo fiber which is increased by 13.51% while the compressive strength was found to decrease beyond 1wt% bamboo fiber content comparatively to control sample. Similar trends were observed for flexural strength of the bamboo fiber reinforced cement mortar composites and the value increased by 7% at 0.5 wt% bamboo fiber as compared to the control sample. The water absorption capacity of the composites was increased with increase in bamboo fiber content and no of days.

Keywords: bambusa balcoa, cement, flexural and compressive strength.

SUSTAINABLE CLOTHING: ANALYZING THE NEED AND VIABILITY OF THIS GREENER ALTERNATIVE

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Environmental Cost due to Textile Industry is rising with ever growing demand in apparel sector. The textile industry, ranked as the sixth most environmentally damaging globally, is facing heightened scrutiny due to its expanding role in the fashion sector. The surge in consumerism, coupled with the rapid growth of the apparel industry, has led to the entry of both domestic and international clothing brands in India. This surge is notably driven by the prevalence of fast fashion trends and the increasing purchasing power of individual households. Fast fashion operates on large-scale production of inexpensive garments designed for short-term use, contributing to significant environmental repercussions. The consequences manifest in various forms, including substantial environmental impact, escalating waste generation from discarded clothing, discharge of harmful dyes, excessive water resource exploitation, and a growing carbon footprint. This paper delves into the environmental impact attributable to the textile industry, exploring sustainability practices within the textile and fashion sectors. Emphasis is placed on utilizing renewable feedstocks as raw materials and leveraging cutting-edge technology for proactive production planning. Additionally, the study compares per capita textile waste across different countries, shedding light on the factors contributing to India's comparatively lower per capita textile waste generation. Furthermore, the article presents case studies highlighting emerging eco-friendly fabric alternatives such as hemp, bamboo, and recycled fabrics. The conclusion suggests that sustainable fashion not only mitigates adverse environmental effects but also paves the way for developmental and innovative opportunities through the exploration of new products and designs.

Keywords—Sustainable, textile, waste generation, renewable feedstock, circular economy, carbon footprints.

PREPARATION AND CHARACTERIZATION OF LOW COST CERAMIC MEMBRANE USING DIFFERENT PORE FORMERS

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Rapid industrial expansion has resulted in the generation of substantial quantities of wastewater. Industries such as textile, petrochemical, dairy, and tannery produce wastewater containing elevated levels of oils, fats, and heavy metals. Consequently, the need to treat industrial effluents before their discharge into water bodies has become crucial to preventing water contamination. Ceramic membranes are becoming more and more popular these days due to their superior mechanical, chemical, and thermal qualities over polymeric membranes. In this work, low-cost raw materials are used for the preparation of membranes that are obtained from industrial and natural waste. The uniaxial pressing method has been used to create membranes utilising fly ash, clay, and pore formers (sawdust, calcium carbonate, and sucrose) as raw materials. The membrane from these composite mixtures (fly ash wt. 50%, clay wt. 45%, and remaining wt. 5% pore former) has to be sintered at different temperatures. Membranes shaped like circular discs were employed in characterization studies involving XRD and SEM analyses. The purpose of these studies was to evaluate how membrane structure, porosity, and mechanical integrity were affected by sintering temperature and pore formers. The results contribute insights into how distinct pore formers influence the morphology, crystal structure, and separation efficiency of the membrane. This research enhances the development of ceramic membrane technology by introducing a cost-effective fabrication method with customizable properties. It plays a vital role in the separation process, providing a sustainable and cheap solution for membrane production.

Keywords: low cost, ceramic membrane, membrane fabrication, microfiltration.

SYNTHESIS AND CHARACTERIZATION of BIOMASS DERIVED ACTIVATED CARBON-BASED SULFUR CATHODE FOR LITHIUM SULFUR BATTERIES

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Lithium sulfur (LiS) battery research has been quite active due to the growing need for sustainable and effective energy storage options. Given this, activated carbon generated from biomass appears to be a potential material for improving LiS battery performance and environmental sustainability. This work describes a new sulfur cathode that was made as a precursor utilizing activated carbon obtained from biomass. This study offers a novel method for producing activated carbon by using banana leaves and sawdust from teak wood as sustainable raw materials. A precise carbonization and activation procedure produced a porous, high-surface-area structure in the carefully created activated carbon. Sulfur was successfully integrated into the activated carbon matrix, as demonstrated by detailed characterization using SEM, TEM, BET, and XRD. This highlights the material's potential for improved conductivity and electrochemical stability. The cathode was meticulously made using a precisely balanced composition of active materials (sulfur and activated carbon), conductive material (acetylene carbon black), and binder, in a methodically optimized ratio of 8:1:1, respectively, guaranteeing a well-organized and effective electrode design. The activated carbon to sulfur ratio ranged from 10% to 90%. The cathode with 70% sulfur emerged as the ideal composition, producing the most advantageous and promising outcomes.

Keywords: Activated carbon; biomass; cathode; energy Storage; LiS batteries.

ANALYSIS OF FLOW CHARACTERISTICS OF SAND WATER SLURRY THROUGH HORIZONTAL PIPE

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The slurry transportation system is widely accepted as it reduces the environmental pollution, ecological disturbance and saves energy consumption. River water from upstream of reservoir is usually collected and conveyed through penstocks to turbine for hydroelectric power generation. Sediments are usually presents in river water and the flow dynamics of sediment laden water is different from flow characteristics of only water. The efficient movement of required quantity of water through a pipeline having various transitions requires proper designing of slurry transportation system. The present study used the pilot plant test loop to investigate flow dynamics of sand water slurry having different particle sizes and solid concentrations during the flow through a horizontal pipeline. The experiments were performed with different particle sizes (<53 and 53-106), for solid concentration ranging from 10 to 40% by mass and influx velocity ranging from 1-4 m/s. The pressure loss during the flow is observed to be a function of slurry concentration, particle size present in slurry and influx velocity. The pressure drop during the flow of sand water slurry in a horizontal pipe flow was also calculated through Wasp model and it is observed that the trend obtained for both the methods are similar, but the pressure drop obtained through Wasp model is over predicting in comparison to pressure drop obtained through experimentally for all conditions.

The flow of sand water slurry having different particle sizes and solid concentrations is also investigated numerically with or without additives and it is observed that results obtained by using SST k-omega turbulence model agrees with experimental results. It is observed that the increase in volume fraction due to increase in concentration of sand water slurry from 10-40% during the flow through a horizontal pipeline lies in a range of 15-367%.

Key Words: Additives; Pressure drop; Slurry transportation system; SST k-omega turbulence model; Wasp model.

LOW COST BIO-ADSORBENT FOR CR(VI) CONTAMINATED EFFLUENT DISCHARGE FROM LEATHER INDUSTRY

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Water pollution, which has become a severe worry due to the rapid growth in industrialisation, urbanisation, and population over the past several decades. Among various contaminants discharged from industries heavy metals are the principal pollutants in wastewater among these contaminants since they can annoy living things and persist in the environment. There are several techniques for eliminating these heavy metals from aquatic sources including membrane filtration, electrocoagulation, chemical precipitation, electrolysis, coagulation, ion exchange and adsorption. Compared to other available technologies, adsorption is a straightforward, economical, environmentally benign, and sustainable method of treating wastewater. Accordingly, this article highlights the effective removal of hexavalent chromium (Cr VI) one of the major contributors causing water pollution discharged from tanneries was removed by using a low cost synthesized biowaste adsorbent derived from Watermelon leaf (WML). High removal efficiency for heavy metals (HMs) was found in the results, particularly when employing acid treated WML. The WML adsorbents achieved more than 90% removal percentage in synthetic wastewater especially Cr(VI) at pH 2 ± 0.5 with a chromium metal content of 15 mg/L when compare to pristine WML.

Keywords: Adsorption; Contamination; Environment; Sustainable; Wastewater.

DEVELOPMENT OF EPOXY/FLYASH COMPOSITES CONTAINING HALLOYSITE NANOTUBES: THERMAL STABILITY ANALYSIS AND DEGRADATION KINETICS

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This research article investigates the development of a composite material by inclusion of halloysite nanotubes (HNTs) into an epoxy/flyash matrix using a solution blending technique. The HNTs were added at different loadings and were effectively dispersed throughout the epoxy/flyash matrix. The thermal properties and degradation kinetics of developed epoxy/flyash composites were studied using a thermo-gravimetric analyzer (TGA) at varied heating rates (10, 15, 20, and 25 °C/min). The thermal stability increased and the degradation process was slow with increasing heating rates. The thermal degradation activation energy (E) was calculated using the model-free Kissinger Akahir-Sunose (KAS) and Flynn-Wall-Ozawa (FWO), and it was revealed that epoxy/flyash composites containing 3wt% HNTs have the highest E, 77.52 and 79.89 kJ/mol, respectively.

ALKALINE EXTRACTION OF PHENOLICS FROM RICE STRAW AND THEIR CHARACTERIZATION

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The optimization of alkaline hydrolysis of rice straw was done for the extraction of phenolic compounds from rice straw using Response Surface Methodology via Central Composite Design (CCD). Phenolic content (PC) of 14.58 mg g⁻¹ of rice straw (RS) was obtained by treating dried powdered biomass of rice straw at 3 M NaOH concentration, 30 min time and L: S ratio of 26.62: 1 mL g⁻¹ at 121°C. The extract obtained under optimal conditions was analyzed using Gas chromatography mass spectrometry (GC-MS), ¹H-Nuclear magnetic resonance spectroscopy (¹H-NMR), and Fourier transform infrared spectroscopy (FT-IR). Fourteen compounds were identified by GC-MS which were classified into G (guaiacyl), S (syringyl), and H (p- hydroxyphenyl) type of phenolics and furan derivatives. A significant GC-MS area percentage for Ethanone, -1, (4-hydroxy-3, 5, -dimethoxy phenyl) - (32.44%), 2- methoxy- 4 - vinyl phenol (23.71%), and 2, 3- Dihydrobenzo furan (31.72%) were observed which were mainly formed due to cleavage of ether (β -O-4) bonds in lignin. The extract exhibited considerable levels of 2, 2'-azino-bis (3-ethylbenzothiazoline-6-sulfonic acid) (ABTS) (IC₅₀ (half-maximal inhibitory concentration) = 0.45 mg mL⁻¹) radical scavenging activity, which was comparable to ascorbic and gallic acids. Rice straw biomass, a source of nearly 22% lignin, can be a good feedstock for biorefineries to produce compounds such as phenolics for application in food and pharmaceutical industry. This study offers an economically viable approach for obtaining phenolic compounds under milder conditions, in contrast to harsh methods employed in other studies.

Keywords: Rice straw; Lignin; Depolymerization; Phenolics; Antioxidant.

OPTIMIZING P-NITROPHENOL (PNP) DEGRADATION WITH IRON-DOPED BISMUTH AND TUNGSTEN PHOTOCATALYST: EXPLORING WATER MATRIX EFFECTS FOR ENHANCED PERFORMANCE

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P-nitrophenol (PNP) is a well-known and pervasive contaminant of the nitroaromatic family that is utilized as an intermediary in the production of explosives, fungicides, insecticides, therapeutic drugs, and dyes. It is known to impair the functioning of reproductive system and endocrine system in humans. Therefore, it is imperative to effectively remove the PNP from the water. Advanced oxidation processes (AOPs) offer a promising solution for the degradation of PNP. In the last two decades, attention has been paid to the heterogeneous catalytic technology for eliminating recalcitrant chemicals. This study investigates the degradation of PNP using iron doped bismuth and tungsten-based photocatalyst. A modified hydrothermal technique was used to synthesize the heterojunction Bi₂WO₆ photocatalyst with varying amount of iron. The synthesized photocatalysts were characterized for structural and spectral characteristics using various techniques. Within a specially constructed photoreactor chamber, the PNP was degraded. HPLC was used to quantify the degradation of PNP. A syringe filter was used to extract a 100 µL sample from the reactor at predetermined intervals into an HPLC vial. One millilitre of HPLC-grade methanol was then added as a quencher. Well-crystalline structure of the photocatalyst was observed through XRD. UV-DRS analysis showed that the synthesized photocatalyst was activated in the UV-C spectral range. PNP degradation under UV irradiation alone was observed to be ~ 8 %. With the addition of Fe₂O₃/Bi₂WO₆ photocatalyst under UV irradiation, the degradation was improved to ~ 45%. Adding hydroxyl radical generating, H₂O₂ to this reaction mixture significantly enhanced the PNP degradation to more than 90%. The various components of the water matrix had different effects on the degradation of PNP; some components improved the degradation, while others reduced the degradation of PNP.

Keywords: Bismuth; tungsten based photocatalyst; p-nitrophenol (PNP); degradation; radicals.

UTILIZATION OF NON-CONVENTIONAL AND WASTE MATERIALS IN PRODUCTION OF CONCRETE

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A huge amount of waste is produced globally each year and a very limited amount gets recycled. Recycling of waste material consumes a lot of energy and results in environmental pollution. Also, improper disposal of non-recycled waste further harms the environment. Utilizing waste and non-conventional materials in concrete production can fulfil the objectives of eliminating waste, producing sustainable construction materials, and preventing environmental pollution. The paper is divided into two broad categories: replacement of natural aggregates and cement with non-conventional waste materials. Under each category impact on the compressive strength of concrete produced after different types of waste replacement is summarised to comprehend the effectiveness of the replacement. The increase in compressive strength of concrete was observed when natural coarse aggregates replaced with tiles and sanitary ceramics, brick aggregates, waste tyres and rubber, metal slag and recycled coarse aggregates up to 25%, 50%, 10%, 30%, and 30% respectively. Whereas, a decline in strength was observed when coarse aggregates were replaced with glass, polyethylene terephthalate, and agricultural waste up to 30%, 20%, and 10% respectively. On the other hand, the compressive strength of concrete increased upon partial replacement of cement with silica fume, rice husk ash, fly ash, and glass up to 15%, 15%, 20%, and 10% respectively and it decreased when the cement is partially replaced with ceramic waste up to 20%. Overall, studies indicate that the replacement of aggregates and cement with waste materials is effective and suitable for plain cement concrete. The study highlights the contribution and change in properties that waste materials bring into the concrete after replacement. Further, the collation of such vast amount of information may provide valuable insight for further research.

Keywords: Fly Ash, Concrete, Waste Materials, Sustainable Construction Material, Pavement Quality Concrete.

HETEROGENEOUS $\text{Mn}_3\text{O}_4/\text{CuO}$ INTERFACE FOR BOOSTED OXYGEN EVOLUTION REACTION: A COMPREHENSIVE STUDY

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A lot of effort has gone into finding noble metal-free electrocatalysts for the oxygen evolution reaction that are inexpensive, easy to produce, and highly efficient. In this scenario, a highly crystalline $\text{Mn}_3\text{O}_4/\text{CuO}$ nanocomposite electrocatalyst was successfully prepared via in-situ hydrothermal synthesis route. The phase formation was confirmed with X-ray diffraction (XRD) analysis. The formation of the nanocomposite was also confirmed with Raman spectra and Fourier Transform Infrared Spectra (FTIR). The morphological features of the prepared samples were examined through Field Emission Scanning Electron Microscopy (FESEM). The oxygen evolution reaction performance of the $\text{Mn}_3\text{O}_4/\text{CuO}$ nanocomposite was studied using electrochemical workstation. The working electrode was prepared by depositing $\text{Mn}_3\text{O}_4/\text{CuO}$ nanocomposite on the Ni mesh for electrochemical studies. To evaluate the electrodes performance, cyclic voltammetry (CV), linear sweep voltammetry (LSV), and electrochemical impedance spectroscopy (EIS) were conducted for all the prepared samples in alkaline medium. The prepared composites showed higher stability, as confirmed from stable CV cycles. The value of the charge transfer resistance (R_{ct}) for oxygen evolution reaction was also determined for the prepared composite. This study has shown that $\text{Mn}_3\text{O}_4/\text{CuO}$ nanocomposite has the ability to substitute noble metals in oxygen evolution reactions.

Keywords: energy conversion and storage; nanocomposites; electrocatalyst; oxygen evolution reaction; impedance spectroscopy.

BEYOND GRAPHENE: NOVEL PERSPECTIVES ON 2D MBENE STRUCTURES

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Boron being light element can create multi-centred bonds to compensate for its electron deficiency, which makes it highly reactive with metal ions. Thus, the presence of boron is favourable for the development of complex multi-dimensional boride structures. Such structures are developed when there is a bond between boron and other less electronegative elements. Recently, MBenes have emerged as a new class of 2D materials based on boron. 2D MBenes are prepared by etching metal layer from their 3D counterpart, i.e., MAB phases. Herein, we report a synthesis approach for the formation of MoAlB MAB phase. The samples were prepared through conventional heat treatment in the presence of Argon gas. The synthesis temperature, composition and holding time is varied to obtain MAB phase. The phase formation was determined by using X-ray diffraction (XRD) technique. The morphological features of the prepared sample were examined through Field Emission Scanning Electron Microscope (FESEM). The prepared samples could be used as the precursor to prepare 2D MBenes.

Keywords: energy conversion and storage; Boron; 2D materials; MAB phases; MBenes.

FABRICATION AND SYNTHESIS OF BIODEGRADABLE FILMS FROM CELLULOSE WASTE TO BE USED AS AN ALTERNATIVE FOR PLASTIC

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Renewable and biodegradable materials that are derived from cellulose waste are considered a better alternative for plastic materials. The materials used in this paper are rice husk, sawdust, and sugarcane bagasse. Two methods are used to make the film: first, carboxymethyl cellulose (CMC) is synthesized from the biomass mentioned above using mercerization and etherification. Next, CMC is added to starch for crystalline structure, and glycerol and citric acid for mechanical properties and flexibility, to create a biodegradable film. The second approach involves breaking down the biomass using deep eutectic solvents (DES) to create a homogenous cellulose-lignin slurry. Gelatine is then added to the resulting material, and casting is carried out to create the biodegradable film. XRD and FTIR have been used to analyze the produced film. Physical characterization tests, including those for strength, flexibility, water absorption, and deterioration, have also been conducted. Compared to the CMCs obtained from the rice husk and sawdust the CMC obtained from the sugarcane bagasse is more suitable for film production since it possesses flexibility, degradability, and mechanical properties.

Keywords: Biodegradable film; Carboxymethyl cellulose; Cellulose-lignin; Deep eutectic solvents; Glycerol.

DEGRADATION OF CONGO RED DYE BY USING SYNTHESIZED PHOTOCATALYSTS OF CdO-SnO₂ NANOCOMPOSITES IN THE PRESENCE OF VISIBLE LIGHT

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The precipitation method was used to create CdO, SnO₂ and CdO-SnO₂ photocatalysts, which were then calcined at 400 °C for an hour and crushed to create nanoparticles. Then, the synthesized CdO, SnO₂ and CdO-SnO₂ nanocomposites was characterized by XRD, SEM, XPS, FTIR, PL and EDAX. Based on the findings, the XRD pattern showed that SnO₂ has a tetragonal structure, while CdO has a cubic structure. In addition, hydroxyl groups and peaks present in the Cd-O and Sn-O bonds may be verified by Fourier transform infrared spectra. Furthermore, the Congo red (CR) disappears, after a 90-minute irradiation period was used to assess the photocatalytic activity of CdO, SnO₂, and CdO-SnO₂ nanocomposites. According to the process, the main reactive species involved in the degradation of CR dye are (OH) and (O₂•(-)) radicals. CR dye confirms the validity of the measured rate constant (*k*) and follows the pseudo-first-order kinetic model. After a 90-minute irradiation period, the CdO-SnO₂ nanocomposite eventually significantly reduced the color of the dye solutions, indicating its higher photocatalytic activity compared to the pure CdO and SnO₂ catalysts. Because of the efficient charge carrier separation and decreased electron-hole pair recombination rate, the composite may have a higher photodegradation efficiency. Moreover, azo dye was broken down by the composite more quickly than by the fluorescent cationic dye.

Keywords: Nanocomposites, CdO-SnO₂, Photocatalytic activity, Degradation, Congo Red.

SUSTAINABLE LEAD REMOVAL FROM PRODUCES WATER USING CASSAVA STEM BIOCHAR: AN ECO-FRIENDLY APPROACH

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Produced water is the one which comes out of oil and gas wells during the crude oil production. The discharge of produced water poses a significant environmental challenge due to the presence of heavy metals, including lead, which can have detrimental effects on ecosystems and human health. In this study, we explore an eco-friendly and cost-effective approach for lead removal from produced water by adsorption using cassava stem biochar. Cassava stem biochar, derived from agricultural waste, serves as a promising adsorbent due to its porous structure and rich carbon content. The biochar was prepared by pyrolysis process, optimizing temperature and residence time to enhance its adsorption capacity. Batch experiments were conducted to investigate the influence of various parameters like pH, contact time, initial lead concentration, adsorbent dosage, mixing speed, temperature and particle size on the removal efficiency. Our findings demonstrate that cassava stem biochar exhibits a remarkable affinity for lead ions, with good removal efficiency. The adsorption process was found to follow pseudo-second-order kinetics, suggesting a chemisorption mechanism. Equilibrium data were well-fitted by the Langmuir isotherm model, indicating monolayer adsorption with a maximum adsorption capacity of 12.180 mg/g. The thermodynamic parameters, including changes in enthalpy (ΔH), and entropy (ΔS), were evaluated to gain insights into the spontaneity and feasibility of the adsorption process. The negative ΔH values indicated that the adsorption of Pb onto CSB was non-spontaneous and exothermic in nature. The negative ΔS value suggested the adsorption is feasible. Furthermore, the study evaluates the reusability and regeneration potential of the cassava stem biochar, ensuring a sustainable and economically viable solution for lead removal. This research contributes to the growing body of knowledge on green technologies for water remediation and emphasizes the potential of cassava stem biochar as a sustainable tool to reduce impact of heavy metals in produced water.

Keywords: Adsorption; Isotherm; Kinetics; Pyrolysis; Thermodynamics.

APPROACHES FOR THE SYNTHESIS OF SILICON NANOPARTICLES: A SURVEY

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Silicon nanoparticles, often called silicon quantum dots, play an important role in nanotechnology and have gained researchers' attention because of their potential in electronics, photocatalysts, and bioimaging. Studies on silicon quantum dots began after the 1990s when Canham discovered the emission of visible light from silicon because of the porous nature of silicon. The size of these quantum dots provides silicon with some unique properties. Bulk silicon is inadequate as an active metal but after quantization, they have a wide scientific and technological application in optoelectronics and electronics. The increasing demand for eco-friendly and cost-effective methods in silicon nanomaterial synthesis has spurred interest in exploring alternative sources, and agro waste stands out as a promising candidate due to its abundance and renewability. This review paper aims to provide all the information related to the synthesis, structural properties, and applications of Si-NP's obtained from agricultural waste. Agro-waste is often considered a burden on the environment but is now a valuable resource for sustainable nanomaterial synthesis. Furthermore, we will discuss their behavioral change by varying the source of origin, size of particles, and surface passivation.

Keywords: Silicon Nanoparticles.

EMERGING TRENDS IN THE DESIGN OF NON-BIOACCUMULABLE FLUORINATED SURFACTANTS AND ITS APPLICATION

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Fluorinated surfactants have emerged as a super surfactant due to their exceptional properties and plethora of applications. Fluorinated materials play an essential role in modern industry and technology, and their importance is expected to grow even more in the near future. Fluorinated surfactants comprises of two structural components, a hydrophobic and oleophobic perfluorinated carbon chain and a hydrophilic component. These properties make them a material of choice for several applications including but not limited to fire-fighting, oil-field, foaming, coating, and paints. Despite these attractive properties, environmental concerns like persistence and toxicity associated with legacy fluorosurfactants are major hurdles to be addressed. There have been many attempts to find suitable replacements for conventional long chain fluorinated surfactants ($>C_6$). Various strategies have been designed by researchers in order to develop non-bioaccumulable fluorosurfactants which focus on shortening of fluorinated alkyl chain, incorporating heteroatom and introducing branching in the hydrophobic tail. Our work extensively covers the various methodologies for the synthesis of potentially non-bioaccumulable fluorosurfactants that show potential for improved degradation.

Keywords: Bio-degradable; Fire-fighting Foam; Fluoro-surfactants; Synthesis; Toxic.

SUSTAINABLE TRENDS IN THE DEVELOPMENT OF FIREFIGHTING FOAM

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Firefighting foams are aggregates of small bubbles which efficiently combat class B fires by forming a vapor suppressing blanket over the fuel surface. The fire-fighting efficacy of the aqueous film forming foams (AFFF) can be attributed to the presence of perfluorinated molecules as the major ingredient. Following the regulations imposed on the use of long chain fluorinated compounds, researchers have been actively engaged in developing safer and more sustainable alternatives to legacy firefighting foams. However, researchers face a major challenge in developing advanced and better alternatives as composition of these film forming foams remains a trade secret of the industries. Our work aims to investigate the components of the commercially available aqueous film forming foams. Additionally, the release of firefighting foams into the soil and water bodies arises as a matter of environmental concern. Keeping this in view, fluorine free foams have emerged as better alternatives to the traditional AFFFs in terms of environmental persistence. Therefore, researchers have been trying to develop fluorine free foams that do not contain toxic perfluorinated compounds. However, meeting the exceptional performances of these compounds remains largely an unaccomplished challenge. We have also made an attempt to discuss the relevance of AFFF in the era of emerging fluorine free foam.

Keywords: AFFF; environmental persistence; firefighting; fluorosurfactants; fluorine free.

REVEALING URBAN ATMOSPHERE DYNAMICS: UNDERSTANDING NEW PARTICLE FORMATION IN THE KANPUR REGION THROUGH ANALYSIS OF FRESH EMISSIONS AND LONG-RANGE TRANSPORT

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The production and expansion of atmospheric aerosol particles play a crucial role in air quality and climate dynamics. Despite the global significance of this phenomenon, only a limited number of studies have explored the occurrence and characteristics of new particle formation (NPF) events in the Indo-Gangetic Plain (IGP) region, setting it apart from other regions worldwide. In this paper, we present the outcomes of an extensive investigation into NPF utilizing scanning mobility particle sizer (SMPS) measurements conducted at the urban background of Kanpur in the IGP Plain (26.46 N, 80.32 E), during the months of October and November in 2021. Our study period encompassed 30 days, during which we observed a total of 10 distinct NPF events. To categorize these events systematically, we identified three types: Type 1, Type 2, and events that could not be clearly defined (Undefined, U). Days characterized by the absence of particle formation or size increase were classified as non-event days (NE). Our analysis revealed a correlation between high concentrations of sulphur dioxide (SO₂) and global radiation with the frequent occurrence of NPF events. Interestingly, elevated concentrations of fine particulate matter (PM_{2.5}) were associated with a lower probability of NPF events. To gain deeper insights into the spatial distribution and origins of NPF events in the IGP Plain, we employed particle number size distribution (PNSD) data and air mass return trajectories. The Nano Map approach allowed us to examine the geographical extent and source locations of NPF events. This comprehensive investigation sheds light on the unique characteristics of NPF in the Kanpur urban atmosphere within the IGP Plain, contributing valuable information to the broader understanding of atmospheric aerosol dynamics.

Keywords: New Particle Formation, Particle Number Concentrations, Particle Size Distribution, Formation rate, Nano Map.

SUSTAINABLE INTEGRATION OF RICE STUBBLE BIOCHAR IN CEMENTITIOUS COMPOSITES: A NOVEL APPROACH TO MITIGATE STUBBLE BURNING IMPACT

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Stubble burning, a widespread agricultural practice in India, has significant environmental repercussions. This conference paper explores an innovative solution by converting stubble waste into biochar and investigates its sustainable integration into construction materials. This study examines the impact of using Rice Stubble (RS) biochar on the properties of cementitious composites. The research employs an extensive testing regimen, including compressive strength, water absorption, drying shrinkage, workability assessment, and SEM/XRD analyses. Furthermore, cost analysis and environmental impact assessment are carried out to assess the overall sustainability of RS biochar. This study's findings highlight RS biochar's positive impact on compressive strength, demonstrating significant improvements across all testing ages, with a maximum increment of 18.7% at 7 days. However, no RS biochar-containing mix showed higher flexural strength than the control mix. Mortars containing RS biochar exhibited higher water absorption than the control mix, suggesting altered moisture-related behavior. Also, incorporating RS biochar into mortar formulations leads to a decrease in workability. SEM and XRD analyses confirm the compatibility of RS biochar with cementitious composites, indicating no substantial disruption to the binding material's structure. This study provides valuable insights into the sustainable utilization of stubble waste biochar, emphasizing the positive impact of RS biochar on cementitious composite properties. By incorporating RS biochar into construction materials, this research contributes to mitigating the environmental challenges posed by stubble burning and promoting sustainable practices in the construction industry.

Keywords: Bio-based material; Construction material; Environmental impact; Stubble burning; Workability.

FABRICATION AND CHARACTERIZATION OF COST EFFECTIVE CERAMIC MEMBRANES UTILIZING SAWDUST AS A PORE-FORMING AGENT FOR OIL WATER EMULSION SEPARATION

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The contamination of water resources by oily wastewater poses a significant threat to human health and water ecosystems; arising from both industrial expansion and everyday activities. Traditional treatment methods, such as adsorption, chemical precipitation, and dissolved air flotation, suffer from drawbacks like low separation efficiency and the risk of secondary pollution. Membrane separation processes have emerged as a leading technology for treating oily wastewater, owing to their notable effectiveness in removing oil produced by diverse industries, including oil refineries, petrochemicals, metallurgy, food and beverage, dairy, and kitchen wastewater. Ceramic membranes have drawn a lot of interest in a variety of industrial applications because of their extraordinary characteristics, including long-term toughness, great chemical and thermal stability, high fouling resistance, and mechanical strength. This study aims to develop cost-effective ceramic membranes using sawdust as a pore-forming agent, applied in the separation of oil-water emulsions. The precursor mixture comprises fly ash, clay, and sawdust. The uniaxial pressing method is employed, and the resulting membrane is sintered at 900°C for 8 hours. Circular disc-shaped membranes (51 mm diameter and 4 mm thickness) undergo characterization through XRD, SEM, and FTIR analysis. Porosity, average pore size, acid-alkali test and pure water flux were conducted to examine the membrane's performance. The membrane's capability is evaluated in separating oil-water emulsions from kitchen wastewater, showcasing an oil rejection rate of up to 98.2%.

Keywords: Ceramic membrane; Fly ash; Membrane; Oil water emulsion; Sawdust.

BIO-BASED SURFACTANT FROM BIOFUELS: PRODUCTION AND OPTIMIZATION USING RESPONSE SURFACE METHODOLOGY

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Surfactants are amphiphilic in nature and act as wetting agents to lower the surface tension of liquids or the interfacial tension of two-phase systems due to adsorption at the surface or interface, respectively. Surfactants have been widely used in every industrial area ranging from household detergents to drilling muds and food items to pharmaceuticals. Therefore, production, optimizations, and utilizations of the surfactants for various applications have been the research interest among researchers of various fields. However, due to increasing environmental concerns, the hazardous nature of surfactants posed due to their non-degradable properties is of great concern. For combating such problems, bio-based surfactants could definitely serve as a potential alternative. In the present study, bio-based surfactant was produced in two step process and the reaction was optimized using response surface methodology. A Box-Behnken Design (BBD) was used to study the effect of different parameters viz. mole ratio, temperature and time on the yield of the reaction. In first step, waste/used cooking oil (UCO) was converted into biodiesel (UCOME) and in the second step, this UCOME was used to synthesize methyl ester sulphonate (MES) by incorporating a sulphonate group. The synthesized MES was characterized using Fourier Transform Infrared (FTIR) and Nuclear Magnetic Resonance (Proton NMR) Spectroscopy techniques and all the peaks are in agreement with the literature values. Further, the surface tension properties were measured to calculate the CMC values of the developed surfactant. Thus, utilization of bio-based surfactants will be the next step towards cost-effective thus creating sustainable economics and promoting sustainable development goals.

Keywords: BBD; Biodiesel; Methyl Ester Sulphonate, Sulfonation, Used Cooking Oil, UCOME etc.

FUNGI-BASED MICROBIAL FUEL CELLS (MFCS) AN ALTERNATIVE SOURCE OF ENERGY AND WASTE MANAGEMENT

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Microbial fuel cells (MFCs) have garnered a lot of attention in the past few years owing to their versatility in terms of usage of a variety of substrates, microorganisms, and various configurations which allow for the production of value-added compounds, bioelectricity and bioremediation capabilities. A typical MFC functions by the degradation of organic material in the anodic chamber, the electron released subsequently then move to the cathodic chamber via the anode and bioelectricity is generated. Recent years have shown an increasing interest in the use of fungi in MFCs for biodegradation and energy production. Fungi can be used in either the cathode or anode and also in varied MFC configurations. This study aims to explore the potential of fungi in MFCs for bioelectricity generation by investigating the biodegradation of filter mud from jaggery producing cottage industry using a fungal culture isolated from the waste in a single chambered membraneless MFC (SCM-MFC). Filter mud is a by-product of jaggery production and management, handling and disposal of this waste is a major concern for industries as it has a high moisture content, foul odour and it attracts pests and pathogens. 18s rRNA sequencing showed that the isolated fungi was *Mucor pseudolusitanicus*. The experiments were conducted in batch mode for up to 25 days with a working volume of 150 mL. The maximum power density and the maximum current density obtained were 10 mW/m² and 127.39 mA/m² respectively. Thus, fungal-MFCs hold the potential to be a sustainable, cost-effective and environment-friendly technology to reduce pollution, help clean up the environment and generate bioelectricity.

Keywords: Bioelectricity; bioenergy; green energy; microbial fuel cell; waste management.

FABRICATION OF CYANIDE CHEMOSENSOR AND ITS STUDY AS LIVE CELLS IMAGING REAGENT FOR DETECTION OF CELLULAR UPTAKE OF CN⁻ ION

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Designing of anion sensors is a newly emerging field and quite challenging. Among various dangerous chemicals, cyanide is the most hazardous or lethal anion. In the present work a sensor **A** has been synthesized and characterized by various techniques as ¹HNMR, ESI-MS and FT-IR spectroscopy. The sensor **A** has anion sensing ability through selective binding in a mixed solvent system of H₂O/DMSO (1:1 v/v) with CN⁻ ion (as tetrabutylammonium salt) having sharp change in its fluorescence and UV-Visible spectrum. OH and NH groups of sensor **A** bind CN⁻ ion bidentately utilizing hydrogen-bonding interaction which is supported by ¹H NMR. Acute change in fluorescence and UV-Visible spectra is used for calculating the binding constant of sensor **A** with CN⁻ ion. Remarkable change has been observed in the colorless solution of **A** to yellow due selective binding of CN⁻ ion in H₂O/DMSO (1:1 v/v). The sensor **A** has been also studied as live cell imaging reagent for intra-cellular detection of CN⁻ anion in SW480 cells.

Keywords: 4-chlorophenylhydrazine; Cyanide Ion; Fluorescence Spectra; Naphthol Derivatives; SW480 Cells Imaging.

IMPACT OF DIFFERENT PREPROCESSING TREATMENTS ON THE GERMINATION OF BOTTLE GOURD SEEDS FOR THE APPLICATION OF HYDROPONIC

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The bottle gourd (*Lagenaria siceraria*) seeds are a valuable resource with prospective uses in agricultural and food processing due of its nutritional and therapeutic qualities. We examine the effects of four different preprocessing methods on bottle gourd seeds in this study namely: A) Scarification, B) Soaking the seeds in water for ten hours, C) Soaking the seeds in water for twenty-four hours and D) Stratification. This study assesses how various treatments affect the viability of the seeds, the rate at which they germinate, and the growth of the seedlings. It has been observed that the preprocessing treatments impact the seed germination significantly. It is also found that the scarification and soaking the seeds for 10 hours enhances the germination quality however prolonged soaking and stratification deteriorates the germination property of the bottle gourd seeds. Finally, after germination the plants of bottle gourd the continuous growth of the plants has been monitored and corelated with different pretreatment.

CHARACTERIZATION AND MORPHOLOGICAL STUDY OF FOULANT IN WATER COOLED HEAT EXCHANGER OF A PHARMACEUTICAL INDUSTRY

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Fouling is one of the major problems occurring in process industries leading to significant heat losses. Fouling is an accumulation of unwanted materials, attaching themselves to the heat transfer surfaces due to high process temperature and/or increased humidity. It affects the thermal performance like heat transfer leading to a drastic reduction in steam economy and an increase in steam consumption. It is important to identify the composition and morphology of such foulants. This study presents the characterization of selected fouling samples collected from shell and tube-type forward-feed water-cooled heat exchangers taken from the pharmaceutical industry. Scanning electron microscopy (SEM) was carried out to characterize the sample foulant physically. It was observed that the size and shape varied between samples due to varying working temperature ranges in the heat exchangers. The samples for their composition were measured by energy dispersion x-ray microanalysis and 25.02% Ca, 16.04% S, 5.95% C, 2.08% Fe, 1.76% K, 1.73% S and 1.28% Cl with traces of Al, Mg, Na and some other salts were observed. X-ray diffraction (XRD) revealed the presence of anhydrite, gehlenite and olivine with traces of thenardite and calcite. CHNS revealed the presence of Carbon, Hydrogen, Nitrogen and Sulphur in the foulant sample. Thermal conductivity and thermal diffusivity were measured using experimental and theoretical means. The result obtained in this study provides useful information on understanding the composition and morphology of foulant. This information is useful in designing the mitigating technique for removal or controlling the fouling. Keywords: Fouling, SEM-EDS, Thermal conductivity, Thermal diffusivity, XRD.

PLASTIC ROADS: A SUSTAINABLE SOLUTION FOR WASTE DISPOSAL

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Plastic pollution is a huge environmental problem, leading research into novel solutions such as road construction utilising plastic trash. This method has various advantages, including increased durability, decreased maintenance requirements, and cost-effectiveness. However, further study is needed to fully grasp the potential of waste polyolefin pyrolysis products for bitumen modification, which is critical for optimising the quality and performance of binders in such applications. It is critical to investigate the relationships between pyrolysis process parameters and final product quality. Additionally, techniques for modifying these goods and improving their performance in diverse applications should be investigated. This technique might be extended beyond road building to provide sustainable solutions in other areas. The growing garbage creation emphasises the importance of addressing disposal techniques. One intriguing option includes combining plastic garbage with asphalt to generate durable road surfaces. This may drastically reduce plastic litter in landfills and seas while also reducing asphalt consumption and providing a more environmentally friendly road building option. However, difficulties remain. Concerns about the dependability, safety, and potential harmful environmental effects of plastic roads persist. The production and use of plastic materials for road building generate legitimate concerns regarding their overall environmental impact. Despite these obstacles, the continued investigation and deployment of plastic roads represent significant advances in sustainable infrastructure innovation. More research, stricter standards, and coordinated efforts are required to reduce dangers and maximise the advantages of this method.

Keywords— Waste plastic roads, Plastic waste, Road construction, Bitumen, Asphalt, Collection, Sorting, Shredding, Composite material, Laydown, Maintenance, Durability, Safety, Innovation I.

QUANTIFICATION OF URBAN GROWTH DYNAMICS USING SPATIAL METRICS AND RELATIVE ENTROPY IN ROORKEE PLANNING AREA

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This study emphasizes the utility of Spatial Metrics and Relative Entropy in quantifying the configuration and composition of the urban growth of the study area. The structure of a metropolitan area is significant for properly managing any human habitat development. Haphazard urban growth without proper planning or understanding of the dynamics associated with urbanization might lead to sustenance issues for all the species on earth and negative urban footprints on the environment. In the proposed study, Remote Sensing and GIS are used to produce urban growth maps of the Roorkee planning area. These maps are evaluated using nine spatial indices to define the statistics of urban growth dynamics. The information was validated using the Relative Entropy index to give complete information on different aspects of the landscape, providing an insight into the sprawling process and dynamics of change. The study results are used to establish the forms, patterns and types of urban growth, which is quantitatively evaluated using these indices. It is observed that there is clustered continuous growth in and along the city center and across the different patches of the whole landscape. The built-up, however, is evenly and equitably distributed along the road corridors. But as the distance increases perpendicularly away from the center of the roads, the urban growth shrinks. Hence, a ribbon-sprawl development with linear aggregated growth is observed in the study area. This information is beneficial to put forward appropriate policies and monitoring mechanisms on urban growth and guide sustainable development plans for the future.

Keywords: Geographic Information System (GIS); relative entropy; spatial metrics; sustainable development; urban growth.

SONOCHEMICAL CUS SYNTHESIS: UNVEILING ITS IMPORTANCE IN THE REMEDICATION OF MALACHITE GREEN

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Utilizing copper sulfide (Cu_2S) in photocatalysis emerges as a strategic and environmentally conscious approach to wastewater treatment, highlighting its potential to alleviate the adverse impacts of organic contaminants. This investigation concentrates on the sonochemical synthesis of Cu_2S under diverse pH levels, to evaluate their influence on the morphological and structural characteristics of Cu_2S nanoparticles and its application in malachite green degradation. X-ray diffraction (XRD) analysis revealed the crystalline nature of the synthesized material, encompassing two distinct phases— Cu_2S and CuS . Subsequent scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDAX) analyses affirmed the material's spherical morphology and offered insights into its elemental composition respectively. UV-Visible diffuse reflectance spectroscopy (DRS) indicated the material's suitability for operation within the visible range of 1.9-2.4 eV. These findings substantially contribute to a deeper comprehension of the sonochemical synthesis process, laying the groundwork for the controlled fabrication of Cu_2S nanoparticles which exhibit over 90% photocatalytic degradation of malachite green within one hour.

Keywords: Cu_2S , sonochemical, pH dependent, photocatalysis.

UTILIZING KAPOK SEED FOR MN (II) REMOVAL FROM SYNTHETIC WASTEWATER SOLUTION THROUGH ADSORPTION

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At the current scenario, clean environment is essential to living world. However the usage of heavy metals in different industrial process for decades has increased the amount of residual metallic compounds in wastewater, which poses serious health hazards to living things. This study addresses water pollution concerns by focusing on cost-effective and efficient removal of contaminants from wastewater, particularly heavy metals through adsorption process. Adsorption is a cost-effective method of treating wastewater effluent with features such as low operating costs, environment friendly and ease of usage. In the present research work the potential of Kapok seeds as a low-cost adsorbent for Mn(II) removal from wastewater has been studied. The prepared Kapok seed powder (KSP) was characterized by using XRD, FTIR, BET, and SEM. The batch experiments were conducted by varying the different parameters such as pH, agitation time, initial metal concentration, adsorbent dosage and temperature. As per the results the adsorption process was best at 6.00 pH with Mn (II) removal efficiency of 92.5%. Different kinetic models and adsorption isotherm were also applied for predicting the behavior of the adsorption process. On the basis of results, it can be accomplished that the KSP may be used as a better adsorbent for removing metals from wastewater.

Keywords: Adsorption isotherm, Bioadsorbent, Heavy metals, Kinetics Model, Wastewater Treatment

UNRAVELLING ARSENIC CONTAMINATION IN THE GANGA-GHAGHARA BASIN, INDIA: RECONNAISSANCE USING GEOGRAPHICAL INFORMATION SYSTEM

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Freshwater resources constitute 2.5% of the global water supply, with one-third of it being groundwater. The contamination of groundwater is occurring because of the rise in unregulated human activity and the probable consequences of climate change. Arsenic exposure in humans occurs through contaminated groundwater and food chains, leading to a significantly higher intake of arsenic in their diet compared to the recommended maximum tolerable daily intake (MTDI) of 2.1 µg/kg/BW/day defined by JECFA in 2011. This study introduces an innovative method that utilizes cutting-edge Geographical Information System (GIS) techniques to evaluate the presence of arsenic pollution in the soil and groundwater of the Ganga-Ghaghara Basin, Uttar Pradesh. The study aims to investigate the possible increase in existing groundwater pollution sources and the resulting risks to human health caused by changes in flow patterns and increased human activities. The Ganga-Ghaghara basin is a populated and agriculturally intensive region that is susceptible to the impacts of climate change on groundwater resources and human health hazards. The collection of primary data was achieved through field sampling at the study sites, while secondary data was obtained through a literature survey. The application of spatial analysis was utilized to understand and identify the specific area with a high concentration of arsenic in the Ganga-Ghaghara Basin of Uttar Pradesh. More than five times the permissible limit of 50 µg/L, significant arsenic contamination is affecting 20 districts in Uttar Pradesh. The arsenic levels in the groundwater of Ballia, Ghazipur, and Varanasi have above the World Health Organization's recommended limit of 10 µg/L.

Keywords: Arsenic hotspot. Environmental constituent; Groundwater; Heavy metal; Spatial analysis.

WATER FOOTPRINT ASSESSMENT OF PADDY BASED CROPPING SYSTEM IN HIGH ALTITUDE TEMPERATE ZONE OF KASHMIR, INDIA

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Water scarcity has become one of the most serious issues in the world. The water footprint for rice that is the total amount of water used for the production of rice from sowing to harvesting in terms of volume, has been calculated for three different regions involving three different districts of Kashmir Valley from 2010 to 2022. The results show that the lowest green water footprint was 850ltrs/kg in 2016 in Central Kashmir's Budgam District and the highest green water footprint was 2280 ltrs/kg in 2022 in Baramulla District. While the lowest blue water footprint was in 10.9ltrs/kg in 2010 in Budgam District and the highest blue water footprint was in 3478 ltrs/kg in 2019 in Baramulla District. The average WFP in the three districts was 2003 ltrs/kg in Budgam, 3358 ltrs/kg in Baramulla and 2300 ltrs/kg in Anantnag District. It is also observed that with passing time, there is decrease in the area under paddy cultivation as more and more farmers in Kashmir valley are converting their agricultural land into horticultural areas due to the untimely rainfall in Kashmir valley. There is increase of irrigation dependent on canals, so the production of rice has decreased in the valley.

Keywords: Green Water Foot Print, Blue Water Foot Print, Rice cultivation, Evapotranspiration, Crop Water Usage.

GREEN SYNTHESIS OF COPPER OXIDE NANOPARTICLES USING SOLANUM TUBEROSUM EXTRACT TO MEDIATE PHOTOCATALYTIC DEGRADATION OF METHYLENE BLUE

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The dimensional reduction and manipulation of matter establish a high surface-to-volume ratio along with distinctive optical, magnetic and electrochemical properties. Such characteristic features have found immense relevance in environmental remediation, the pharma industry, biomedical devices, biosensor platforms, etc. Synthesis of nanoparticles employs several top-down and bottom-up approaches thus facilitating a wide range of possibilities with improved and favorable synthesis techniques. However, green synthesis has gained momentum in the scientific community owing to local availability and abundance of plant sources. Moreover, with specific regard to environmental concerns, the additional features exhibited by green synthesis are non-toxicity, eco-friendly processing, less byproduct generation, etc making them the most preferred and economical alternative. Herein, we reported the green synthesis of copper oxide nanoparticles (CuONPs) from *Solanum tuberosum* (potato) extracts to assist the photodegradation of methylene blue dye. Various characterization studies were performed to ascertain the nanoscale particle generation. The UV-visible spectroscopy data demonstrated in terms of surface plasmon resonance (SPR) peaks indicated the formation of the nanoparticle with the highest intensity peak at 273nm with 25mM salt concentration of copper salt and 5mL of reducing agent in a volumetric ratio of 5:1. The morphological characteristics were assessed via scanning electron microscope (SEM) data, which demonstrates spherical particles with an average size range of 336nm. The Fourier transform infrared (FTIR) spectrum revealed distinct peaks with corresponding functional groups responsible for nanoparticle synthesis and present on the surface of synthesized nanoparticles. The synthesized and characterized metal nanoparticle was tested for photocatalytic degradation of methylene blue in the presence of stimulated light (100 watts). The degradation data displayed visual alteration in reaction mixture color from blue to lightest blue or colorless. The resultant change in the color was spectrophotometrically quantized by recording the absorbance of the reaction mixture against the untreated methylene blue at 668nm. The homogenous dispersion of CuONPs generates electron-hole pairs which enhances the charge transfer thus facilitating successful dye degradation with a degradation efficiency of 76.1%. Results from the data suggest its potential application in nanoparticle-based remediation of toxic pollutants from environmental samples.

Keywords: Green synthesis; Metal Nanoparticle; Photocatalyst; Reducing Agent; SPR.

TECHNO-ECONOMIC ANALYSIS FOR THE INNOVATIVE PROCESS OF PRE-TREATMENT OF PADDY STRAW FOR BIOFUEL PRODUCTION

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In recent years, biofuel has become an alternative source of energy for its application in transportation, power generation, heating and other industrial applications. Biofuels are produced from biomass, which is produced through photosynthesis using carbon dioxide and water in the presence of sunlight. Therefore, it is considered as reservoirs of renewable energy, which is yet to be properly harnessed. Many technologies have been developed and demonstrated, which could not be commercialized due to economic viability. Techno-economic analysis (TEA) plays an important role for any developing technology to identify the operation, feasibility, costs of specific equipments, bottlenecks, etc., and it helps to improve overall process. TEA helps to analyse both technical and economical details of a specific project or process technology. The present work is focused on economical pretreatment of paddy straw for 2G ethanol production followed by simultaneous saccharification and fermentation. Comparative experiments were performed using water, deep eutectic solvents (DES) using choline chloride and urea in water, and choline chloride in cow urine at 121°C for 1 h. Further, the impact of these pretreatment experiments was analyzed by enzymatic saccharification using Cellic CTec2 and Cellic HTec2. The results revealed the maximum sugar recovery in choline chloride with cow urine. Further, the process model will be developed using SuperPro for techno-economic analysis.

Keywords: Paddy straw, TEA analysis, pretreatment, enzymatic saccharification, SSF.

TREATMENT OF TEXTILE AND PHARMACEUTICAL WASTEWATER USING LOW-COST ADSORBENTS FOR SINGLE AND MULTI-COMPONENT SYSTEMS

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The use of inexpensive adsorbents in single- and multi-component systems for the remediation of wastewater produced by the textile and pharmaceutical industries is critically examined in this review study. For environmental engineers and academics, treating such complex effluents is a substantial challenge that necessitates sustainable and affordable solutions. This study summarises the most recent developments and discoveries in this important area in response. The first section of the article provides an outline of the growing issues surrounding the wastewater from textile and pharmaceutical industries, including the release of heavy metals, organic compounds that are difficult to break down, and other dangerous materials into natural water bodies. It emphasises how crucial it is to have creative and economical wastewater treatment methods. A thorough examination of inexpensive adsorbents is provided, covering a wide range of materials including industrial byproducts, naturally occurring compounds, and agricultural leftovers. The composition, surface properties, and adsorption capacity of each class of adsorbent are carefully examined, with an emphasis on how well it can handle the pollutants found in textile and pharmaceutical effluents. The paper then delves into the complex dynamics of adsorption processes in single and multi-component systems. It breaks down the kinetics, thermodynamics, and adsorption isotherms that control these systems, providing important insights into the processes behind adsorption phenomena. The study offers a basis for improving operating conditions and low-cost adsorbent efficiency in various wastewater treatment situations by combining this information. The assessment also emphasises the sustainability benefits of using inexpensive adsorbents, such as their capacity for regeneration and reusability. Adopting these materials can help wastewater treatment procedures dramatically minimise their environmental impact, which is in line with the goals of global sustainability.

Keywords: Adsorption; Low-cost adsorbents; Pharmaceutical wastewater; Remediation Textile wastewater.

A GREEN APPROACH TOWARDS SYNTHESIS AND WASTE MANAGEMENT OF BIOLOGICAL MOF FOR DRUG DELIVERY APPLICATIONS

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Biologically active MOFs and metal complexes are actively being recruited for biomedical applications. MOFs have proven ability to acquire bioactivity by slowly releasing the drug molecule and efficacy with drug molecules of small molecular weight. In this work, Cu and amino acid are employed to synthesize MOFs with a purpose to have biologically active, biocompatible and porous nanomaterials for drug loading ability. As a part of preliminary studies, the purity of synthesized MOFs will be analyzed through fundamental characterization techniques. To ascertain the biological activity, drug loading and unloading will be performed on the MOF materials. The synthesis of Cu based MOF was done using Cu acetate and L-Glutamic acid taken in 0.5:1 ratio in aqueous solvent and stirred at room temperature for 2 h. The resulting MOF $[\text{Cu}(\text{Glu})_2(\text{H}_2\text{O})_2] \cdot \text{H}_2\text{O}$ had orthorhombic crystal structure with space group of P212121. $a = 7.2036$ (5) Å. $b = 10.2696$ (7) Å. $c = 11.0023$ (8) Å. $Z = 4$. Loading of drug molecules was performed in green conditions. Further more, FTIR and UV-vis absorption spectra gave confirmation regarding the coordination of ligand to metal centre and tagging of the drug molecules. The study shall help understand the mechanistic details of working of the mof based drug release especially for specific diseases. The methodology applied for synthesis and drug loading experiments is completely green and does not produce toxic waste leading to a positive solution to pollution free synthesis.

Keywords: Green synthesis; drug delivery systems; porous materials; biological mofs; solvent waste.

FACILE SYNTHESIS OF NICKEL OXIDE NANOPARTICLES AND ITS APPLICATION PHOTOCATALYTIC REMEDIATION OF CRYSTAL VIOLET DYE

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The green synthesis of nanoparticles has many benefits over the physical and chemical synthesis processes. The benefits include decreased waste production, simple synthesis, scaling, and affordability. Metallic nanoparticles offer a wide range of applications because their surface can be modified to make them biocompatible, stable, and functional. Nickel oxide nanoparticles (NiO NPs) is one such important metallic nanoparticles. In the current study NiO NP was synthesised using green tea leaf extract and then it was examined with the help of UV visible spectroscopy, FTIR, TEM and XRD. The synthesised NiO NPs was functionalized using cysteine, a basic and cationic amino acid. By changing one parameter at a time, the synthesis of NiO NPs procedure was optimized. The best optimized parameters for the Ni NPs synthesis were the 70° C temperature, 25mM nickel-sulphate salt concentration, 11 pH, 20:30 volumetric ratio of nickel-sulphate solution: reducing agent (green tea leaf extract) and 30 min reaction incubation time. The optimal cysteine concentration for functionalization of NiO NP was 2mM. To degrade crystal violet solution, functionalized NiO NP was used because of their photocatalytic properties. For different concentrations of crystal violet solution were tested, the degradation efficiency within 15 minutes was found to be 55 to 60%. The effect of pH and light intensity on the efficiency of photocatalytic degradation of crystal violet by functionalized NiO NP was also tested.

Keywords: Degradation; Green tea; Metallic nanoparticles; Photocatalysis; Plant extract.

AN EXPERIMENTAL STUDY OF PHOTOCATALYTIC REDUCTION OF CARBON DIOXIDE TO METHANOL

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Increasing percentage of presence of carbon dioxide in the atmosphere is one of biggest challenge for environment. For past a decade, a vast amount of research has been conducted on conversion of CO₂ into some value added product and these researches have shown that CO₂ can be utilized by converting into methanol. This conversion of carbon dioxide into methanol does not only solve the problem of high concentration of CO₂ in atmosphere but also provide an alternate path to produce energy source to us which will be a big break through to our second biggest problem which is energy crisis. Photo catalytic conversion of anthropogenic CO₂ gas in presence of water into methanol not only solves the problem of green house effect of carbon dioxide but also provide cheapest and innovative way to energy crisis. This work describe conversion of CO₂ into methanol using different photo catalysts like TiO₂, ZnO and graphene oxide and also the loading of copper oxide on these metal oxides. To increase the photo catalytic activity of catalyst a suitable reactor system was designed which provides suitable reaction conditions for maximum conversion of carbon dioxide and selectivity of methanol as desired product. The photo catalytic conversion of carbon dioxide was observed as 20.68% for graphene oxide as catalyst under UV radiation which is very high as compared to other catalysts but for copper oxide loading, zinc oxide provides better conversion into methanol around 33.13% which was much higher as compare to the other catalysts with copper oxide loading under UV radiation.

Keywords: Environment pollution; Greenhouse gas; Methanol production; Photo catalysis; Renewable energy.

HYDROTHERMALLY TREATED TEA WASTE: A NOVEL APPROACH FOR METHYLENE BLUE PHOTODEGRADATION AND ANTIMICROBIAL TREATMENT OF WASTEWATER

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Water pollution stands out as a major environmental challenge in the present era. Use of synthetic dyes in textile industries and living pathogens are primarily polluting the adjoining water bodies. The presence of these contaminations in water bodies are giving rise to a range of health hazards, including but not limited to carcinogenic risks, reproductive issues, skin allergies, and other potential health hazards. In the recent times, the uses of waste management play an important role to remove the various types of contamination from waste water through the photo degradation process. In this study, green method synthesized tea waste (TW) hydrochar was produced first by hydrothermal process at 180°C for 10 hours. TW hydrochar were characterized by x-ray diffraction (XRD), scanning electron microscopy (SEM), and Fourier transform infrared spectroscopy (FTIR). Further, the synthesized hydrochar was used to remove the dye-based contamination from pure (Milli Q water) and river based wastewater through photo degradation process. The experimental results demonstrated that methylene blue (MB) dye degraded from pure and wastewater with the degradation efficiency of 90% and 75% in 3 hours under UV light irradiation, respectively. Hydrochar was also used for the removal of *E. coli* bacteria from the waste water under the antimicrobial photo degradation process with a removal efficiency of 99.64%. The results revealed that hydrothermal treated tea waste hydrochar can be a good photo catalyst for waste water treatment especially considering degradation of MB and antimicrobial activities.

Keywords: Water pollution; Hydrochar; Photo degradation; MB; Antimicrobial degradation.

WASTE TO WEALTH: CARBON COATED ZrSiO_4 NANOCOMPOSITES FROM RECYCLED PLASTIC AND NATURAL ZIRCON FOR OXYGEN EVOLUTION CATALYSIS

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The conversion of waste plastic into valuable materials has gained significant attention due to the environmental concerns associated with plastic waste disposal. In the present work, carbon coated ZrSiO_4 Nanocomposites are synthesized using waste materials. The source of Zirconium i.e. ZrSiO_4 (Zircon sand) is obtained from sea beach and waste plastic (Plastic bags) as a source of carbon are used as precursors for synthesis and the synthesis is carried out in Auto Clave by using heat treatment routes. Analyses of the synthesized samples were carried out using a variety of techniques including X-ray diffraction (XRD), Field emission scanning electron microscopy (FESEM) and to check the electrolyte behaviour of the samples electrochemical studies are carried out. XRD depicts a mixed-phase formation of carbon and ZrSiO_4 . FESEM reveals that agglomeration of particles. The as-synthesized samples shows good electrochemical performance towards OER. The sample synthesised at 800°C for 15 hours possess a low Tafel slope of 94.8 mV/dec . A low over potential of 628 mV (vs. SCE) is required for a current density of 10 mA/cm^2 . Moreover, it also shows optimal stability up to 2000 Cyclic Voltammetry cycles in alkaline medium. This study provides a new route for utilizing waste to wealth and a low-cost electrolyte for electro catalysis.

Keywords: Nano Composites, Waste Plastic, Zirconium Carbide, Electro catalyst, OER.

EFFECT OF SBA-15 LAYER ON PERFORMANCE OF POROUS ALUMINA SUPPORTED PALLADIUM/PALLADIUM-ALLOY MEMBRANES FOR HYDROGEN SEPARATION

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Palladium/palladium-alloy based membranes supported on porous alumina are widely used for separation of hydrogen from mixture gases. However, the selectivity of such membranes are largely depends on support characteristics. A wide pore size distribution of support requires a thick layer of the Pd/Pd-alloy deposition. This increases the cost of such membranes and reduces its commercial application. In current study, a layer of SBA-15 doped with metal is deposited on surface of the porous alumina support to minimize the pore size distribution. The SBA-15 loaded with Pd, Pd-Cu, and Pd-Ag is deposited as an intermediate layer on the surface of the α -Al₂O₃. The intermediate layers are prepared using the hydrothermal method technique. The characterization of the synthesized intermediate layer is performed by using TEM, XRD, SEM and BET. The performance of the prepared membranes are tested in a specially designed membrane separator assembly. The experiments are performed by using pure gases, hydrogen and nitrogen, as well as mixture gas composition. The tests are performed at a temperature of 250, 300, 350, and 400 °C with pressure driving forces ranging from 0.5 to 2.0 bar. The effect of temperature and pressure on hydrogen permeation flux and hydrogen selectivity are calculated.

Keywords: Surface modification, SBA-15, Palladium, Hydrogen Separation, Electroless pore plating.

ECO-FRIENDLY MANAGEMENT SYSTEM FOR NON-BIODEGRADABLE WASTES

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In an effort not only to highlight how essential is non-biodegradable waste management but also, to address some of the challenges involved and propose approaches collectively. Following the guidelines and principles of the United Nations' Sustainable Development Goals (SDGs). Menstrual hygiene and early childhood care represent crucial life stages that generate substantial waste in the form of non-biodegradable sanitary pads and diapers. This specific type of waste is the primary focus here, as not only it poses health risks but also places immense strain on waste management systems. Sanitary pads and diapers share a common trait: they are predominantly composed of non-biodegradable materials, primarily plastic, which took centuries to decompose. To tackle the issues of menstrual and diaper waste disposal in tandem, this approach aligns with SDGs related to Good Health and Well-being (SDG 3), Sustainable Cities and Communities (SDG 11), Clean Water and Sanitation (SDG 6), and Responsible Consumption and Production (SDG 12). Effective management of menstrual and diaper waste is vital amidst increasing natural disasters. Both types of waste require specific attention due to the challenges posed by privacy, hygiene, and environmental considerations. Improper disposal methods, like flushing products down toilets or discarding them carelessly, contribute to environmental pollution and sewage system blockages. Comprehensive waste management strategies are in progress, like developing and implementing safe, discreet, and environmentally friendly disposal methods, such as designated bins and incinerators. This paper highlights the challenges related to women's menstrual hygiene as well as sanitary pads and diaper disposal management. By embracing SDG9 (Industry, Innovation, and Infrastructure), we suggested ICT-based solutions (Blockchain, IoT and Machine learning, etc) for managing menstrual hygiene and early childhood products' life cycle in an eco-friendly manner.

Keywords: Eco-friendly management, Hazardous waste, Health and hygiene, ICT, Non-Biodegradable Waste, Sustainability.

EVALUATING THE EFFECTIVENESS OF EGGSHELLS IN REMEDIATION OF LEAD METAL, IONS FROM AQUEOUS SOLUTION; ISOTHERMS AND KINETICS STUDY

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The occurrence of lead metal ions in wastewater has become one of the most significant concerns for enhancing water quality and mitigating its impact on the environment, wildlife, and public health. This paper primarily addresses the removal of lead metal ions from wastewater using chicken eggshell (CES), which is a readily available and cost-effective food waste. In order to achieve the most favorable operational conditions for lead remediation, a series of batch tests, i.e., particle size, pH, biosorbent dose, temperature, contact time, initial lead concentration, and agitation speed, were conducted. Also, the characterization of XRD, SEM, and FTIR was done for crystal structure, morphological examinations, and functional group information of modified CES, respectively. To investigate the equilibrium behavior of adsorbent, Langmuir, and Freundlich models were employed. Additionally, the application of pseudo-first order and pseudo-second order kinetic models was used to analyze the kinetic behaviors of the experimental data. The findings demonstrated that the pseudo-second-order kinetic model and the Langmuir isotherm model exhibited the highest level of agreement with the experimental data. According to the Langmuir model, the maximum adsorption capacity of 454 mg/g is achieved in lead removal, suggesting the occurrence of monolayer adsorption at homogenous sites. Eggshell, an industrial food waste, has been proven to be an effective adsorbent for the treatment of lead-containing wastewater.

Keywords: Food waste; Eggshell; Heavy metals; Lead adsorption; Wastewater.

EFFECTS OF RECYCLED SAW SLAG AND THE PROCESS PARAMETERS ON THE CHEMISTRY OF STAINLESS STEEL CLADDINGS

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Submerged arc cladding of stainless steels finds frequent applications in the manufacturing industry. This process produces an industrial waste known as welding slag, which being useless is thrown away. A technology was developed for the recycling of submerged arc cladding slag for its reuse as flux in the same process. The cost of recycled slag was found to be only 27% of the original flux. This technology also helped in conservation of the natural resources and the environment. The quality of submerged arc claddings primarily depends upon their chemical composition, which in turn is strongly dependent on the element transfer behavior of the welding flux. Therefore, the element transfer behavior of the recycled slag with respect to the variation in the process parameters was studied. Statistical mathematical models were developed by central composite rotatable design (CCRD) to predict the element transfer of various elements like Carbon (C), Silicon (Si), Manganese (Mn), Nickel (Ni), Phosphorus (P) and Sulfur (S) between the slag and the deposited metal. The results showed that the welding current has a positive effect on the transfer of Mn and a negative effect on the transfer of C, Si, Cr, P and S to the deposited metal. The open circuit voltage increased the transfer of C and decreased the transfer of Mn to the weld pool. Whereas, the welding speed decreased the Si and Ni transfer to the deposited metal. The transfer of other elements remained unaffected of the variation in the process parameters.

Key words: sustainable manufacturing; recycling; metal chemistry; element transfer.

EXPERIMENTAL INVESTIGATION OF BI-COMPONENT PARTICLE SEGREGATION STUDIES IN A SPIRAL CONCENTRATOR

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Spiral concentrators are commonly used in various industries, including mineral and coal processing, to efficiently separate materials based on their density and size. In these concentrators, a mixture of solid particles and fluid (usually water) is introduced as feed at the top of a spiral channel. As the mixture flows down the spiral, centrifugal and gravitational forces act on the particles, causing them to stratify based on their density and size. Spiral flows exhibit complex fluid dynamics, and interactions involve multiple phases and components in the process. Understanding the behavior of these phases within the spiral concentrator is crucial for achieving efficient separation. An experimental bi-component particle interaction study is conducted in this work utilizing magnetite (heavier density) and silica (lighter density) with different proportions processed in the spiral concentrator. The observation separation reveals that denser particles accumulate towards the inner region of the spiral trough, while a significant concentration of lighter particles are found close to the outer edge. The 5th turn of the spiral trough is partitioned into five zones to achieve a comprehensive distribution analysis of bicomponent particle segregation. Samples are then gathered from these individual streams using an in-house sample collector, and subsequent analysis is conducted to assess component segregation. Along the trough, there was a decline in the concentration of coarser particles, accompanied by an increase in the concentration of lighter particles. The segregation pattern indicates that the heavier coarse component accumulates in the inner zone, whereas the lighter fine component collects in the outer zone. The middle zone primarily consists of heavier fine particles and lighter coarse particles. The zone-wise results reveal that there is a significant fraction of segregation occurs in inner and middle zones. Finer magnetite and silica particles predominantly accumulate in outer zones with the smallest fraction of segregation.

Keywords: Bi-component; Multiphase flow; Particle segregation; Spiral concentrator.

ADVANCE CORROSION RESISTANT COATINGS IN RECENT ERA: A REVIEW OF ADVANCES AND PERSPECTIVES

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The field of corrosion prevention has long been a challenge for various industries. However, this review is an attempt to explore a promising solution that brings together advanced materials and intelligent sensing technologies - smart corrosion coatings. The smart corrosion coating system employs sensors to detect early signs of corrosion and triggers a responsive mechanism for timely intervention. Key findings demonstrate the enhanced protective capabilities of the smart corrosion inhibitory coatings compared to traditional alternatives. The integration of smart technologies not only enhances the lifespan of infrastructure but also contributes to sustainable practices by minimizing the frequent repairs and replacements. By delving into the innovative realm of smart corrosion coatings, this study opens up new avenues for exploration and development in the field. The implications extend beyond corrosion prevention, influencing the way we approach materials science and paving the way for future advancements in various industries. In conclusion, this review paper sheds light on the potential of smart corrosion coatings to address one of the industry's persistent challenges.

Keywords: anticorrosion; coatings; future advancement; smart; sustainable.

EFFECT OF CERIUM OXIDE AS ADDITIVE IN BIODIESEL ON THE PERFORMANCE AND EMISSION CHARACTERISTICS OF BIODIESEL FUELLED ENGINE

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Fossil fuels are the major sources of energy. But the stock of these fossil fuels is limited and the increasing demand is lowering these stocks rapidly. And also, the excessive use of fossil fuels is harming the environment. The burning of fuels causes the production of CO, CO₂, hydrocarbons, and NO_x. Biodiesel have shown a reduction in CO, CO₂, and hydrocarbons emission. But biodiesel have slightly lower performance and higher emissions of NO_x. Adding additives to biodiesel can solve this issue. Nanoparticle fuel additives have gained popularity among researchers because of their interesting properties. Nanoparticles have shown a reduction in emissions and an improvement in combustion characteristics of CI engines. In the present work, nanoparticles are used as additives in biodiesel. Nanoparticles have been added in the quantites of 50 ppm, 100 ppm and 150 ppm in pure biodiesel with the ultrasonication process, and B100Ce50, B100Ce100 and B100Ce150 blends have been prepared and tested for performance and emission characteristics. Blends of cerium oxide show better performance characteristics and lowered the emissions.

Keywords: Cerium Oxide; Biodiesel; Emission; Nanoparticle Fuel Additives; Performance characteristics

HIGHLY ECONOMICAL AND EFFICIENT POLYETHYLENE SEPARATOR FOR REDOX FLOW BATTERY

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Low-cost and negligible protonic resistance of the porous PE separator makes it a potential alternative to the high cost Nafion membrane. However, severe vanadium ion crossover through the micron size pores of the pristine PE separator needs to be addressed. In the present study, pores of the PE separator have been modified using the heat treatment technique with the separator exposing to high temperatures in the range of 130°C to 250°C. Separator characterization such as the FTIR, TGA, tensile strength, SEM, BET, ionic resistance and vanadium permeability are discussed in detail. BET results shows that the surface area and pore volume of the PE separator decreases with the increase in the temperature. Vanadium permeability of the pristine PE separator ($217 \times 10^{-7} \text{ cm}^2 \cdot \text{min}^{-1}$) reduced significantly to $9 \times 10^{-7} \text{ cm}^2 \cdot \text{min}^{-1}$ for the treated PE separator owing to decrease in the pore size. Consequently, the self-discharge time of the treated PE separators increased with the increase in the temperature. PE separator heated at the 170°C exhibited maximum self-discharge time of 94 h compared to the Nafion117 membrane (70 h) and pristine PE separator (24 h). As a result, the CE of the separator reached up to 98% at $80 \text{ mA} \cdot \text{cm}^{-2}$ owing to the decrease in the vanadium ion cross-over. VRFB cell equipped with the heat treated separator showed better capacity retention with a capacity fade rate of $0.259\% \text{ Ah} \cdot \text{cycle}^{-1}$ than the Nafion117 membrane ($0.549\% \text{ Ah} \cdot \text{cycle}^{-1}$). Overall, the results demonstrates that the modified PE separator is suitable for the VRFB system.

Keywords: Capacity retention; Heat treatment; PE separator; Redox flow battery; Vanadium permeability

TURING WATER INTO WEALTH: CONVERSION FROM SYZYGIUM CUMINI SEED POWDER INTO READY TO EAT NUTRITIONAL COOKIES

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Upcycling the fruit-based by-product has gained attention within the framework of circular economy, to reduce environmental landfill and resource waste. Using the tons/year of Jamun fruits entering into landfill, approximately 1/3rd of the *syzygium cumini* seeds are discarded which causes resource waste. Therefore, this research work aims to provide nutritious and ready-to-eat cookies out of *syzygium cumini* waste to cure diabetes and achieve net zero waste. In this context, *syzygium cumini* seed powder (SCSP) was incorporated into wheat, finger millet, and multigrain flour at 10% to formulate the cookies and compare with 100% of wheat, finger millet, and multigrain cookies as control. Besides, the formulated cookies were evaluated for the effect of incorporation of SCSP on the physical, proximate composition, sensory attributes, and storage study. In conclusion, the incorporation of SCSP into wheat, finger millet, and multigrain flour provides nutritious and ready-to-eat anytime healthy snack cookies which further contribute not only to minimize waste and preserve the environment but also recover nutrients and provide a variety of food products. Thus, the use of this agro-waste is a realistic example of upcycling and is technically feasible in providing healthy cookies.

Keywords: Circular economy, Waste to wealth, Sensory evaluation, Zero waste, Fruit by-product, *Syzygium cumini* seed powder.

OPTIMIZATION OF MICROWAVE PYROLYSIS PARAMETERS TO ENHANCE YIELD AND BET SURFACE AREA OF MSH CHAR USING RSM METHODOLOGY

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Mustard husk (MSH) is an efficient agricultural waste to be used in various energy storage applications. The feasibility of the MSH can be enhanced by converting it into biochar by using different conversion techniques. The present study uses microwave pyrolysis to convert MSH into char. Microwave pyrolysis process parameters highly influence the generation of char and its BET surface area due to its volumetric and rapid heating technique. Microwave heating and its end product depends upon various parameter. Different combination of process parameters produces different end products viz., biochar biofuel and syn-gas etc. The effect of microwave power (MWP), radiation time (RT), and nitrogen flow rate (NFR) was studied using central composite design. To optimize the process parameters for maximizing MSH char yield and BET surface area, Response Surface Methodology (RSM) was employed. The selected process variable was optimized with the intention to maximize both MSH char yield and MSH BET surface area. MWP shows prominent effect on MSH yield while NFR shows noticeable effect on BET surface area of MSH char. The study developed regression models for MSH char yield and BET surface area, which demonstrated high Fischer test values and low p-values, indicating their reliability and applicability. Analysing the experimental data using ANOVA provided the optimal process parameters to achieve a maximum MSH char yield of 51.78% and a BET surface area of 303.26 m²/g subsequently, the predicted results were validated, and it was found that the experimental data deviated only slightly, with a 5.99% difference in yield and a 6.34% difference in BET surface area from the predicted values. This confirms the reliability and accuracy of the developed empirical models, which can accurately predict MSH char yield and its BET surface area.

Keywords: Biomass, Microwave pyrolysis, Optimization, Process parameters, RSM

COMPARATIVE ANALYSIS ON THE EFFECT OF AERATION AND AGITATION ON QUALITY AND QUANTITY OF ALGAL BIOMASS FOR ITS FURTHER VALORIZATION TO VALUE-ADDED PRODUCTS

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In the last decade, socio-economic awareness has reached to its peak, which makes the world shift towards bioresources to minimize their dependency on organic and inorganic natural sources. Microalgae can be potential source of oils and other essential molecules that can be used as feedstocks in the production of biofuels and other high-value products, potentially becoming a significant renewable energy source. The objective of this research is to analyse the effect of intensive aeration and agitation on the cultivation of indigenous microalgae for its further biorefinery to extract various value-added products. In this research, low-strength domestic wastewater was used as the potential source of nutrients for algal cultivation having nitrates and total phosphorous around $21.6 \pm 0.39\text{mg/l}$ and $12.7 \pm 0.8\text{ mg/l}$, respectively. Indigenous algal consortia having concentration of 5% (v/v) was cultivated under the illumination of 6000 lux with 14:10 light - dark period at the temperature of $24^\circ\text{C} \pm 4^\circ\text{C}$ for 20days with two different approaches: one with agitation at 240 rotation per minute (rpm) and another with aeration at 2.0 litres per hour (lph). Microalgae was quantified by analysing the cell count with a haemocytometer under a microscope, optical density at 680nm, chlorophyll content every alternative day. Further the harvested algal biomass from both the approaches was analysed for its dry weight and lipid content. The results revealed that dry weight ($348.4 \pm 10.5\text{mg/l}$) and lipid content (39.7%) of aeration mediated microalgae cultivation was observed to be higher than agitation induced microalgae cultivation, which proves that aeration favours the microalgal growth.

Keyword: aeration; agitation; bioresources; feedstock; value-added products

GROUNDWATER QUALITY ASSESSMENT AND MANAGEMENT STRATEGIES IN MATHURA: INSIGHTS AND IMPLICATIONS

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Mathura, is a one of the oldest religious cities, attracting pilgrims from far places every year. Moreover, the increasing number of devotees and the resulting growth in urbanisation, industrialisation, and other human interventions have presented a significant risk to the quality and quantity of groundwater in the area. Mathura relies mostly on groundwater for drinking and irrigation. Hence, it is imperative to conduct monitoring and evaluation in order to enhance the quality of groundwater. In order to assess the appropriateness of groundwater in Mathura, UP, India for drinking and irrigation reasons, a total of 27 samples were taken from various places and analysed. The assessment of quality is conducted by estimating parameters such as temperature, pH, turbidity, EC, TS, TDS, TSS, total hardness, Ca^{2+} , Mg^{2+} , Na^{+} , K^{+} , Cl^{-} , SO_4^{2-} , NO_3^{-} , NO_2^{-} , NH_3 , Zn, Cu, Fe, PO_4^{3-} , F⁻, and total alkalinity. Cluster analysis (CA) was used to categorise 27 sample locations into two distinct clusters (low and high pollution) based on their shared features. The use of Discriminant study (DA) successfully identified six water quality measures that achieved a 100% accuracy rate. The introduction of TOPSIS included the use of entropy weights and rough set theory to provide a dependable and impartial assessment of the overall pollution levels at each sample point. The analysis of hydro chemical using Piper diagram indicates that NaCl, CaNaHCO_3 , NaHCO_3 , and Na^{+} , K^{+} are major water type dominant. CAI-1 and CAI-2 shows that 100% water samples reveal negative values, indicating an active ion exchange process between Na^{+} and K^{+} in water containing Ca^{2+} and Mg^{2+} from rocks. This research has the potential to provide valuable insights to the local government and decision-making authorities on the control and management of pollution, as well as the enhancement of groundwater quality protection and implementing efficient water resource management.

Keywords: Cluster Analysis; Groundwater; Hydrochemical Facies; Piper Diagram; Principal Component Analysis; TOPSIS;

USE OF SPENT BALLAST AS A COLUMN MATERIAL FOR ENHANCING LOAD CARRYING CAPACITY OF SOFT CLAY BED: NUMERICAL ANALYSIS

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One technique for stabilizing soft soil by using waste is the use of stone columns, often referred to as granular piles, which are typically used to boost slope stability and carrying capacity. Along with lowering the compressibility of loose and finely graded soils, it also hastens the consolidation effect by enhancing pore drainage. During seismic events, water pressure dissipation lowers the risk for soil liquefaction. Recognizable "natural" ground treatment technique or foundation system now in use is arguably the use of stone columns. Spent Ballast is a waste of various sizes are used to build stone columns. In this paper, a quantitative examination has been carried out to study the behavior of a virgin clay substrate in terms of spent ballast column installation, namely a regular stone column (OSC). Various parameters were considered for the study to analyze the behavioral changes following stone column placement. Also, a comparative inquiry was conducted to understand the working behavior of geosynthetic encased stone column (GESC). The geogrid-encased stone column (GESC) technology has been designed to enhance the load-carrying capacity of stone columns through boosting the confinement effect. The findings demonstrate that the geogrid encasement's additional confinement boosted the spent ballast column's rigidity and decreased the settlement of the soft ground. Additionally, it was noted that the GESC bulged precisely beneath the geogrid encasement's base. Even with partial encasement, the performance of GESC showed a noticeable improvement. A detailed comparison among virgin soft clay bed, OSC and GESC have been represented.

Keywords: Bearing capacity; Geosynthetics; Settlement; Soft clay bed; Spent ballast column.

APPLICATION OF DEMOLISHED CONSTRUCTION WASTE AS A COLUMN INFILL TO IMPROVE SOFT SOIL'S STRENGTH: NUMERICAL ANALYSIS

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In the field of civil engineering, sustainability can be attained in a number of ways, such as by using recycled building materials, and can help achieve a much wider range of sustainability goals, such as those pertaining to economic development, climate change, and other issues. Over the years, numerous studies have been carried out to improve soil strength. Various engineering characteristics of low shear strength soil are enhanced by the application of ground improvement techniques, such as the use of Portland cement, lime, fly ash, ground freeing, jet grouting, and vertical drains. An important consideration in the ground improvement can be done by the use of vertical columns which are now favored as a cutting-edge and contemporary method. In this study, vertical columns are made up of demolished building materials which is a waste material as compared to the conventional column. The column are also encased with natural geotextile in order to utilize it as an effort to use environmentally friendly materials and achieve sustainability. In this paper a numerical analysis is conducted to study the vertical settlement and lateral deformation of soft clay with ground improvement method (use of vertical columns encased in natural geotextile) and without any ground improvement method and further result has been discussed.

Keywords: bearing capacity; geotextile; ground improvement; reinforced soft soil; sustainability

UTILISATION OF FLY ASH AS A TOP LAYER OF FOUNDATION UNDER EMBANKMENT LOADING: NUMERICAL STUDY

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As an environment friendly and high-performing alternative to sandy soil, fly ashbased building materials have a lot of promise in the current scenario. A fine powder called fly ash is produced in thermal power plants when pulverized coal is burned. When this industrial waste is discharged into the atmosphere, serious environmental issues arise. The great majority of the material produced year was once stored in ash dams or other similar landfills, but in more recent times, a variety of methods have been created to enable the useful application of fly ash in construction industry. The goal is to identify new applications for fly ash that will boost its beneficial uses and lessen environmental pollution. This study offers a thorough analysis of pertinent research to assess the durability and settling of fly ash as a foundation material in order to promote sustainable development and the advantageous use of waste fly ash in the construction sector. The sandy soil layer at the top layer of foundation of an embankment serves as the drainage layer for the embankment. Fly ash is a material which is used in place of sandy soil in embankments as a drainage layer in our study. According to our research, fly ash settles at a rate that is around 22% slower than sand's. Similarly, this method can be used in the field analysis on a large scale.

Keywords: Drainage; Settlement; Soft soil; Sustainable material; Waste fly ash

Advancements in Energy Engineering (AEE)

PHASE CHANGE MATERIAL BASED PASSIVE COOLING OF A SINGLE POUCH LI-ION CELL UNDER DIFFERENT AMBIENT CONDITIONS

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The rise in greenhouse gas emissions and environmental pollution are contributing to global warming thereby emphasizing on the urgency to take initiatives for the production and adoption of electric vehicles (EVs) in transport sector worldwide. EVs predominantly employ the rechargeable lithium-ion batteries (LIBs) as a power source due to their superiority in high energy density and longevity. However, their large scale usage is limited by the huge heat generation during charging and discharging. This leads to an uneven temperature distribution inside battery which may result in catastrophic thermal runaway under extreme circumstances. Therefore, the implementation of an efficient battery thermal management system (BTMS) is imperative to maintain a safe working temperature (within 20-40°C) for avoiding such disasters. Our current work employs a single 3D pouch cell configuration incorporating passive cooling by the phase change materials (PCMs) of different melting temperature at the ambient temperatures of 300K, 305K, and 310K. We also analyzed the cooling performance at various convective heat transfer coefficients such as $h = 5, 10, \text{ and } 15 \text{ W/m}^2\text{K}$ with varying thicknesses (1, 2, and 3 mm) of PCM encapsulation over the cell. The thermo-coupled electrochemical problem integrating with PCM melting was tackled using the multi-scale multi-dimensional (MSMD-NTGK) model in ANSYS Fluent across high discharge rates of 3C, 5C, and 7C. The pertinent parameters such as the maximum cell temperature, cell voltage, PCM liquid fraction, and thermal uniformity inside the battery were examined. Notably, for the conditions as discharge rate 5C, ambient temperature 300K, and $h = 5 \text{ W/m}^2\text{K}$, the maximum temperature was substantially dropped by $\sim 18.2\text{K}$, 23.5K , and 23.7K by employing 1 mm, 2 mm, and 3 mm PCM (n-octadecane) thickness, respectively, when compared to the battery system without PCM. As observed, beyond a critical thickness, the PCM no longer reduced the cell temperature.

Keywords: Discharge rate; Heat generation; Pouch lithium-ion cell; PCM cooling, Ambient conditions.

EFFECTS OF SEGMENTATIONS IN COMPOSITE PHASE CHANGE MATERIAL ON MELTING PERFORMANCE OF TRIPLEX-TUBE THERMAL ENERGY STORAGE SYSTEMS

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A critical approach is essential to tackle the demand-supply discrepancy of energy requirements. Latent heat thermal energy storage (LHTES) can be an effective technology to eliminate the challenges of renewable energy such as intermittency and unpredictability. In this work, a numerical evaluation of the melting performance of phase change material (PCM) filled inside a triplex-tube latent heat storage unit has been investigated. To enhance the melting performance, the porous Cu metal foam was embedded inside PCM (termed as composite PCM). Alternative segments of pure PCM and composite PCM have been allocated in such a way that both the pure PCM and composite PCM occupy the equal annular area (i.e., equal volumes). Influence of increasing number of segments and heterogeneity of metal foam have been delineated on the melting rate, complete melting time, and thermal energy storage rate. The comparisons have been drawn with reference to the limiting cases of complete filling (100%) of pure PCM and composite PCM together with the additional effects of metal foam anisotropy and random porosity distribution. The results show that the model containing 32 segments with alternate allocations of PCM and composite PCM has a faster melting rate than other models. However, the model filled with composite PCM only (full porous) has the most rapid melting rate but with the low thermal energy storage density. Moreover, the segmentation of composite PCM helps in increasing the average thermal energy storage rate thereby reaching up to $\sim 70\%$ of the TES rate corresponding to the full composite PCM model. With 16 segments of metal foam, the full melting time reduced by $\sim 22\%$ with respect to the case with one segment only. Lastly, the anisotropic metal foam predicts an augmented melting rate over both the random and isotropic distributions of metal foam.

Keywords: Energy storage; Phase change material; Triplex-tube; Heterogeneous Metal Foam; Segmentation.

LIFE CYCLE ASSESSMENT OF BIODIESEL PRODUCED FROM CEREAL BYPRODUCTS

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This study conducts a comprehensive life cycle assessment of biodiesel production from crop by-products, specifically focusing on Dried Distiller's Grains (DDGS). Methodologies encompassing data collection, calculations, and key theoretical concepts are detailed for the life cycle assessment. The process involves fermenting DDGS with yeast to produce biodiesel. DDGS, a by-product from ethanol plants, exhibits 2 to 3 times higher nutrient content than corn. The production process entails fermenting corn with yeast to yield ethanol, followed by processing the ethanol into Olefins through dehydration. Ethylene is then separated from water molecules, undergoes dimerization to form propylene, and is further polymerized to create a gasoline blendstock. Notably, using sugarcane, corn, or other biomass as raw materials renders the resulting fuel a renewable energy source. This study sheds light on the environmental implications and sustainability aspects of biodiesel production from crop by-products, providing a detailed insight into the entire life cycle assessment process.

Keywords: Lifecycle; Energy Analysis; Biodiesel; Cereal byproducts; Crops.

FABRICATION AND CHARACTERIZATION OF SHAPE-STABLE COMPOSITE PHASE CHANGE MATERIALS SUPPORTED BY BIOMASS-DERIVED ACTIVATED CARBON FOR ENERGY STORAGE

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Phase change materials (PCM) have emerged as a potential solution for addressing energy imbalance and promote sustainable energy storage system. Paraffin wax is the most common and efficient PCM among other organic PCMs exhibiting high latent heat of energy but their application is limited by the incorporated leakage issue. Herein porous activated carbon derived from easily available groundnut de-oiled cake (GDO) is a promising support matrix for the improvement in shape stability and thermal conductivity. In this work a novel shape stable composite PCM is developed by using GDO derived activated carbon as a support matrix for paraffin wax. The activated carbon was synthesized by carbonization followed by chemical activation with zinc chloride ($ZnCl_2$). The paraffin wax was successfully incorporated into pores of activated carbon by a vacuum impregnation method. The composite was prepared by specific composition of PCM (paraffin wax), support matrix (activated carbon) and conductive material (graphene nanoplatelets) in an optimized ratio of 9:0.8:0.2, respectively for effective composite design. The chemical compatibility of paraffin wax/activated carbon was characterized by FTIR, thermal performance by DSC and surface morphology by SEM, BET and XRD. The characterization shows the enhancements in shape stability and thermal conductivity of composite PCM. Composite with varying composition of paraffin wax to activated carbon ratio was prepared ranging from 90 wt% to 10 wt%. The composite with 80 wt% paraffin wax shows enhanced thermal conductivity and excellent thermal stability as compared to pure paraffin wax.

Keywords: Graphene nanoplatelets; Groundnut de-oiled cake; Paraffin wax; Support matrix; Vacuum impregnation.

EFFICIENT THERMAL MANAGEMENT SOLUTIONS FOR HIGH-SPEED MACHINING USING PELTIER-BASED SPINDLE COOLING AND AIR- COOLING SYSTEMS

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In high-speed machining spindle thermal management system is important to improve the performance of the high-speed spindle. Also, in high-speed machining thermal management at the tool workpiece interface plays an important role in improving the tool life and workpiece surface characteristics. Hence this research work focuses on the development of a thermoelectric-based cooling system for high-speed spindle for set temperature and a thermoelectric-based air coolant system for temperature control during micromachining is capable of controlling air temperature -10°C.

Keywords: High-speed machining; Peltier device; Pump; Thermal management system; Temperature sensor.

INVESTIGATION OF SHUNT ACTIVE POWER FILTER FOR POWER QUALITY IMPROVEMENT IN DISTRIBUTION SYSTEM

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In distribution systems, the load has been a unexpected increase or decreases and it is like as nonlinear loads so the load draw non-sinusoidal currents from the utility mains and causes the reactive power load, harmonics and high neutral currents that occurs the power quality problem in electrical distribution systems. Most power quality problem created in distribution systems are due to the nonlinear load and fast switching of power electronic devices. Shunt active filter based on current controlled pulse width modulation (PWM) converters are seen as a most viable solution. In this paper, a shunt active power filter (SAPF) with PI controller is used. Current controlled voltage source converter topology is used in this SAPF. The quality and performance of SAPF depend on design of power converter, DC voltage control strategy, the method implemented to generate reference command and generation of switching pulses. For reference command generation unit Active Power Component Theory (APCT) is used and for DC voltage control PI controller is used. A hysteresis current controller is used to generate the switching pulses. The detailed simulation study of work has been carried out in MATLAB/Simulink environment.

Keywords: APCT; Distribution System; Power Quality; PWM; SAPF.

EFFECT OF CERIA MORPHOLOGY ON HYDROGEN PRODUCTION VIA METHANE STEAM REFORMING FOR MEMBRANE REFORMER

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Membrane reformers emerge as an efficient technology for 'on-site' production of ultra-pure hydrogen. Hydrogen is a potential energy carrier in comparison to the conventional fuels due to its high energy content. Natural gas is an attractive source for the 'on-site' production of hydrogen by using membrane reformer due to its low cost. However, such reformers are not well studied and high temperature operation of natural gas steam reforming makes the integration difficult. Further, the main product of natural gas steam reforming is CO and H₂ in which CO has inhibition effect on membrane separation process. Therefore, it is vital to synthesize a low temperature and low CO selective catalyst for a suitable integration and long-term operation of membrane reformer. Nickel-based catalyst is widely used for methane steam reforming due to its low cost and high catalytic activity. Commonly, alumina is used as a support for methane steam reforming due to high surface area. However, it enhances the side reactions due to its acidic nature. Therefore, researchers have used different supports like CeO₂, ZnO, ZrO₂, MgO, etc. Amongst these CeO₂ is widely used as it mobilizes the lattice oxygen and reduces the coke formation and CO selectivity. Though several studies are reported on CeO₂ based support, the effect of CeO₂ surface morphology is not studied well. In the current work, CeO₂ of different shapes (cube and rod) are synthesized. The complete characterization of the support was performed by using BET, XRD, SEM, and TEM. Thereafter, Ni metal is doped on the ceria support. The effect of support shape on methane steam reforming is performed at different temperatures in a packed bed reactor. For each condition conversion, hydrogen selectivity, CO selectivity and hydrogen yield are calculated. The results show the CeO₂ morphology has considerable effect on CO selectivity and hydrogen yield.

Keywords: catalyst; hydrogen production; methane; shape; steam reforming.

FORCED CONVECTION HEAT TRANSFER FROM CONFINED CIRCULAR/ SEMI-CIRCULAR HEATERS AND COOLERS WITH VARIOUS ORIENTATIONS

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The combined implementation of porous medium and hybrid nanofluid with heaters and coolers can be an effective technique to improve the efficiency of several electric equipments. In this regard, the present study has been conducted to analyze the forced convection heat transfer of Al₂O₃-CuO water-based hybrid nanofluid in a porous channel with pairs of heaters and coolers of various shapes. The circular and semi-circular heaters and coolers with distinct orientations have been considered. The Peclet number ($25 \leq Pe \leq 200$), Darcy number ($10^{-6} \leq Da \leq 10^{-1}$), porosity ($0.1 \leq \varepsilon \leq 0.9$) and volume fraction of hybrid nanoparticles ($0.02 \leq \phi \leq 0.08$) are chosen as the governing parameters for flow and heat transfer. The acquired results exhibit that the heat transfer from the heaters and coolers enhances by decreasing ε and Da for all the cases and values of Pe and ϕ . The lowest heat transfer has been obtained by the circular heaters and coolers (case 1). The higher heat transfer is observed in two cases when the flat surfaces of heater/cooler are perpendicular to the flow direction (case 2) and the flat surfaces are along the flow direction (case 4). Case 4 shows the minimum drag coefficient for all the values of the operating parameters. Moreover, it also exhibits the maximum heat transfer at the highest values of the governing parameters.

Keywords: Darcy number; nanoparticles; Nusselt number; Peclet number; porosity.

THERMODYNAMIC ANALYSIS OF SCO₂ RECOMPRESSION CYCLE FOR CRITICAL ANALYSIS OF RECOMPRESSION FRACTION AND PRESSURE RATIO TO OPTIMIZE PRESSURIZED WATER REACTOR PLANT EFFICIENCY.

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Supercritical carbon dioxide recompression cycle proved to be highly efficient cycle compared to different cyclic configuration. Numerous studies prove recompression cycle provide most acceptable results with different heat source with different plant complexity. This paper aims to provide the critical analysis of optimizing parameter for pressurized water reactor power plant. Study is focused to analyse recompression fraction, pressure ratio dependency on cycle efficiency. Operating condition are taken from established pressurized water reactor plant to provide realistic platform to thermodynamic analysis. This paper will define the changing pattern and dependency of operation parameters on source and sink temperature. Study is limited with the used of industrial turbomachinery for the ease of defining standard equipment efficiency and related data. Results to this analysis helps to provide basis to design plant machinery for higher thermal efficiency for recompression cycle. Results and findings of this analysis indicates that operational parameters have direct dependency on the inlet temperature of recuperator. Software tool Aspen hysys is used for thermodynamic evaluation and to represent relational charts.

Keywords: Power generation; Turbomachinery; Recuperator; Aspen Hysys; Flow parameters.

Materials in Engineering (MAT)

EFFECT OF SILICATE UNITS ON OPTICAL AND STRUCTURAL PROPERTIES FOR VARIOUS AUTOMOTIVE WINDSHIELD GLASSES

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The Windshield glasses of the Maruti Alto 800 (A800), Hyundai i20 (Hi20), and Maruti Suzuki Eeco (MSE) are taken from an automotive repair shop to investigate their structural and optical properties to check their durability. Fourier transform infrared (FTIR) was used to study the fundamental mode of vibrations and structural units present in available windshield glasses. The UV-visible spectroscopy was used to determine the transparency of these windshield glasses. The Makishima-Mackenzie model has been used to study mechanical properties such as elastic modulus (E_m), bulk modulus (B_m), shear modulus (S_m), longitudinal modulus (L_m), hardness (H), and Poisson ratio (σ). It is found that the structure of glass A800 is highly depolymerized due to the presence of the most prominent Q2 unit which leads to the lowest hardness and constant transparency throughout the variable range as compared to other windshield glasses.

Keywords: Windshield glasses; Hardness; Transparency; FTIR spectroscopy.

AGRO-FOOD WASTE-DERIVED BIOGLASSES: ENGINEERED STRUCTURES FOR OPTIMAL BONE REGENERATION

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The hybrid resources, including agro-food ashes and conventional chemicals, are used to synthesize the bioglasses using the melt-quench technique. Before in-vitro testing, the glasses undergo various characterization techniques to get an idea about their suitability as bioactive materials. The substitution of MgO for CaO in these glasses modifies their structural, physical properties and other properties in such a way that it reduces the bioactivity of the glasses. Notably, the introduction of MgO gradually decreases the leaching of ions from the glass samples, indicating slow dissolution in the SBF solution. Utilizing biowastes addresses waste management-related problems and boosts eco-friendly and cost-effective approaches to synthesize diverse bioproducts.

Keywords: Agro-food waste, Bioactive glasses, XRD, in-vitro testing, dissolution, Hydroxyapatite.

SYNTHESIS OF COAL-DERIVED GRAPHENE OXIDE AND ITS APPLICATION IN GLASS-FIBER BASED POLYMER COMPOSITES FOR IMPROVED MECHANICAL PERFORMANCE

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The primary objective of our study was to compare and comprehend the role of different precursor-based GO nanofillers in enhancing the mechanical behavior of the E-glass fiber reinforced epoxy resin composites (EGFPs). To this end, we dispersed varying amounts of GO derived from bituminous coal (BC-GO), and graphite (Gr-GO) into an epoxy resin matrix. The mixture, after homogenization, sonication and mixing with a hardener, was combined with E-glass fiber (EGF) mats to fabricate nanocomposites using vacuum-assisted resin infusion molding technique. The results show a significant improvement in the mechanical properties, including tensile strength, flexural strength, and impact strength of the EGFP composites upon the addition of both the nanofillers. However, the most notable improvements were observed in the BC-GO reinforced EGFPs at 0.25 phr loading of BC-GO. These were, 38.9% in flexural strength, 22.9% in tensile strength, and 21.6% in impact strength. The corresponding percentage were quite higher than those for Gr-GO reinforced EGFPs. XRD, SEM, TEM and FTIR analyses were conducted to explain the differences observed with different GO precursors. The indication is that the presence of more abundant functional groups in BC-GO as compared to those in Gr-GO, and their interaction with the epoxy resin and hardener contributes to the observed improvement in mechanical properties for the BC-GO composites. The enhanced bonding facilitates efficient stress/load transfer from the epoxy matrix to the GO reinforcement, thus resulting in a significant increase in the mechanical performance of the composites. Our findings underscore the potential of BC-GO as effective reinforcements for polymer composites for enhanced mechanical properties, including high strength, lightweight characteristics, and cost-effectiveness, making them desirable for deployment in various sectors such as aerospace, automobile, marine, sports equipment, and defence.

Keywords: Coal; Graphene oxide; Graphite; Mechanical properties; Vacuum-assisted resin infusion molding.

EFFECT OF SUPERPLASTICIZER DOSAGE ON FRESH AND MECHANICAL PROPERTIES OF BLENDED SELF-COMPACTING GEOPOLYMER CONCRETE

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Self-Compacting Geopolymer Concrete (SCGPC) is a sustainable approach to concrete production that incorporates waste materials as binding agents, contributing to environmental consciousness within the building sector. The main aim of this study is to optimize the dosage of a superplasticizer, a crucial constituent that substantially influences the rheological and mechanical properties of concrete. The present study meticulously examines the effects of varying superplasticizer dosages (3%, 5%, and 7%) on different percentages of binder content. Due to the absence of recognized design standards for self-compacting geopolymer concrete (SCGPC), a rigorous approach was employed to develop meticulous mix designs by amalgamating the specified components. Fifteen distinct formulations of self-compacting geopolymer concrete (SCGPC) were prepared. All samples were treated with an identical activator solution of neutral grade. The solution-to-binder ratio was maintained at a constant range of 0.5 to 0.55. The composition of the mixture included 60% GGBFS (ground granulated blast furnace slag), 20% fly ash, and a binder content of 400 kg/m³. Additionally 15% water was added to the mixture. Numerous rheological tests were conducted on the recently prepared concrete mixture. The tests encompassed slump flow, T_{50cm} slump flow, V-funnel for filling ability, J-ring for passing ability, and V_{5MINUTES}-funnel for segregation resistance. The results suggest that all recently acquired traits adhere to the guidelines set forth by EFNARC, hence demonstrating the effectiveness of SCGPC. The gradual increase in the quantity of superplasticizer adversely affects the mechanical characteristics of self-compacting geopolymer concrete (SCGPC).

Keywords: Compressive Strength, fly ash, Granulated Blast Furnace Slag (GGBS), Micro Silica (MS), Rice Husk Ash (RHA), Single Activator Solution.

EFFECT OF NANO-MODIFIED EPOXY ON THE BOND BEHAVIOR OF OLD-NEW CONCRETE INTERFACE

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In strengthening or retrofitting concrete structures, the transition between old to new concrete is the most susceptible zone, often leading to interface or adhesive failures in the composite. Epoxy is a widely used adhesive for the bonding of old and new concrete however, its low strength and inadequate roughness of concrete surface hamper the effective composite action and result in poor concrete-to-concrete bonding. Further, graphene oxide (GO) modified epoxy showed an improvement in the bond properties at very low concentrations with a plain concrete interface. Moreover, the formation of grooves between old and new concrete interfaces solely increased the bond strength under various loadings. However, the effect of other nanofiller-modified epoxy along with the grooved concrete surface on the bond performance of old-new concrete has not been fully studied. Hence, this study investigates the bond performance of old-new concrete under the effect of four nano-modified epoxies, with GO, reduced GO, coal-derived GO, and MWCNT as nanofillers having a concentration of 0.05 wt% of epoxy resin. The nano-modified epoxy is applied over plain and grooved concrete surfaces, with exposed sound concrete at the interface in both cases. The FESEM analysis confirmed the uniform distribution of the nanofillers within the epoxy and demonstrated the formation of the bond between cement hydrates and the nanofiller at the interface. The bond strength is assessed by conducting split tensile, bi-surface shear, and slant shear tests. The GO-modified epoxy on grooves shows superior performance in the form of strength enhancement and required failure pattern, i.e., failure in either of the concretes. Thus, epoxy modified using 0.05 wt% GO proved to be a robust adhesive for strengthening, retrofitting, or repairing the existing concrete structures with fresh overlay concrete.

Keywords:

Bond strength; Carbon nanotubes; Failure patterns; Graphene derivatives; Grooves; Nano-modified epoxy.

EXPERIMENTAL INVESTIGATION OF SUSTAINABLE LIGHTWEIGHT PRECAST STRUCTURAL ELEMENTS

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Concrete stands as the most extensively employed synthetic material in diverse construction projects. The current research primarily centers on the advancement of sustainable lightweight concrete. The construction sector's rapid consumption of raw materials has led to a persistent shortage of building materials and consequential environmental degradation. Typically, coarse aggregates account for 30–45% of the concrete volume. This sustainable lightweight concrete is achieved by incorporating Agro-Industrial waste. In the precast and construction industries, a key challenge lies in the dead weight of structural members and the associated environmental impact from cement production. To alleviate the structural weight and reduce environmental harm, Palm Oil Shell is introduced as a replacement for coarse aggregates. Additionally, to mitigate greenhouse gas emissions from cement production, Palm Oil Fuel Ash and Ground Granulated Blast Furnace Slag are introduced as partial replacements. These replacements are implemented at levels up to 50% for both coarse aggregates and cement. Furthermore, 0.5% ECR-Glass fibers are added, resulting in the production of M30 Grade Concrete for all replacements. Mechanical properties, durability, microstructural characteristics, flexural behavior of beams, slabs, and portal frame actions (both monolithic and precast frames) are thoroughly examined for all concrete mixes. These investigations are compared with conventional concrete, considering various curing periods. Based on the outcomes of this study, it can be inferred that replacements are successfully implemented up to 30% in both coarse aggregates and cement. The utilization of Palm Oil Shell as a coarse aggregate and GGBS, POFA as cement substitutes holds promise for reducing environmental pollution. Embracing these waste materials as alternative construction materials in the building industry has the potential to decrease energy consumption and CO₂ emissions.

Keywords: Durability studies, Flexural behavior, Mechanical properties, Palm Oil Shells, Palm Oil Fuel Ash. Precast Structural Elements, and Portal frame actions.

STUDY OF CHEMICALLY TREATED PET FIBER WITH MICRO SILICA IN M25 GRADE CONCRETE MIX

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The PET fibers are non-biodegradable waste which contribute to huge environmental pollution. Solution of this problem lies in recycling the plastic wastes as PET fibers in concrete. In this study, the workability of concrete is increased by treating the PET fibers with NaOH solution and partial replacement of cement with micro silica is done, since PET fibers have lower workability with concrete. Also, micro silica helps in providing early strength to the concrete and improves its rheological characteristics. The studies also represent that the tensile strength of concrete mix also gets improved due to incorporation of fibers. In this paper, we have analysed the impact of varying percentage of plastic fibers and micro silica on compression strength of M25 concrete mix. Also, chemically treated plastic fibers mixed with M25 grade concrete are analysed for their plastic workability and adhesive properties. The analysis results represent that an increase in the compressive strength occurs with increase in micro silica content to 15% by weight of cement, as micro silica is a highly reactive pozzolana and provides better bonding due to smaller particle size. It was also deduced that increase in content of plastic fibers (whether untreated or NaOH treated fibers) increases the compressive strength of concrete, but the improvement in strength is higher for chemically treated solution as it provides better interlocking with aggregates because of increased roughness of fibers. It was also found that the workability of concrete mix is reduced by increase in percentage of plastic fiber but it adheres the bond matrix of concrete mix.

Keywords: Chemically treated fibers; compressive strength; concrete; micro silica; PET fibers.

FABRICATION AND CHARACTERIZATION OF GRAPHENE-REINFORCED MMC BY FSP FOR LIGHTNING STRIKE PROTECTION OF AIRCRAFT

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Lightning strike protection (LSP) is a critical concern for the aviation industry. Every aircraft has to deal with the possibility of lightning striking during its takeoff and landing. The industry invests a massive amount of money in damage repair and protection systems. In this study, the requirement of lightweight material with high mechanical, electrical and conducting properties, was fulfilled by fabricating reduced graphene oxide (rGO) reinforced metal matrix composite (rGO-MMC) of aviation grade AA7075 plates via solid state friction stir processing (FSP) technique. The suitability of the fabricated rGO-MMC for its intended application of LSP was checked through a detailed characterization of its mechanical properties and metallurgical features. Textural change with respect to the different zones i.e., thermo mechanical heat affected zone, heat affected zone and nugget zone is represented by optical micrographs. Grain's internal structure including grain shape and size, grain boundary distributions, grain misorientation angle, etc. Is explored with the help of electron backscattered diffraction technique. X-ray diffraction and Energy-dispersive X-ray spectroscopy techniques are used in the identification of elemental precipitates and phases. Moreover, the microstructural evolution, grain refinement and dynamic recrystallization are discussed at different proportions of reinforcement in MMC. As compared to the parent material, the MMC showed a significant increase in mechanical and physical properties like microhardness, tensile strength, impact energy and electrical conductivity up to the required specifications of the aviation industry. This research paves the path of green technology involvement in solid state welding and opens up new avenues for sustainable manufacturing processes.

Keywords: AA7075; aviation industry: dynamic recrystallization: FSP: Graphene.

STATE-OF-THE-ART REVIEW ON RHEOLOGY TYRENE-BUTADIENE-STYRENE MODIFIED BITUMEN

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Rheological characterization of binder has been proved as a preliminary tool to ensure the post-construction pavement performance. While modifying binders, the rheological studies are extensively carried out on aged samples to predict the modified binder's performance for field application. Among polymeric modifiers, Styrene-Butadiene-Styrene (SBS), is the most abundantly used elastomer to enhance the engineering properties of bituminous binder. Though numerous articles on SBS modified binder are available, the literature still lacks in a comprehensive review on rheological characteristics of SBS modified binder. At present, a thorough review on the rheological characteristics of SBS modified binder has been offered. The research findings are grouped for different tests such as frequency sweep test, temperature sweep test, multiple stress creep & recovery test, linear amplitude sweep test, bending beam rheometer test and some other specialized tests. Impact of SBS dosage, bitumen's grade, properties of SBS and its co-additives, and mixing parameters have been analyzed. Applicability of SBS modified binder has been found evident in high temperature regions while in low temperature areas it is still questionable. Supplementation of nanomaterials has reflected an improvement in performance parameters as well as aging behavior of modified binder. The necessary inferences drawn in this review provides the recommendations for field application and highlights the existing research gaps for the peer researchers.

Keywords: modified binder; nanomaterial; performance parameters; rheometer; SBS polymer.

VARIATION OF CHIRALITY CAUSE SIGNIFICANT THERMODYNAMIC CHANGES IN MULTIWALLED CARBON NANOTUBES: A COMPUTATIONAL STUDY

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The nanoscience and nanotechnology is playing vital role in each and every interdisciplinary fields. The product made by nanomaterials are trending now a day. There are huge applications of nanomaterials in agricultural, pharmaceutical, chemical, cosmetics, petrochemical, automobile, etc. including all other allied industries. In the same sequence the nanocomposites are the multiphase solid materials, which represents many interdisciplinary applications. Therefore, it is important to understand the scientific and technical aspects of nanocomposites. Hence, we have chosen carbon nanotubes (CNTs) which belongs to the same family of nanocomposites for our study. It is frequently used in many significant industrial applications including waste water treatments. In this study, we have performed a thermodynamic analysis on multi walled carbon nanotubes (MWCNTs) as it is more durable and efficient in comparison to single walled carbon nanotubes (SWCNTs). We have used molecular simulation techniques in combination of "Muller-Plathe" method followed by LAMMPS open source software. Based on the generated data (i.e. thermal conductivity) for differently oriented and structured MWCNTs [i.e. Chirality (5,5), Chirality (10,10), and Chirality (19,10)] we have performed a significant analysis and found that, the thermal conductivity increases while increasing the length of MWCNTs for each orientation. These are interesting results and able to confirm us that, the MWCNTs with high thermal conductivity are more efficient and durable for multiple applications. Also, the changes (i.e. increments) in thermal conductivity are irrespective of Chirality. This is an interesting finding and may be helpful to design and develop the qualitative MWCNTs for desirable purpose.

Keywords: Chirality; Thermal Conductivity; Multi Walled Carbon Nanotubes (MWCNTs).

STYRENE-BUTADIENE-STYRENE POLYMER MODIFIED BITUMEN: A REVIEW ON COMPATIBILITY AND PHYSICO-CHEMICAL PROPERTIES

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The triblock elastomeric copolymer Styrene-Butadiene-Styrene (SBS) improves the engineering properties of bitumen and the performance parameters of flexible pavements. Many researchers have conducted extensive investigations to determine the characteristics of SBS modified bitumen. Although research investigations have been conducted worldwide, the literature lacks a comprehensive review to provide the up-to-date research status in the field. The current review summarises the research findings on the compatibility, stability, physical properties, microscopic characterization and chemical characterization of SBS modified bitumen. Quantitative analysis of physical test results concerning major pavement distresses shows improvement, particularly in high-temperature zones. In addition, physical and chemical characteristics of aged binder have been reviewed. Ageing indices (residual penetration ratio, change in softening point, viscosity ageing ratio, residual ductility ratio) didn't show any consistent trend, which indicates that SBS modified bitumen further require additives to substantially improve ageing deterioration. The review highlights the need for studies to explore suitable nano-additives to improve the storage stability at high temperature and the performance of aged SBS modified bitumen. This review has drawn essential conclusions and highlights existing research gaps for peer researchers and field engineers.

Keywords: chemical characterization; empirical tests; microscopic characterization; SBS modified binder; stability.

DEVELOPMENT AND THERMOMECHANICAL ANALYSIS OF HYBRID EPOXY NANOCOMPOSITE REINFORCED WITH REDUCED GRAPHENE OXIDE AND HNT

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This study focuses on the synthesis and characterization of a novel hybrid nanocomposite by incorporating Halloysite Nanotubes (HNT) and reduced Graphene Oxide (rGO) as reinforcing materials within an epoxy matrix. The objective is to exploit the unique mechanical, thermal properties of HNT and rGO, creating a synergistic enhancement in the overall performance of epoxy-based composites. Epoxy matrix, renowned for its exceptional adhesion, low shrinkage, and chemical resistance, is often limited by its inherent brittleness. To overcome this limitation, nanofillers such as HNT and rGO are introduced to enhance mechanical strength and extend the range of epoxy applications. Halloysite Nanotubes (HNT) with their natural tubular structure and high aspect ratio, serve as effective reinforcing agents and also the Reduced Graphene Oxide (rGO) widely recognized for improving mechanical properties and thermal stability which is derived from graphene. The oxygen-containing functional groups on rGO sheets facilitate strong interactions with the epoxy matrix, enhancing the overall material properties. This study aims to synergize the unique attributes of GO with HNT to achieve a well-balanced enhancement in the resulting nanocomposite. Results indicate a significant improvement in mechanical properties, with increased tensile and flexural strength compared to pure epoxy. The enhanced thermal stability, as revealed by thermogravimetric analysis (TGA), is attributed to the synergistic effects of HNT and GO, manifesting in an elevated decomposition temperature. Microscopic analysis via scanning electron microscopy confirms the homogeneous dispersion of HNT and GO within the epoxy matrix, emphasizing the importance of proper dispersion for achieving the desired property enhancements. This nanocomposite contributes valuable insights into tailoring nanocomposites for specific applications, showcasing improved mechanical strength, thermal stability.

Keywords: Halloysite nanotubes (HNT); Reduced graphene oxide; mechanical properties, tensile strength, flexural strength, scanning electron microscopy.

COMPARATIVE STUDY ON VARIOUS LENGTH SCALE AND CHIRALITY OF SINGLE WALLED CARBON NANOTUBES BY USING MOLECULAR SIMULATION APPROACH

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The carbon nanotubes (CNTs) are a unique class of nanomaterials with exceptional mechanical, electrical, and thermal properties. They have the potential to revolutionize a wide range of fields, including electronics, energy, and medicine. However, the full realization on their potential requires a better understanding on their fundamental properties and behaviour. In this study the CNTs are constructed at three different lengths i.e. 5nm, 10nm, 15nm by using Visual Molecular Dynamic (VMD) software with a chirality (5,5), (10,10), (15,15) of each length scale. The overall construction is done at two different aspect ratios (length to diameter ratio) which chosen as 9.1 and 4.5 respectively. The Molecular Dynamic (MD) simulation were performed on these constructed structure of CNTs by employing AIREBO potential model on open source software "LAMMPS" at 300K temperature. The canonical ensemble were chosen for NVT system which were maintained by Nose Hoover thermostat. From the extracted simulation results a comparative study were conducted to examine the properties of carbon nanotubes (CNTs) under varying CNTs lengths with chirality, and aspect ratio conditions. This investigation was aims to uncover the intricate relationship between these factors and the resulting properties of CNTs. The overall predicted results are compared with the literature and found consistent. Overall this comparative study on CNTs may provide valuable insights into the relationships between different CNTs properties followed by their structure. This study may further be applicable and helpful to design unique CNTs with specific properties for desired applications.

Keywords: Chirality; Length Scale; CNTs; Molecular Dynamics; NVT.

ANTIBACTERIAL ACTIVITY INVESTIGATION OF ZNO-AG NANOCOMPOSITES PREPARED BY THERMAL DECOMPOSITION METHOD AGAINST *E. COLI* AND *S. AUREUS*

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The thermal decomposition method was used to prepare twelve different types of ZnO-Ag nanocomposites. First, three different ZnO nanoparticles were synthesized using three different methods: solid-state thermal decomposition, homogeneous precipitation, and wet chemical. Subsequently, Ag nanoparticles were deposited on the surface of the ZnO nanoparticles by thermal decomposition of silver acetate in diphenyl ether. ZnO-Ag nanocomposites were characterized by XRD, FE-SEM, EDXA, TEM, SAED, TGA, FT-IR, and DRS. XRD results confirm the presence of Ag nanoparticles in all the ZnO-Ag nanocomposites. FE-SEM and TEM results confirm the presence of Ag on the surface of the ZnO nanoparticles in the ZnO-Ag nanocomposites. DRS results show characteristic surface plasmon resonance absorption due to Ag nanoparticles in the ZnO-Ag nanocomposites. Finally, the antibacterial potential of ZnO-Ag nanocomposites against *E. coli* and *S. aureus* was investigated.

Keywords: ZnO-Ag nanocomposites; thermal decomposition; antibacterial activity; *E. coli*; *S. aureus*.

SURFACE MODIFICATION OF TITANIUM ALLOY USING GRAPHENE OXIDE FOR BIOMEDICAL APPLICATIONS

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Titanium and its alloys are widely used in biomedical devices such as artificial joints, spinal fixators, screws, and bone fixators, attributed to their impressive biocompatibility and excellent corrosion resistance. Titanium does not exhibit antimicrobial properties; however, bacterial films developed on the implant surface, leading to infection and implant failure. Surface modification of titanium-based implants with nanomaterials can be an effective approach to induce an antibacterial effect on the implant surface. In this work, graphene oxide (GO) was employed as a coating material on the surface of titanium alloys (Ti6Al4V) to enhance mechanical characteristics and biocompatibility. The titanium surface was chemically functionalised using an APTES (3-aminopropyltriethoxysilane) transition layer and DA (Dopamine Hydrochloride) as an adhesive layer to coat GO on metal surface using the Dip coating method. GO was synthesized using the modified Hummer's method. The degradation rate of the material was analysed for 21 consecutive days in phosphate buffer saline (PBS) solution under controlled conditions. The structural and physiochemical properties of synthesised GO and graphene-coated plate were investigated using FESEM, EDS, XRD, Raman, and Fourier transform infrared spectroscopy (FTIR) techniques. SEM images revealed thin sheets of graphene oxide, while D and G Raman vibrational bands were observed at 1350 and 1580 cm^{-1} , respectively. Furthermore, the effect of surface microstructure and GO coating on the biofilm were analysed. The results showed that GO coated surface exhibited a lower degradation rate but higher antibacterial activity than uncoated and chemically functionalised titanium surface. These results demonstrate nano-modifications have potential for enhancing the surface bioactivation of materials that can be used on implant surfaces to reduce surgical site infections in biomedical applications.

Keywords: biofilm; coating; degradation; Graphene oxide; Ti6Al4V surface.

SOFT, FLEXIBLE AND SKIN-COMPATIBLE ELECTRODES FOR MONITORING ELECTROPHYSIOLOGICAL PARAMETERS

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Heart diseases, encompassing a wide array of conditions that affect the heart's structure and function, stand as a formidable global health challenge. Diagnosing heart disorders involves a comprehensive approach, drawing upon a range of methods and technologies, for example, Electrocardiography (ECG) records the heart's electrical activity, aiding in the identification of irregular rhythms or structural abnormalities. One of the most essential components of ECG is the electrodes, which are strategically placed on the skin to detect and record the heart's electrical impulses. The present work reports the fabrication of textile-based ECG electrodes, where either the textile material is immersed into a conducting polymer solution or conducting threads are woven into the textile. Microfibre cloth and cotton-based fabric have been used as substrates and coated with three different types of materials - Carbon, Conducting Polymer, and Conducting Polymer mixed with ionic liquid. All these materials are skin-compatible, easy to fabricate as well as inexpensive. The electrodes thus fabricated have been tested for their skin-impedance values, where the impedance values $< 10\text{ K}\Omega$ confirm the suitability for ECG measurements. Moreover, the electrodes have been successfully demonstrated to be employed for ECG monitoring.

Keywords- Soft, skin, electrode, textile, ECG.

CORROSION INHIBITION OF MILD STEEL USING *AZADIRACHTA INDICA* LEAF EXTRACT IN AN ALKALINE MEDIUM CONTAINING CL-IONS

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Mild steel (MS) corrosion is a very serious concern in manufacturing industries. Neem (*Azadirachta indica* – AZI) leaf extract was used as a green corrosion inhibitor (GCI) for MS with 1 wt% calcium hydroxide and 0.5 wt % sodium chloride (alkaline medium) and studied via weight loss technique, potentiodynamic polarisation studies, EIS study and adsorption isotherm study. Weight loss experiments were conducted at an immersion time of 10 to 50 min. The findings show that the AZI extract inhibits MS corrosion in a 1 wt% calcium hydroxide and 0.5 wt % sodium chloride solution. In the presence of Cl⁻ ions, passivation breaks down and metal gets corroded at a higher rate. GCI is introduced to mitigate the problem of corrosion. Therefore, the usage of GCI is essential for preventing metal from corrosion in an alkaline environment with chloride ions. The efficiency of inhibition decreases as immersion time increases. Inhibitor adsorption showed spontaneous, physisorption and was best fitted by the Frumkin adsorption isotherm. From the potentiodynamic polarisation study, results showed that the I_{corr} value rises with a rise in the concentration of calcium hydroxide. From the EIS study, the highest inhibition efficiency was found to be ~78%. SEM image confirms a layer of protection with an inhibitor appears over the metal surface. AZI is used as green corrosion, as it is biodegradable, cheap, easily available and economically viable.

Keywords: Green technology; plant extract; Tafel plot; XPS; XRD.

SYNTHESIS OF NANO MEMBRANOUS COMPOSITE MATERIAL BASED ON POLYANILINE/MWCNTS/CELLULOSE ACETATE FOR SELECTIVE ELECTROCHEMICAL DETECTION OF ASCORBIC ACID AND CATECHOL

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A multilayered porous nano membranous composite material made of POLYANILINE, MULTIWALLED CARBON NANOTUBES, and CELLULOSE ACETATE is prepared and characterized in this study. We compare the structure, morphology and electrochemistry of PANI/CA and PANI/MWCNT/CA. The structural information of PANI/CA and PANI/CA/MWCNTs systems was studied through FTIR, Raman, and UV-Visible spectroscopy. Morphology of the material was studied by SEM, HRSEM, TEM and AFM microscopy. Cyclic voltammetry and differential pulse voltammetry were used to measure the current responses with respect to the concentration of different analytes MWCNTs act as nanofiller and facilitate the integration of PANI/CA layers, development of new redox active centers and charge transfer paths and enhance the overall surface area of the material as a result a higher response current was observed in CV experiments for catechol detection. It is safe to conclude that the more significant charge transferability, nano-size morphology, and large aspect ratio of MWCNTs added electrocatalytic advantages to the material. The inherent electroactive nature of PANI facilitated mediated electrochemical charge transfer and electrocatalytic oxidation of catechol at the lower electrode potential at the PANI/MWCNTs/CA modified carbon paste electrode surface. CA unfolds and is fragmented as nano flat sheet-like structures, which act as a substrate for the deposition of PANI fibers, so we observed a composite nano-membranous material formed (confirmed by microscopic studies). CATECHOL and ASCORBIC ACID are good electroactive analytes that are electro-catalytically oxidized at electrode potential. PANI/CA/MWCNTs modified carbon-based electrode was used for CATECHOL and ASCORBIC ACID electrochemical sensing. We observed that the material is more sensitive to CATECHOL than ASCORBIC ACID and facilitates electrocatalytic oxidation at a lower peak potential than the bare electrode surface. Thus, we conclude safely that PANI/MWCNTs/CA is better electrode material than PANI/CA, in terms of better charge transfer, redox activity, and higher sensitivity for CATECHOL sensing. The developed carbon paste-based CATECHOL sensor exhibited a wide linear range (20 M to 20 mM), a quick response time (4 s), and a low detection limit [22.51 M (ASCORBIC ACID) and 19.59 M (CATECHOL)]. The developed Polyaniline polysaccharide-based composite material can be used for many other applications, such as selective filtration, adsorption, selective membrane formation, and other sensor devices, the work is in progress and will communicate soon.

Keywords: Catechol; Cellulose Acetate; Cyclic Voltammetry; Electrochemical Sensor; Polyaniline; Porous Composite material.

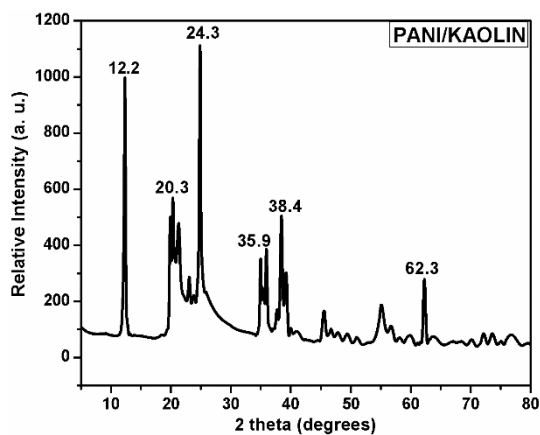
SYNTHESIS AND CHARACTERIZATION OF NOVEL CRYSTALLINE POLYANILINE/KAOLIN COMPOSITE

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This study attends to the crystalline behaviour of PANI/Kaolin composite. Composite was fabricated by in-situ chemical polymerization technique with ammonium peroxodisulphate oxidizing agent. Kaolin such as clay minerals particle incorporated within the network of Polyaniline to improve the morphology and other properties of the prepared composite. The composite material is characterized by using different instrumental techniques such as Fourier Transform Infrared Spectroscopy, Ultraviolet Visible Spectroscopy, Raman Spectroscopy, X-ray diffraction, Scanning Electron Microscopy, Transmission Electron Microscopy, Cyclic voltammetry and amperometry. A capillary electrode was fabricated with carbon paste with the PANI/Kaolin composite. Redox behaviour of the composites are analyzed by comparing cyclic voltammogram at different scan rate (200, 100, 50, 10 mV/s) and at different pH (PBS buffer solution at 1, 2, 3, 4, 5, 6, 7).



Keywords: Composite, Polyaniline, FTIR, SEM, TEM, Cyclic Voltammetry.

PARAMETRIC STUDY, OPTIMIZATION AND RANKING ANALYSIS OF GREWIA OPTIVA FIBRE/DOLOMITE BASED HYBRID COMPOSITES USING HYBRID MCDM TECHNIQUE

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Researchers are currently working to replace synthetic fiber with natural fiber since the latter is more affordable, lightweight, renewable, biodegradable, and environmentally benign. This study investigates the potential of utilizing Grewia optiva natural fibers and dolomite filler as reinforcing agents in epoxy composites, aiming to enhance their mechanical and wear characteristics. Grewia optiva, a plant native to Himalayan regions, offers a sustainable and eco-friendly alternative to synthetic fibers. Dolomite, a mineral combination of calcium magnesium carbonate, is chosen as a filler to further augment mechanical strength and wear characteristics of the composites. Using a hand layup process, the polymer composites were prepared with a variation of dolomite (0wt%, 5wt%, 10wt% and 15wt%) and Grewia optiva (20 wt%) fixed. Preliminary results demonstrate a meaningful improvement in the hardness and impact strength of the composites. In order to understand the composite degradation over time, wear rate was obtained at varying speed and load conditions. The composites with higher dolomite content proved to be more wear resistant than their counterparts. To identify a potential mechanism for sliding wear in hybrid polymers with varying dolomite contents, morphological examination was done. Lastly, the compositions were ranked according to their merits using the AHP ranking system. EGD15 hybrid polymer composites have demonstrated a comparatively optimal overall performance, according to observations.

Modeling and Simulation of Processes (MSP)

THERMODYNAMIC MODELING OF BINARY BINDERS WITH AGRICULTURAL RESIDUES FOR HYDRATION ANALYSIS

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Supplementary Cementitious Materials (SCMs), such as fly ash, blast furnace slag, rice husk ash (RHA), and sugar cane bagasse ash (SBA), are increasingly used to promote sustainable construction practices. However, understanding how the chemical composition, fineness, reactive phases, and pore solution content of SCMs affect the reaction process and hydration of cement is challenging. Researchers are developing models, including thermodynamic modeling (TDM), to better understand these effects. TDM is a useful tool for predicting the composition of pore solution and understanding the composition of hydrated cement and SCMs. This study investigates two types of ordinary Portland cement (OPC1 and OPC2) with varying chemical and mineral compositions mixed with two SCMs (SBA and RHA). Using TDM, the impact of cement content on hydrate generation in binary mixes is studied. The results reveal that OPC1 and OPC2 hydration models predicted CSH gel, ettringite, hydrotalcite, calcite, and portlandite. In comparison to OPC1, OPC2 predicts 21% more CSH gel and 26% less hydrogarnet. The study also found that jennite-kind CSH transforms into tobermorite-kind CSH with an increase in RHA, and a decrease in portlandite is observed in SBA blended systems. The TDM results were validated using experimental data from the literature and it provides valuable insights into the type and amount of hydrates that develop during cement hydration and its blends with SCMs.

Keywords: Sustainable concrete making materials.

DETECTION OF H₂S GAS USING METAL-DECORATED AND METAL-DOPED GRAPHENE USING DENSITY FUNCTIONAL THEORY

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One of the most difficult challenges for a gas sensor is achieving good selectivity when detecting the presence of specific gases in media containing other gases. The creation of sensing materials with the ability to distinguish between different analytes in a mixture is an easy fix. Modified graphene has a very high tendency to adsorb hydrogen sulfide gas H₂S which makes it an optimum choice for separating H₂S gas from industrial waste streams and acting as an H₂S gas sensor. In this paper, the adsorption of H₂S gas molecules on zinc-decorated, and zinc-doped as well as copper-doped graphene nanosheets have been investigated based on Density Functional Theory (DFT). The structural properties of H₂S gas molecules on metal-doped and metal-decorated graphene nanosheets have been discussed. The electronic transport results show the average responses of nano-sensors constructed from Zn-doped and Cu-doped graphene along the armchair and zigzag directions. The present study indicates that Zn-doped graphene can be potentially applied to H₂S gas sensing with high selectivity and sensitivity.

Keywords: Band Structure; Density Functional Theory; Density of States; Doping; Functionalization.

POLYGENERATION PROCESS FOR RECOVERING WASTE COLD ENERGY IN LNG REGASIFICATION TERMINAL

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Liquefied Natural Gas (LNG) has been a rapidly growing energy source due to its low carbon emissions. LNG is stored at cryogenic conditions and is regasified before it can be piped to end consumers. Regasification is carried out using sea water, resulting in a wastage of the available LNG cold energy [1]. LNG regasification terminals have certain specification w.r.t. the heating value of the imported LNG. If the fraction of heavy hydrocarbons (ethane, propane, etc.) is high then the heating value will exceed the desired limit. This necessitates certain modifications in LNG terminals to adjust heating value of LNG. One strategy is to separate a portion of heavier hydrocarbons from LNG using distillation. In this context, LNG cold energy can be used to provide the condensation load for distillation [2]. However, the amount of LNG cold energy utilized in recovering heavy hydrocarbon is much less, thus resulting in the wastage of remaining cold energy. Here comes the potential of implementing a polygeneration process with the heavy hydrocarbon recovery process to maximize the utilization of LNG cold energy. In this study, we have performed process simulations using Aspen HYSYS to investigate the thermodynamic feasibility of different waste cold recovery processes integrated with heavy hydrocarbon recovery. The integrated process encompasses an organic Rankine cycle to generate electricity together with other options to produce different grades of refrigeration. Our analysis shows that the polygeneration process results in a net electricity of 1022.3 kWh, 484.7 kW of refrigeration duty at -50°C (~ 138 tons of refrigeration/hour) for cold storage warehouse cooling, and 5905.6 kW of cooling duty at 0°C (~ 1685 tons of refrigeration/hour) for data centre cooling. The polygeneration process utilizes 87.3% of cold energy resulting in an avoided CO_2 emissions of 1089.3 kg CO_2 kWh. Thus, the proposed process can aid in enhancing the sustainability of LNG regasification terminal.

Keywords: Integrated process; LNG cold energy; Organic Rankine cycle; Process simulation; Refrigeration.

HYDRODYNAMIC FORCES ON A JANUS CYLINDRICAL PARTICLE: A CFD STUDY

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Flow past bluff bodies is a fundamental problem in fluid mechanics. Various interesting phenomena such as flow separation, stagnation point gets affected due to the no-slip boundary walls in the vicinity of the bluff bodies. Recently, a novel concept to design materials known as “Janus” particle has shown tremendous potential in several applications. In this work, flow structures and hydrodynamic forces on a circular Janus cylinder, positioned centrally in a confined channel at low particle Reynolds number ($Re_p = 1$ to 100) for various channel to circular Janus cylindrical particle (H/D) configurations is investigated. The comparison of the flow parameters is also made with the conventional uniform surface particle (stick cylindrical particle). Drag coefficient is observed to vary less appreciably after a certain H/D configuration. The magnitude for lift coefficient is found to be higher for low H/D ratios. Moment coefficients obtained for such particles also follow the similar trend as for drag. Empirical correlations are also proposed for drag, lift and moment coefficients for the Janus cylinder, when it is placed in the centreline of the channel for various H/D ratios.

Keywords: Correlation; Drag; Free slip; Flow streamlines; Lift.

NUMERICAL INVESTIGATION ON AN IN-LINE ARRAY OF TRIMMED ANNULAR FINNED TUBES FOR WEIGHT REDUCTION IN A CROSSFLOW HEAT EXCHANGER

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Heavy weight and larger space requirements can significantly limit the applicability of a heat exchanger. The demand for compact and cost-effective heat exchanger designs is increasing. One way to save the fin material and make a heat exchanger compact is to trim the fins on a heat exchanger. The downstream section of a fin contributes the least to heat transfer; hence, the fin's downstream region can be trimmed. This study investigates the effect of fin trimming on the performance of a crossflow heat exchanger using the CFD approach. A four row in-line array of finned tubes whose downstream region has been removed are used. The characteristics of these fins are compared with the benchmark case of the untrimmed finned tube bundle. The Reynolds number varies from $4330 \leq Re \leq 8790$. A finite volume based methodology is used. The turbulent stresses are modelled using Transition SST equations, and the PISO scheme is implemented for pressure-velocity coupling. The results show that fin trimming significantly reduces the pressure drop due to reduced skin friction. Additionally, trimmed fins show a minimal reduction in heat transfer rate compared to untrimmed fins. The formation of a horseshoe vortex is observed at a fin-tube intersection. This horseshoe vortex enhances the mixing, thereby directly affecting the overall heat transfer. The size of this horseshoe vortex is dependent on the Reynolds number. This study suggests that trimmed fins can give an considerable reduction in weight of heat exchanger than their untrimmed counterpart.

Keywords: CFD; Fin trimming; Forced convection; Material saving; Turbulence.

IDENTIFICATION OF UNIFORM THERMODYNAMIC EQUILIBRIUM IN TETRAHEDRAL LIQUIDS AT LIQUID-LIQUID PHASE TRANSITION TEMPERATURE

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The liquids where a central atom forms four bonds and directed towards the corners of a regular tetrahedron are usually known as tetrahedral liquids. The substance such as silicon, water, and germanium from group 14 of periodic table carried the structure of tetrahedral liquids at some specific thermodynamic conditions. All these substances are very useful in many industrial applications (i.e. chips solar cell, etc.) in liquid, solid, or gaseous states. The existence of liquid, solid, and gaseous states of these substance will depends on the equilibrium conditions which can be any dynamical parameters (i.e. Temperature, Pressure, Volume, etc.). Further, as these substance shows very interesting and challenging behavior in supercooled region (i.e. below the melting point) especially at liquid-liquid transition temperatures due to quick/rapid transformation. Hence, this study has been focused to understand their thermodynamic behavior followed by the thermodynamic equilibrium phenomena at liquid-liquid transition temperatures (T_{ll}) as $1060 \pm 12\text{K}$, $202 \pm 12\text{K}$, $1312 \pm 12\text{K}$ for silicon, water, and germanium respectively. Based on the predicted results in terms of thermodynamic properties the existence of the equilibrium and non-equilibrium states of these substance at or near liquid-liquid transition temperature (T_{ll}) has been identified. This study may be helpful to analyze the equilibrium phenomena of many such substances in supercooled region at laboratory and industrial scale.

Keywords: Tetrahedral Liquids; Phase Transformation; Phase Equilibria.

SIMULATION ON PROPANOL REFORMING FOR HYDROGEN PRODUCTION: A THERMODYNAMIC ANALYSIS

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Oxidative steam reforming as a thermochemical conversion technique is employed to examine the thermodynamic feasibility for the production of hydrogen by using propanol as biomass substrate. For process simulation, R-Gibbs reactor tool software in Aspen Plus is used to check the effects of process variables such as pressure (1 to 10 atm), temperature (300 to 1000 K), and molar feed ratio (f_1 = steam to propanol = 1 to 10 and f_2 = oxygen to propanol = 0.25 to 1) on desired product of reforming reaction as hydrogen production. After sensitivity analysis, the maximum hydrogen is found at atmospheric pressure, 573 K, $f_1=10$ and $f_2=0.25$. Under these conditions, the generation of other products such as methane, carbondioxide, carbon monoxide, and carbon (solid) are also analyzed. The same optimal operating conditions of process variables are also used to obtain thermal efficiency of suggested reforming system.

COMPARATIVE STUDY ON SUPERCOOLED SILICON BY USING DIFFERENT FORCE FIELD MODEL

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Silicon is an important chemical element that represents a tetrahedral structure in which a central atom is connected with four triangular faces. This element exists as an element of group 14 in the periodic table. The element shows many industrial applications including the semiconductor, solar cell, and chips industry. It usually exists in the liquid, solid, or gaseous states. However in supercooled region specifically close to the stability limit the liquid phase of this element itself distinguish between high-density liquid (HDL) and low-density liquid (LDL) phases. Apart from this phase change phenomenon it shows many interesting thermodynamic changes including anomalies such as heat capacity (C_p) during transformation process in supercooled region. Due to its anomalous behavior and unique applications it consider as challenging element to study in supercooled region (i.e. close to stability limit). Hence, we focused our study to perform a significant analysis on this element (i.e. Silicon) close to the stability limit. In this work, we followed the molecular simulation technique by using LAMMPS open source software with Noose Hoover thermostat. This computational study on silicon has been performed in supercooled region at liquid-liquid transition temperature by employing the three different force field model (i.e. Stillinger-Weber Potential, Environment-Dependent Interatomic Potential (EDIP), and Tersoff Potential models) from the prospective of comparison. All the reported results over this study are consistent and able to provide a scientific insight on silicon in supercooled region for all three force filed model as mentioned above with comparison.

Keywords: Supercooled; Tetrahedral; Phase Equilibria.

COMPARATIVE ANALYSIS OF FLUID FLOW BEHAVIOR IN HELICAL COIL, CYLINDRICAL PIPE, AND POROUS MEDIA THROUGH CFD SIMULATIONS

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Fluid flow analysis holds paramount importance in engineering and industrial applications, as it plays a pivotal role in optimizing both design and performance. This study comprehensively examines fluid flow behaviour in helical coils and cylindrical pipes to reveal distinct characteristics. Helical coils find diverse engineering use due to their unique design and excellent heat transfer. ANSYS Fluent is used for CFD analysis, comparing flow patterns, velocities, and pressure drops in both geometries. To establish a reference point, a specific geometrical configuration is chosen as the reference coil. Variations in geometrical features follow, including coil diameter, pipe diameter, pitch-to-diameter ratio, and turns. The obtained results are then contrasted with those of the equivalent cylindrical counterpart. The models are split into two groups: one with constant volume variations, and the other with volume changes tied to geometrical features. The fluid flow analysis is conducted with and without the incorporation of porous media. The fluid flow is treated as incompressible, turbulent in this model. The chosen viscous model is the k-epsilon model, which assumes a consistent turbulence viscosity throughout the flow field. In general, helical coils show higher pressure drops than cylindrical counterparts, with exceptions at certain Reynolds numbers. This observation is accompanied by nearly identical outlet velocities between the two geometries.

APPLICATION OF MODELING TO ANALYSE THE PROPERTIES OF CARBON NANOTUBE- POLYMETHYLMETHACRYLATE COMPOSITES: A REVIEW

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Carbon nanotube-polymethylmethacrylate (CNT-PMMA) composites have emerged as promising materials due to their exceptional properties. However, fully harnessing their potential requires a comprehensive understanding of their behavior, which can be achieved through effective modeling approaches. This critical review explores the modeling techniques employed to investigate CNT-PMMA composites and their contributions in uncovering the underlying mechanisms governing their properties. The review covers continuum-based models, atomistic simulations, and multiscale modeling, highlighting their strengths and limitations. Furthermore, the computational tools and software utilized for modeling are discussed, including finite element methods and Monte Carlo simulations. The review also delves into the insights gained from modeling in understanding the structure-property relationships of CNT-PMMA composites, particularly in interface effects, reinforcement mechanisms, and electrical percolation. The challenges and future directions in the field, such as experimental validation of modeling predictions and the incorporation of defects and imperfections, are identified. Additionally, the potential for developing multifunctional composites and the sustainability considerations in the context of CNT-PMMA composites are discussed. Overall, this critical modeling review provides a comprehensive overview of the current state of research, shedding light on the opportunities and challenges in unlocking the full potential of CNT-PMMA composites.

Keywords: CNT, PMMA, Multiscale modeling

OPTIMAL DESIGN OF COOLANT JACKET FOR CRYOGEN TRANSFER PIPELINES

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Cryogenic liquids such as liquid oxygen and liquid hydrogen are extensively used in many process, and manufacturing industries. In these industries, transfer of cryogenics via pipelines is a routine phenomenon. As the boiling points and latent heats of cryogenics are low, excessive vaporization of these cryogenics is innate. Therefore, it is challenging to ensure that the cryogen reaches the utility in its liquid form. In the case of liquid hydrogen and liquid helium, the pipelines are jacketed with a high boiling cryogen like nitrogen. The idea is to dump most of the heat into cheap nitrogen to limit the loss of precious hydrogen or helium. From a heat inleak point, it is advantageous to maximize the amount of nitrogen in the jacket, by choosing large cross-sectional areas. Also, larger flow cross sections would lead to lower pressure drops and therefore lower pumping costs. However, such a choice would add to the mass of the pipeline. Increase in the mass of the pipeline increases the need for better structural support of the pipeline assembly. Therefore, the design of cryogen jackets for limiting heat inleak is a multi-objective optimization problem. In this work, we model: the heat inleak into the hydrogen via the nitrogen jacket, the pressure drop of liquid nitrogen, and find the mass of the pipeline assembly. Then, we optimize the design of nitrogen jackets fitted over hydrogen pipelines. We employ the evolutionary optimization technique, genetic algorithm (GA) to perform this optimization. The objectives of this optimization are to minimize the heat leak into hydrogen, minimize the pressure drop and, to minimize the mass of the pipeline assembly. The results of this work will be useful for process industries transferring cryogenic liquids in their plants.

Keywords: Cryogen; Genetic Algorithm; Heat inleak; Liquid hydrogen; Optimization.

AN EXPERIMENTAL AND COMPUTATIONAL STUDY OF INDIAN SUB-BITUMINOUS COAL FOR PYROLYSIS OF VITRINITE

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Coal is an organic sedimentary rock that composed of heterogeneous components, which have different physical and chemical properties. It is a non-renewable fossil fuel and is the largest source of energy. Usually for metallurgical applications, coal is converted into coke, which is nearly pure carbon produced when soft coal is heated in the absence of air. Conventionally, one ton of coal will produce 0.7 ton of coke in this process. Coke holds greater importance in the industry because it has a heat value higher than any form of natural coal. The total coal reserve in India is approximately 125.91 BT but only 18.4 BT is coking coal, which is a major concern. Several attempts in the past have been made by the researcher to study the property of coal, its mechanism and kinetics during the combustion. Still the study of structural properties of coal is a big challenge because of the large diversity in the coal structure is due to the different origin and different degree of coalification. However, with the help of molecular representation of coal, dynamics of the coal to coke conversion can be analyzed. Present work is focus on experimental and computational study of Vitrinite, as its concentration is more than 90% in the coal. Hence, it's contribution is more in Coal to Coke conversion. In present work, Vitrinite maceral of Indian Sub-Bituminous coal (Bhojudih) has been separated from the raw coal by density gradient centrifugation method. The molecular representation of Vitrinite separated from coal has been proposed, based on the elemental analysis and ^{13}C NMR experimental data. For this purpose, a base model of Vitrinite was referred from previous published model. The NMR spectrum of the base model was predicted empirically and was fitted, by using Model Validation approach, to the experimental NMR spectra of Vitrinite by modifying the base model.

Keywords: Coke; Indian coal; Molecular dynamics; Molecular simulation; Pyrolysis.

MODIFIED FRACTIONAL ORDER PID CONTROLLER DESIGN FOR MIXING TANK PROCESS

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Controlling Multi Input Multi Output (MIMO) processes poses inherent challenges due to the presence of interactions between process loops. Consequently, employing a standard PID controller often fails to yield satisfactory results. To overcome this issue in the context of a two-input, two-output mixing tank process, a sophisticated control design scheme called fractional-order PID control has been investigated. The implementation of the fractional PID controller involved designing individual controllers for each loop and obtaining the tuning parameters through the Nedler Med algorithm. To minimize interaction within the control loop, a Relative Gain Array analysis was conducted to determine the optimal pairing. Additionally, a simplified decoupler was developed to eliminate any remaining interaction within the system. To assess performance, the proposed decoupled controller was compared against the Ziegler-Nichols (ZN) and Internal Model Control (IMC) approaches. The simulation results demonstrate that the proposed FOPID controller outperforms traditional ZN and IMC PID controllers, showing a 27.3% and 21.5% reduction in settling time, 38.6% and 25.8% improvement in IAE values and 38.1% and 29.3% decrease in ISE values for hot and cold fluids study respectively.

Keywords: Multivariable Process, Fractional Calculus, PID Controller, Relative Gain Array, Simplified Decoupler.

CFD MODELLING OF LIQUID-LIQUID BATCH STIRRED TANK

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In this study, a Computational Fluid Dynamics (CFD) model was developed to simulate liquid-liquid dispersion in a batch mixer. The focus was on determining the minimum mixing speed required for complete dispersion, as well as analysing the flow field and local holdup distribution during two-phase mixing at high organic to aqueous phase ratios. The simulation utilised a two-phase Euler-Euler model in conjunction with the $k-\epsilon$ turbulence model. To validate the CFD model, comparisons were made with Ultrasonic Velocity Profiler (UVP) measurements and experimental observations. The predicted minimum mixing speed for complete dispersion closely matched experimental results, with an average absolute relative error of 7-10%. The study revealed that impeller type, speed, organic-to-aqueous (O/A) ratios, and the characteristics of the liquid system had significant impacts on the flow field. It was observed that increasing impeller speed led to a monotonic rise in turbulent kinetic energy and energy dissipation within the stirred tank. Additionally, the choice of liquid system influenced holdup distribution uniformity in the tank. Liquid systems 1, 2, and 3 demonstrated approximately 99.9%, 97.86%, and 93.8% uniformity in holdup distribution, respectively. Overall, the developed CFD model provided valuable insights into the complex dynamics of liquid-liquid dispersion in batch mixers, with good agreement between simulations and experimental data.

Keywords: CFD modeling; High O/A ratio; Liquid-liquid extraction, Stripping operation.

COMPARATIVE ANALYSIS ON THE EFFECT OF AERATION AND AGITATION ON QUALITY AND QUANTITY OF ALGAL BIOMASS FOR ITS FURTHER VALORIZATION TO VALUE-ADDED PRODUCTS

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In the last decade, socio-economic awareness has reached to its peak, which makes the world shift towards bioresources to minimize their dependency on organic and inorganic natural sources. Microalgae can be potential source of oils and other essential molecules that can be used as feedstocks in the production of biofuels and other high-value products, potentially becoming a significant renewable energy source. The objective of this research is to analyse the effect of intensive aeration and agitation on the cultivation of indigenous microalgae for its further biorefinery to extract various value-added products. In this research, low-strength domestic wastewater was used as the potential source of nutrients for algal cultivation having nitrates and total phosphorous around $21.6 \pm 0.39\text{mg/l}$ and $12.7 \pm 0.8\text{ mg/l}$, respectively. Indigenous algal consortia having concentration of 5% (v/v) was cultivated under the illumination of 6000 lux with 14:10 light - dark period at the temperature of $24^\circ\text{C} \pm 4^\circ\text{C}$ for 20days with two different approaches: one with agitation at 240 rotation per minute (rpm) and another with aeration at 2.0 litres per hour (lph). Microalgae was quantified by analysing the cell count with a haemocytometer under a microscope, optical density at 680nm, chlorophyll content every alternative day. Further the harvested algal biomass from both the approaches was analysed for its dry weight and lipid content. The results revealed that dry weight ($348.4 \pm 10.5\text{mg/l}$) and lipid content (39.7%) of aeration mediated microalgae cultivation was observed to be higher than agitation induced microalgae cultivation, which proves that aeration favours the microalgal growth.

Keyword: aeration; agitation; bioresources; feedstock; value-added products

Bioprocessing and Food Engineering (BFE)

APPLICATION OF FREEZE DRYING TO EXTEND THE SHELF LIFE OF GOLDEN DELICIOUS APPLES

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Freeze-drying is a promising technology for extending the shelf life of Golden Delicious apples. This method removes water from food at low temperatures and pressures, preventing the growth of microorganisms and enzymes that cause food decay. Freeze-dried foods can be stored for long periods of time without refrigeration or freezing. One of the challenges of freeze-drying apples is their high moisture content. To reduce this, different pre-treatments can be used, such as blanching and osmotic dehydration. Blanching involves briefly boiling or steaming the apples, while osmotic dehydration involves submerging them in a concentrated sugar solution. Both of these methods can reduce the moisture content of apples by up to 50%. Once pre-treated, the apples are frozen to a very low temperature. They are then placed in a vacuum chamber and the pressure is reduced, which causes the water in the apples to sublime, or turn directly from a solid to a gas. Finally, the apples are heated to remove any remaining moisture. Freeze-dried Golden Delicious apples have a number of benefits, including an extended shelf life, preserved nutrients, light weight, and delicious taste. They can be stored for up to 25 years without refrigeration or freezing, and they retain most of their nutrients. Freeze-dried apples are also very light weight, making them ideal for camping, backpacking, and other outdoor activities. More research is needed to optimize the freeze-drying process for Golden Delicious apples. This could focus on developing new pre-treatments, reducing energy consumption, and improving drying efficiency. Additionally, research is needed to evaluate the quality and nutritional value of freeze-dried Golden Delicious apples over time.

Keywords: Food decay; Low temperatures; Moisture content; Nutritional value; Nutrient retention.

EFFECT OF DRYING ON EXTRACTION OF ANTHOCYANINS FROM SYZYGIUM CUMINI (JAMUN)

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Jamun pulp foam was dried using the foam-mat drying technique with the incorporation of egg albumin and methylcellulose. The drying process involved three different techniques: hot air foam drying, vacuum foam drying, and microwave foam drying. Microwave foam drying was selected as the preferred method based on the recovery of anthocyanin, with drying power options of 180W, 360W, and 540W. For hot air foam drying and vacuum foam drying, temperatures of 50°C, 70°C, and 100°C were employed. It was found that microwave drying with 180W had maximum anthocyanin recovery, while vacuum and hot air drying has lesser anthocyanin content. The anthocyanin content obtained at this power level was determined to be 491.42 ± 6.97 mg/100g, indicating a significant increase compared to the initial content. Using lower power levels in microwave foam mat drying has proven advantageous by preventing the degradation of anthocyanins due to decreased exposure to heat. Subsequently, various analyses were conducted, from the extracts obtained from microwave power 180W which included total phenolic content (TPC), 2,2-diphenyl-1-picrylhydrazyl (DPPH), Ferric Reducing Antioxidant Power (FRAP), and antimicrobial assay. Lower values of TPC was found because of breakdown of bioactive compounds due to higher power levels. An increased quantity of DPPH value was observed. The FRAP absorbance was around 0.055 nm, signifying its capacity for reducing power. The extract exhibited notable antibacterial efficacy, although no significant antifungal impact was detected. These characterizations offer valuable information about the composition and potential uses of the anthocyanin extract obtained through microwave foam mat drying at 180W power.

Keywords: Microwave foam mat drying; Hot air foam mat drying; Vacuum foam mat drying; Anthocyanin recovery; Anthocyanin degradation.

ATMOSPHERIC PLASMA MEDIATED QUALITY CHANGES IN FINGER MILLET STARCH AND ITS MECHANISM

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This research was conducted to evaluate the effect of atmospheric cold plasma generated by voltage (170, 200 and 230 Volt) and treatment time (10, 20, and 30 mins) on starch isolated from finger millet. The impact of treatment was studied for various physico-chemical, and functional properties. A significant reduction ($p < 0.05$) in the peak temperature (from 84.1 to 68.42 °C), pH (from 7.2 to 5.57), average molecular weight, and amylose content was noticed. A remarkable increase in water solubility index by 5 times was registered. Depolymerization as analysed by the intrinsic viscosity showed zero-order deterioration kinetics. An increase in reducing sugar at higher voltages also suggested occurrence of severe depolymerization. Reducing sugar, Peak temperature, Relative crystallinity, etc. also confirmed that cross-linking was also observed. Findings of XRD, NMR and other analyses were also documented and suggested valuable findings such as occurrence of oxidation of OH group at C6 position to form carbonyl group followed by carboxyl group. The findings were thus used to report the in-depth mechanisms such as oxidation of anhydroglucose ring, depolymerization and crosslinking, carbonyl carboxyl groups incorporation, and plasma etching enlisted.

Keywords: Cold plasma; Cross-linking; Depolymerization; Emerging technology; Starch modification; Under-utilized finger millet.

DETERMINATION OF THE BOVINE SERUM ALBUMIN SUBJECTED TO MICROWAVE INTERACTION BY LIQUID CHROMATOGRAPHY-TANDEM MASS SPECTROMETRY

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This study aimed to understand the effect of microwave radiation on the bovine serum albumin. Clinical proteomics abundantly uses albumin derived from cows for biochemical applications owing to its biocompatibility, stability, and non-interference in reaction. The applicability of bovine serum albumin as a potential biomarker, in possible treatments of various ailments such as hypervolemia, haemorrhage, hypoalbuminemia, etc. has shown the probability of improvement when microwaving radiation is incorporated into the procedures. The bovine serum albumin samples were subjected to microwave radiation in a multimode cavity working on frequency 2450MHz at different powers and the absorbance and the dielectric changes in the treated protein samples placed in an ice bath were analyzed. The exposure to microwave radiation exhibited changes in the absorbance and exhibited the capability to quantify bio-transformations in the proteins post-treatment in the bovine serum albumin samples. The top-down MS is based on the identification of intact proteins before and after undergoing modification including the benefits of the detection of post-translational modifications and proteolytically processed proteins. This study works as a guideline to understand the non-thermal effects of microwave radiation on the most abundantly available and used bovine serum albumin.

Keywords: Microwave; Bovine serum albumin; LC-MS/MS; Proteomics; spectroscopy.

EXTRACTION OF MANGO (*MANGIFERA INDICA*) SEED KERNEL STARCH: PROCESS OPTIMIZATION FOR EXTRACTION USING RESPONSE SURFACE METHODOLOGY (RSM) AND ITS COMPARATIVE CHARACTERIZATION WITH COMMERCIAL CORN (*ZEA MAYS*) STARCH.

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Objective: Optimisation of the extraction conditions of starch from mango kernel (Totapuri) employing response surface methodology to enhance yield and purity, and comparison with commercial starch (corn starch) to assess its potential application.

Methodology: The Central Composite Design with four independent variables viz., steeping temperature (4-50 °C), steeping time (2-24 h), pH (9.5-10.2), and sample-to-solvent ratio (1:4-1:8) were employed. The optimised mango starch was further characterized using XRD, SEM, FTIR, DSC, Particle size analyzer, and rheometer.

Results and discussion: The optimal conditions of 31.62 °C, pH 10.05 with 1:8 solid-to-solvent ratio at 2h showed the highest starch yield of $35.98 \pm 0.29\%$ with $90.01 \pm 0.26\%$ purity. In comparison with commercial starch, mango kernel starch showed no significant differences in oil binding capacity, paste clarity, particle size, and turbidity till 5th day whereas notable distinctions ($p < 0.05$) were observed in % syneresis, water binding capacity, thermal properties, and gel hydration properties at 70°C. FTIR showed similar spectra for mango kernel and commercial starch showing the presence of A-type starch. The structural attributes of the samples have been adequately confirmed by SEM images. Thus, mango kernel starch showed comparable physiochemical, functional, and thermal properties as compared to commercial starch.

Keywords: Mango kernel starch; optimization; physiochemical and functional properties; potential substitute; response surface methodology.

DEVELOPMENT OF COMPOSITE BIOHYDROGELS USING GELLAN GUM AND GLUCOMANNAN, AND THEIR POTENTIAL APPLICATION IN SUSTAINABLE AGRICULTURE AND CONTROLLED RELEASE OF BIOACTIVES IN THE GUT

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Biopolymers have gained tremendous interest in recent times. Microbial or plant-based hydrocolloids are being preferred as an alternative to synthetic polymers, due to their high biodegradability, low-toxicity, abundance and low-cost. Furthermore, the requirement of chemicals for cross-linking is also minimal, making them highly biocompatible. In this study, gellan gum and glucomannan from the konjac root (KG) were chosen for the fabrication of hydrogels using ionotropic gelation by Ca^{2+} ions. The ratio of the gellan gum and konjac glucomannan was varied for the hydrogel. A 1:1 ratio of glucomannan and high acyl gellan was found to have the highest water holding capacity (>98 %) and the lowest syneresis rate (6.89 %). While low-acyl gellan resulted in brittle gels, the addition of KG enhanced the elasticity and the structural recovery. A comprehensive rheological characterization was performed to understand the underlying interactions between gellan gum and KG; and the impact of the same on the pH-dependent swelling properties. Due to the increasing pollution and discharging of effluents into the soil, the water absorption capacity of the soil is drastically affected; thereby increasing irrigation requirements. Biohydrogels can be an efficient tool in the improvement of soil quality and moisture, by facilitating a controlled release of moisture from the hydrogel matrix into the soil. The composite hydrogels were investigated for their ability to impact the germination and growth of fenugreek seeds; when subjected to a three-day intermittent watering pattern to imitate the semi-arid conditions. Due to the excellent swelling ratio (>3.5) and water holding capacity (>98 %) of the KG-high acyl gellan composite hydrogels, a higher germination percentage and growth rate of microgreens were observed at 10 % loading compared to the control. Similarly, the composite KG-low acyl gellan exhibited a controlled in vitro release of riboflavin in simulated gastric and intestinal conditions.

Keywords: semi-arid regions; in vitro release; heat-set gels; biopolymers; sustainability.

BACTERIAL INGENUITY: SYNTHESIS OF RHAMNOLIPID BIOSURFACTANT UTILIZING GLUCOSE AS A GROWTH ENHANCER AND WASTE ENGINE OIL AS ECONOMICAL SUBSTRATES BY A NON-PATHOGENIC BACTERIUM

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Recently, biosurfactants have gained significant attention because of their advantageous properties, such as low toxicity, biodegradability, and eco-friendliness. However, despite these benefits, their high production and substrate costs make them uncompetitive with chemical surfactants. To overcome this challenge, the utilization of low-cost substrates is crucial as they can account for 30-40% of the total production cost and significantly reduce the overall production cost of biosurfactants. The current study aims to investigate the production of biosurfactant using waste engine oil (WEO) and glucose (growth enhancer) as low-cost substrates. A non-pathogenic bacterial strain *Pseudomonas fluorescens* MTCC 103^T was utilized for this purpose. Batch experiments were conducted in Erlenmeyer flasks, using 0.5% glucose and 0.5% WEO as substrates for 72 hours. The cell-free supernatant samples of the strain showed a significant reduction in surface tension during the stationary phase. Fourier transform infrared (FTIR) and ¹H NMR spectroscopy confirmed the biosurfactant produced was of rhamnolipid type. The biosurfactant yield was recorded to be 3.96 g/L using 0.5% glucose and 1.12 g/L using 0.5% WEO. These results demonstrate the potential for producing biosurfactant using renewable and cost-effective resources. The emulsifying activity of the biosurfactant against different hydrocarbons also suggests its potential application in bioremediation of hydrocarbon-polluted environments.

Keywords: Biosurfactant, Waste engine oil, *Pseudomonas fluorescens*, Bioremediation, Non-pathogenic bacterium.

ARTIFICIAL INTELLIGENCE: SCOPE AND DEMAND IN BIOPROCESS

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Artificial Intelligence (AI) is taking over the major part of the world and the clinical sector is not left unattended in order to attain human like performances. The flowchart and database approach used in a computer system is being applied in the development of artificial neural network in addition with AI and machine learning (ML). Bioprocess playing a major role in rapid transition of bioproducts from lab scale to a cost effective large scale production. Its control and optimization in combination with the latest computer-based strategies will take the production scale at a different level obeying all the required parameters. Artificial neural network (ANN) can help in dealing with the complexity of bioprocess in building the data driven model due to its human brain-like neural network and with the use of machine learning (ML) it can be employed for the optimization of the pretreatment process. Spectroscopy with electrode probe is the most convenient and accurate rapid detection method in the bioprocess for the qualitative and quantitative analysis, while using AI- based detection method approach image analysis has come into action. AI has been employed, along with deep learning, over data driven soft sensors for low cost high accuracy online monitoring, single and multi-variable estimation of critical processes parameters (CPPs). AI control strategies can lead to effective yield productivity and improvement of bioprocess. Dynamic time warping (DTW) technology is being used for accurate understanding of mechanism required for the model construction and optimization of control strategies by using automated processes like robots and high-throughput systems. Thus, all these advances reveal high potential of AI/ML in the advancement of bioprocess development and implementation and therefore further exploitation of AI in this field can lead universal automated bioprocess systems in industries.

Keywords: ANN; autocontrol; bioprocess modeling; technology; machine learning.

MICROBIAL ENGINEERING: A BOON TO ENHANCE FOOD SUSTAINABILITY

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In modern times, various techniques have greatly improved the microbial production of numerous foods, nutrients, dietary supplements, and food additives. In this challenge, genetic and metabolic engineering plays an increasingly important role in producing various genetically modified foods in the food industry. Knowledge of microbial engineering is widely used to improve human health. Products produced by genetically modified microorganisms promote health benefits and prevent malnutrition problems. The power of microbial culture in the competitive world of commercial synthesis can be seen in the fact that even simple molecules are produced by fermentation rather than chemical synthesis. Most natural products are so complex that they probably cannot be produced commercially by chemical synthesis. Strains isolated from nature produce only small amounts of product. The extensive use of (i) metabolic engineering and novel genetic methods to quantify and control metabolic fluxes, including reverse metabolic engineering and transcriptional expression, analysis such as association analysis and massively parallel signature sequencing; and (ii) direct evolution, (iii) molecular breeding, including DNA shuffling; and whole genome shuffling and (iv) combinatorial biosynthesis, can facilitate not only the isolation of improved strains but also the elucidation and identification of new gene targets for use in strain improvement programs. Hence, the immense potential of microbial engineering methods puts forward the ultimate need for further research and promotes food sustainability across the globe.

Keywords- combinatorial biosynthesis; healthcare; metabolic engineering; microbial Culture; modified food.

ARTIFICIAL INTELLIGENCE: ROLE AND SIGNIFICANCE IN PATHWAY DESIGNING

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Shortage and high demand of non-renewable resources has played a vital role in the increment demand of sustainable bio-based chemicals as an alternative. Metabolic engineering has enabled engineers in developing microbial cell factories which provides chemical using renewable resources. To design enzymes and pathways for the target chemicals of interest it is vital to know about the chemicals produced by microbes. Artificial intelligence (AI) is a wide branch of computer science which requires human intelligence in order to perform tasks in a short span of time. AI-aided protein design is a popular approach enabling protein function optimization. Neural network can help in solving the complexity of pattern obtained from large data and in effective machine learning. AI-aided protein engineering can be used to obtain desired protein function and property by reducing the experimental numbers. AI-based De novo protein design graphs a neural network that fold into predetermined shape and hence can be applied for designing of new protein sequences and structures. Thus, with the significant rise in big bio-data, AI data driven techniques facilitates more advanced protein and pathway design.

Keywords: computer aided; enzyme engineering; neural network; optimization.

BIOPROCESS INDUSTRIES: THE FUTURE WITH AI/ML AND DATA MINING

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The bioprocessing industry is undergoing a transformative phase through the integration of artificial intelligence (AI) and machine learning (ML), particularly in the context of Critical Process Parameters (CPPs) and Critical Quality Attributes (CQAs). ML algorithms, including Artificial Neural Networks (ANN) and multivariate data analysis methods such as Process Analytical Technology (PAT) instrumentation measurements, are employed for real-time estimations and control. The utilization of latent variable methods, like Partial Least Squares (PLS), enhances multivariate regressions of PAT instrumentation data. While recursive partitioning algorithms and ANN are common, the majority of applications emphasize pattern recognition. Despite advancements, ML in bioprocessing is still in its early stages, necessitating proper data management and addressing research needs. Challenges include limited access to authentic on-site datasets, the development of online-cyber network-based systems, and the evolution from sequential to hybrid process-driven techniques. Hybrid models, advanced deep learning architectures, and standardization issues pose additional complexities. Future perspectives focus on in-depth research on biosensors, microfluidic sensors, and high-throughput phenotypic platforms, linked with the Internet of Things (IoT) and hybrid modeling. The implementation of wireless soft sensor networks with ML-driven automation for smart product quality monitoring and the development of affordable wireless sensor node solutions are essential. Leveraging IoT scalability for expanding capabilities in online administration of wastewater quality parameters represents a key area of exploration. In conclusion, ongoing research is pivotal for advancing ML applications in bioprocessing, overcoming challenges, and realizing the full potential of these technologies.

Keywords: analytical technology; artificial neural network; CPPs; future aspects.

NUTRITIONAL ENRICHMENT OF SALAD DRESSING WITH FLAX MICROGREENS: FORMULATION AND SENSORY EVALUATION

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Microgreens, known for their youthful and delicate foliage, have become increasingly popular for elevating culinary creations owing to their rich nutritional profile and distinctive sensory attributes. This examination specifically delves into the nutritional aspects of flax microgreens and their integration into a tastefully flavored salad dressing. Flax microgreens were selected for their elevated nutritional composition, boasting significant proportions of fat, protein, total ash, and moisture, rendering them a valuable enhancement to the dressing. The formulation of the salad dressing entailed substituting conventional basil leaves with flax microgreens to introduce nutty undertones and mitigate the inherent bitterness associated with basil. Parmesan cheese and virgin olive oil were chosen as fundamental components due to their synergistic qualities and nutritive attributes. The infusion of various herbs and spices heightened the flavor profile, while sodium benzoate was incorporated as a preservative for prolonged shelf life. Five distinct formulations were crafted with diverse ratios of basil leaves to flax microgreens, and laboratory analysis pinpointed that sample 5, comprising 40 g basil leaves and 60 g flax microgreens, showcased the highest levels of protein and fat. A sensory evaluation by trained food technologists affirmed that sample 5 garnered the utmost overall acceptance, underscoring its superior taste, texture, appearance, and nutritional richness when juxtaposed with alternative formulations. This investigation underscores the promising potential of integrating flax microgreens into salad dressing formulations, promising not only enhanced nutritional value but also an enticing sensory journey for consumers.

Keywords: Flax Microgreens; Microgreens; Nutritional Composition; Salad Dressing Formulation; Sensory Evaluation.

BIOSURFACTANT PRODUCTION BY FREE AND IMMOBILIZED CELLS OF *PSEUDOMONAS AERUGINOSA* USING WASTE ENGINE OIL

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The global predicament of oil contamination necessitates a confrontational approach employing innovative and eco-friendly technologies. One prominent oil contaminant is waste engine oil (WEO), whose intricate composition renders its disposal a source of deleterious repercussions on the atmosphere, terrestrial ecosystems, and aquatic environments. Beyond environmental ramifications, this contaminant inflicts adverse effects on various life forms. To mitigate the deleterious effects of WEO, an environmentally conscientious remediation technique can be implemented, leveraging biosurfactant-producing microorganisms. These biosurfactants exhibit distinct advantages compared to their chemical counterparts, notably characterized by low toxicity, biodegradability, and sustainability. This investigation delves into the production of biosurfactants by both freely suspended and immobilized strains of *Pseudomonas aeruginosa* bacteria, utilizing WEO as the substrate. Batch experiments were meticulously conducted in Erlenmeyer flasks and systematically monitored at 24-hour intervals for both free and immobilized cell systems. Microbial population dynamics were assessed utilizing the plate count method, while hydrocarbon degradation rates were calculated to gauge bacterial activity. The findings revealed a maximum surface tension reduction of 37 mN/m in the free cell system and 35 mN/m in the immobilized cell system. Fourier transform infrared spectroscopy authenticated the biosurfactant as rhamnolipid. Notably, the immobilized system exhibited the highest biosurfactant yield, reaching approximately 18 g/L. Further scrutiny through scanning electron microscopy demonstrated the potential for reusing immobilized cells on polyurethane foam cubes.

Keywords: Biosurfactant, Immobilization, Polyurethane foam, *Pseudomonas aeruginosa*, Waste engine oil.

THERMODYNAMIC EVALUATION OF CALCIUM IN MILK USING GRAPHICAL ANALYSIS

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Milk is the major source of calcium intake for almost all living beings, owing to its easy availability and abundant presence of calcium as a major nutrient. Calcium in milk is present in 2 major forms i.e. bound and ionized forms. The form of calcium which is bound to casein micelles constitutes the majority of calcium absorbed by our body through vital organs. Understanding this correlation of calcium molecules binding with caseins and with other mineral complexes has been derived by calorimetry techniques like ITC, DSC and Ion Selective Electrode, with the obtained data analysed using ORIGIN software to find out the goodness of fit and other suitable variables that allowed us to infer thermodynamic properties like enthalpy and entropy. This analytical approach allows us to ascertain the goodness of fit and identify relevant variables, contributing to a more nuanced comprehension of these molecular interactions. Apart from the inherent properties of milk, we will also provide the tendency of pH variation with temperature changes of milk after undergoing heat treatment methods like UHT and pasteurization. This aspect is crucial for understanding the implications of these widely employed methods in process industries, offering a practical dimension to the study. Overall, these inferences will enable us to deeply understand the co-existence of calcium molecules with other minerals in milk and suggest ways to enhance the parameters for maximum solubility of calcium with the protein molecules and higher bioavailability to ensure adequate calcium intake by our body.

Keywords: Casein ; ITC ; Micelles ; Titration ; UHT.

BIOTRANSFORMATION OF HEXAVALENT CHROMIUM FROM WASTEWATER USING BACILLUS FIRMUS

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Chromium is a naturally occurring element present on the earth's crust, which acts as an adulterant to food, water, soil, air and causes a vulnerable effect on humans. Among the various oxidation states, the hexavalent form has higher toxicity, higher solubility in water, and higher mobility in soil. The maximum recommended permissible limit of Cr in drinking water is 0.01 mg/L by US EPA whereas 0.05 mg/L by BIS & WHO. Several physical and chemical methods have been developed for the removal of chromium from contaminated water. In most of the techniques, there is a higher chance of re-contamination after the disposal of generated sludge. In this study, *Bacillus firmus* have been exploited for in-situ bioremediation of Cr⁶⁺ from wastewater under batch mode. The effect of operating conditions such as pH, initial concentration, RPM, external carbon and nitrogen sources were investigated. The maximum Cr⁶⁺ conversion of 99.9% was obtained at 0.5 and 1 ppm initial concentration, pH 7, 30°C and 120 RPM. Dextrose was found to be the potential C source for enhancing microbial growth and efficient conversion at higher Cr⁶⁺ concentrations. Furthermore, living and dead biomass revealed a comparatively high biosorption of 18.3% at 100 ppm Cr⁶⁺ concentration.

Keywords: *Bacillus firmus*; Bioremediation; Biosorption, Chromium removal; Waste water treatment

AI, Machine Learning and Data Analytics in Process Design/Optimisation (AML)

COMPARISON OF MACHINE LEARNING MODELS TO PREDICT THE PHASE EQUILIBRIUM OF N₂-CO₂ SYSTEM FOR CRYOGENIC CARBON CAPTURE

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Cryogenic carbon capture (CCC) represents a class of methods to capture carbon dioxide by applying subzero temperatures. All the CCC methods need a knowledge about the phase equilibrium characteristics of CO₂-laden gas. N₂-CO₂ is a common mixture that represents major portion of the emission gases from various process industries. The conventional ways of using equations of state (EoS) to predict the phase equilibrium have limited domain of their applications. Moreover, the computational time in solving the EoS is often high. Machine learning (ML) models can provide quicker solutions of the EoS by using statistical techniques. Such models work based on the real-life data from the process under study. Experimental data on phase equilibrium of N₂-CO₂ system in cryogenic domain are limited. Therefore, to train the machine learning models, phase equilibrium data were generated using conventional EoS. In this work, we present a comparison of different machine learning models to predict the phase equilibrium of N₂-CO₂ mixture. Four models were used for prediction; support vector regression-linear (SVR-lin), support vector regression-radial based function (SVR-RBF), multiple linear regression (MLR) and multilayer perceptron (MLP). The models were tested to find out the best model and the best test train ratio that predicted the highest R² value. Multilayer perceptron gave the best R² value of 0.9994 and 0.9983 in the prediction of solid vapor equilibrium (SVE) and vapor liquid equilibrium (VLE) respectively for a test train ratio of 80:20.

Keywords: Carbon capture; Desublimation; Equation of state; Solid vapor equilibrium; Vapor liquid equilibrium.

OPTIMAL SELECTION OF ENERGY STORAGE MEDIUM FOR CRYOGENIC ENERGY RECOVERY USING A MULTICRITERIA-BASED DECISION ANALYSIS

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In the natural gas supply chain, LNG is regasified before distribution to the customers. During regasification, vast amount of ultra-cold (cryogenic) energy (~ 111 K) is generated. It has been suggested to recover this cold energy to meet the cooling requirements in various industrial and domestic applications. The energy recovery may be done using a suitable energy storage medium. The energy may be stored as sensible heat and/or latent heat in a medium. Choice of the energy storage medium is critical in determining the overall thermal performance, capital and operating costs, safety of the system. The storage medium is expected to offer rapid energy storage and release rates, high energy capacity at the source temperature with low costs and without creating any operational hazard. Consideration of these factors often leads to a multiplicity of choices, and thus an optimized selection of the energy storage medium based on some user-defined criteria and constraint(s) becomes necessary. However, to the best of the author's knowledge, no systematic guidelines and framework exist in the open literature to optimally select an energy storage medium for a cryogenic energy recovery system. In this paper, we present a study on the application of multicriteria decision analysis (MCDA) to select an energy storage medium to recover the cold energy during LNG regasification. This work should help the material scientists and engineers formulate novel materials for cryogenic energy storage.

Keywords: Cryogenic energy storage; LNG; sensible heat; latent heat; Phase change material.

MACHINE LEARNING FRAMEWORK FOR PREDICTING IGNITION PROPERTIES OF PURE AND FUEL MIXTURES

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To expedite development of alternate fuels it is important to develop reliable and accurate models that can predict ignition properties of fuels. Using existing data from the literature, we have developed three machine learning models to predict cetane number (CN), research octane number (RON) and motor octane number (MON) for pure and fuel mixtures. Our dataset consists of hydrocarbons, alcohols, esters, ethers, aldehydes, carboxylic acids, ketones, aromatics, etc. as fuel compounds. Each data point is represented using a 32-dimensional feature matrix. The feature set comprises of a mixture of functional group identifiers, functional group position descriptors and fuel reactivity descriptors. To extract these features, an automated feature extraction algorithm is implemented in Python. The input to this algorithm is the IUPAC name of the molecule and the output is the corresponding feature set. The developed feature extraction algorithm relies on PubChemPy and RDKit libraries. PubChemPy is used to convert the IUPAC names to the SMILES representation of molecules, while RDKit is used to extract relevant features from the SMILES notation. Using these features and XGBoost algorithm, we have developed three models to predict CN, RON, and MON of pure and fuel mixtures. XGBoost models for prediction of CN, RON, and MON resulted in R2 values of 0.906, 0.838, and 0.826, respectively. Each XGBoost models was further interpreted using SHAP analysis, an explainable machine learning algorithm. The features obtained from SHAP analysis as most important contributors towards predicting the ignition properties of fuels are indeed found to be the most important features based on the knowledge from existing literature. Thus, insights obtained from SHAP analysis enhances the reliability of our models. To enhance the usability of our proposed models, we have deployed the entire framework online that can be accessed freely using the following link: <https://fuels-xgb.onrender.com/>.

Keywords: Cheminformatics; Cetane number; Decision tree; Interpretable machine learning; Octane number.

ENHANCING OSMOTIC COMPUTING USING MACHINE LEARNING FOR DYNAMIC RESOURCE OPTIMIZATION IN CLOUD ENVIRONMENTS

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Cloud computing is a widely accepted approach that delivers on-demand infrastructure. It faces the challenge of efficiently managing dynamic workloads across the distributed network. This research paper explores a synergistic approach by combining Osmotic Computing with the capabilities of machine learning to enhance system adaptability and optimize resource allocation. The concept of Osmotic Computing is inspired by the process of Osmosis in chemistry. It forms the foundation for the automatic deployment and migration of services in cloud environments. Integrating Machine Learning with OC enables the system to learn from real-time and historical data. It empowers it with predictive analysis to determine patterns, environmental conditions, and resource usage. The paper proposes a novel approach to enhance resource allocation in a cloud-edge environment by integrating Osmotic Computing with a modified KNN algorithm. The effectiveness of the proposed framework is demonstrated through experimental validation, showcasing significant improvements in resource allocation efficiency and system performance.

KEYWORDS : Osmotic Computing, segregation, microservices, macroservices, machine learning, KNN-regression, edge computing.

GUI DEVELOPMENT FOR DATA TREATMENT IN RADIOTRACER BASED RTD ANALYSIS OF INDUSTRIAL PULP DIGESTERS USING DATA SCIENCE

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Residence Time Distribution (RTD) data analysis and GUI development has been done in this work. The radiotracer based RTD investigations of industrial pulp digester is considered for the study. The RTD data is treated as starting point correction, background correction, half life correction, and tail correction is done using the proposed method. RTD is usually done manually and prone to errors and noises. The manual data corrections as stated above are tedious and time consuming tasks. This paper therefore offers a comprehensive approach to utilise Data Science techniques for pretreatment of RTD experimental data for a continuous pulp digester. To increase the efficiency of RTD modelling and expedite data preprocessing, a user-friendly Graphical User Interface (GUI) has been proposed that aims to streamline the process of experimental RTD data analysis. The GUI offers features like uploading of raw dataset, visualisation, and facilitating stated corrections in an accurate way. The GUI gives the features like NULL value treatment, outlier identification, data format correction, missing value corrections, and value predictions. The GUI offers different RTD model graph plotting, and comparison options. The treated dataset can be downloaded for further RDT model prediction using Machine Learning approaches. This research represents a significant step towards harnessing advanced data analytics to optimise complex industrial processes of RTD modelling.

PREDICTION OF RESIDENCE TIME DISTRIBUTION (RTD) MODELS USING MACHINE LEARNING TECHNIQUES FOR CONTINUOUS FLOW PULPING DIGESTER

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This study explores the prediction of Residence Time Distribution (RTD) models in the continuous flow pulping digester, employing radiotracer technology using Machine Learning techniques. The Primary objective is to evaluate the predictive ability of different machine learning models in estimating the degree of fit between the Axial Dispersion Model (ADM) and tank in series with Back Mixing Model (TIBM) with a Plug Flow Component in Series with experimental data. Multiple machine learning techniques, including regression models like linear regression, support vector regression (SVM), random forest (RF), K-Nearest Neighbours (KNN), generalised additive model (GAM) and Time series models were utilised to create predictive models. The study's findings offer a thorough comparison of different machine learning strategies for forecasting the degree of agreement between experimental data and RTD models. By employing mean squared error (MSE) mean absolute error (MAE), and R-squared (R²) score as the evaluation metric, we offer a robust assessment of each model's accuracy in capturing the likeness between experimental data and the ADM and TIBM models.

APPLICATION OF OPPOSITE POINT-BASED DIFFERENTIAL EVOLUTION (OPDE) ALGORITHM FOR MODELING VLE DATA

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The field of chemical engineering has shown significant interest in parameter estimation within thermodynamic models due to its intricate nature, involving challenges such as nonlinearity, a flat objective function near the global optimum, badly scaled models, and equations containing non-differential terms. Recently, an improved version of classical differential evolution (DE) called the opposite point-based differential evolution (OPDE) algorithm has been proposed in our previous work. Unlike differential evolution (DE), the OPDE algorithm uses both the opposite point based and random initialization method to initialize population. In this study, OPDE algorithm is applied to the estimation of parameters in the vapor-liquid equilibrium (VLE) models to fit the experimental data. Parameter estimation based on both the least squares (LS) and error-in-variables (EIV) approaches are considered. Performance results of OPDE are compared for different objective functions. Twenty-three VLE datasets for binary systems at various temperatures and pressures are considered. Furthermore, VLE datasets for three different binary isobaric systems has been merged and the parameters has been estimated for each merged dataset. The results of estimated parameters for activity coefficient models, the deviations in terms of average absolute deviation (AAD) and root mean square temperature deviation (RMSTD) are presented and are compared with those reported in literature. The percentage improvement in RMSTD value (with respect to RMSTD value reported in literature) for fourteen isobaric problems has been found within the range 7% - 66% for LS approach and 91% - 99% for EIV approach. The results indicated the efficiency and robustness of the OPDE with 100% success rate for LS approach. However, EIV approach has been found to be more reliable. In addition, new results obtained for most of the studied thermodynamic models are presented.

Keywords: Evolutionary algorithm; Global optimization; Isobaric; Parameter estimation; Thermodynamic models.

DEVELOPMENT OF NEW CORRELATION FOR THE PREDICTION OF POWER NUMBER FOR CLOSED CLEARANCE IMPELLERS USING MACHINE LEARNING METHODS TRAINED ON LITERATURE DATA

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The accurate estimation of the power number for closed clearance impellers holds significant importance in industries such as chemical, biochemical, paper and pulp, as well as paints and pigments. Existing state-of-the-art correlations for predicting power numbers, however, have certain limitations. They are often specific to impeller types, limited to certain ranges of Reynolds Number, and applicable only within defined impeller dimensions. This underscores the necessity for a more comprehensive correlation that encompasses all types of closed clearance impellers and a broad spectrum of their dimensions. In this study, we compiled a dataset of 1470 data points from 15 research articles in the open literature, covering five types of impellers: i) Anchor; ii) Gate; iii) Single Helical Ribbon; iv) Double Helical Ribbon; and v) Helical Ribbon with screw. Six machine learning models, namely Artificial Neural Networks, Catboost Regressor, ExtraTreeRegressor, Support Vector Regressor, Random Forest, and XGBoost Regressor, were developed and compared. The results revealed that Artificial Neural Networks (ANN) emerged as the most efficient model, demonstrating the highest testing R² value of 0.99 and the lowest testing MAPE of 7.3%. Additionally, a sensitivity analysis was conducted using permutation feature plots. The proposed machine learning correlation proves to be applicable across all five types of impellers and a wide range of their dimensions. It covers the Reynolds Number range for impellers from 0.1 to 106.

Keywords: Closed Clearance Impeller; Data-Driven Modelling; Newtonian Fluid; Power Consumption; Sensitivity Analysis.

SELECTION OF AN ALTERNATE CEMENTITIOUS MORTAR USING CERAMIC TILE DUST WASTE: A HYBRID MCDM APPROACH

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Selection of a suitable alternative material from a pool of alternatives with many conflicting criteria becomes a Multi Criteria Decision Making (MCDM) problem. In the present study, ternary blended mortars were prepared using Ceramic Tile Dust Waste (CTDW), Fly ash (FA) and Ground Granulated Blast Furnace Slag (GGBFS) as binder components. Crusher Dust (CD) was used as fine aggregate component. Binder to aggregate ratios of 1:3 and 1:1 was prepared considering suitable flow. A total of 16 mortar mixes were cast. These mortars were tested for various conflicting criteria compressive strength, flexural strength, porosity, water absorption, bulk density, thermal conductivity, specific heat, thermal diffusivity and thermal effusivity whose weightages obtained were 29.09%, 20.08%, 12.77%, 10.60%, 8.74%, 6.74%, 5.54%, 4.47% and 1.97% respectively as per AHP analysis. Later, considering these different criteria and alternate mortars it was observed that 1:1 mortar with 20% CTD, 30 % FA, 50% GGBFS found to be the suitable mortar with highest relative closeness coefficient of 0.861 and highest net outranking flow of 0.316 with respect to MCDM techniques TOPSIS and PROMETHEE-II respectively.

Keywords: Ceramic Tile Dust Waste, Fly ash, GGBFS, TOPSIS, PROMETHEE-II.

PREDICTION OF REACTANT CONCENTRATION IN BATCH REACTOR THROUGH SOFT SENSOR USING ELECTROCHEMICAL PARAMETERS

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Esters are important industrial products used as flavouring agents, perfumes, solvents, medicinal agents, and sometimes as jet fuel. Esterification is a complex process and analysis of products involves time and material consuming methods. This study involves a two-step experimental process focused on synthesis of ester in batch reactor followed by recording of electrochemical signatures focused on cyclic voltammetry. The collected data is organized in the form of a matrix comprising temperature, molar ratio, and current peak heights. PCA is utilized for classification of data at different time intervals leading to development of a soft sensor using multilinear regression machine learning algorithm. This approach is anticipated to provide insights into the relationship between synthesis parameters and electrochemical responses, enabling the creation of predictive model for process optimization and control in ester production.

Keywords: Ester; butyl butyrate; cyclic voltammetry; Principal component analysis; Machine learning.

A NEW INTEGRATED COLOR BALANCE TECHNIQUE FOR UNDERWATER IMAGE ENHANCEMENT

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This research paper presents a novel Integrated Technique for enhancing underwater images, which are often plagued by color distortion and visibility degradation. Current methods rely on a range of assumptions and constraints to achieve satisfactory results in underwater image enhancement. In contrast, our approach is a single-image method that does not necessitate specialized hardware or expertise in underwater conditions or scene structure. The proposed methodology entails a fusion of conventional contrast enhancement techniques utilizing CBMG and fuzzy histogram equalization techniques. To evaluate its efficiency, certain statistical parameters are employed. The proposed approach involves a synthesis of traditional contrast enhancement techniques and fuzzy-histogram equalization techniques. The experimental findings indicate that the utilization of fuzzy logic techniques is efficacious in augmenting the caliber of underwater images. The purposed method has batter visual effects and color image quality evaluation vision compared with other popular method. Resulted image also compared with underwater color image quality evaluation (UCIQE) compared with other popular methods.

Keywords: Image enhancement, underwater images, fuzzy-histogram equalization, CBMG.

EMBEDDING BASED LINK PREDICTION IN PROTEIN-PROTEIN INTERACTION NETWORKS BY ATTRIBUTED GRAPH EMBEDDING

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This research focuses on protein-protein interaction networks (PPINs) and the prediction of interactions between proteins using an attributed embedding technique. The proposed Ensemble Node Embedding-based Link Prediction (ENELP) framework considers network topology and various node attributes for link prediction, including structural and attribute-wise proximity, vertex features, text features, node network structure and content, node context, and node attributes. The study integrates features of seven state-of-the-art attributed node embedding-based approaches and compares the proposed approach to traditional machine-learning and ensemble-based classifiers. The results demonstrate that the optimized ENELP model outperforms individually attributed node embedding-based link prediction methods when tested on five different well-known PPINs.

Keywords: Protein-protein network, link Prediction, Machine Learning, attributed embedding, Ensemble-based classifier.

COMPARATIVE STUDY OF WIRELESS SENSOR NETWORKS FOR SUSTAINABLE ENVIRONMENTAL SURVEILLANCE

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With the number of environmental problems rising, environmental monitoring is crucial. Wireless Sensor Networks (WSNs), which have the ability to gather and analyses data in real-time, have become an important technological tool for sustainable environmental surveillance. WSN consists of various heterogonous nodes present in different environment conditions. Because of this heterogeneity, WSNs require specific routing protocols in order to effectively manage node-to-node communication and data sharing. In order to evaluate the effectiveness of heterogeneous routing protocols, this research provides a thorough comparative analysis of them. This work offers a comprehensive comparison that evaluates and examines a number of WSN features used for long-term environmental monitoring. The paper examines popular heterogeneous routing protocols such as Cluster-based protocols, on the basis of key performance metrics such as energy efficiency, load balancing, network lifetime, and number of alive nodes in specific interval of time. The outcomes of this comparative study provide valuable insights for researchers and practitioners involved in the design, deployment, and management of WSNs for sustainable environmental surveillance. By identifying the strengths and weaknesses of different approaches, the research aims to contribute to the development of more robust, energy-efficient, and scalable WSN solutions, ultimately advancing the cause of environmental sustainability through effective monitoring and management.

Keywords: Routing Protocols; Energy Efficiency; Clustering; Performance Metrics; Network Structure.



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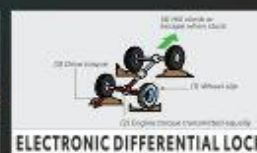
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